An Empirical Analysis of the Interconnection Queue*

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Abstract

Electricity generation projects seeking to connect to the U.S. power grid go through an interconnection queue, where a sequence of engineering studies determines each project's interconnection cost. Most projects that begin the process do not complete it. Using new data, we find that a long queue delays the necessary engineering studies. We develop and estimate a dynamic equilibrium model in which entry, waiting, withdrawal, completion, and queue congestion are jointly determined, and quantify the effects of policy reforms. Speeding up study delivery by 50% raises completed generation capacity by 32%. Increasing the cost of entering the queue has little effect on waiting time or completed capacity, because it primarily deters projects that would spend little time in the queue and therefore contribute little to congestion.

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1 Introduction

Meeting climate goals requires large-scale investment in low-carbon electricity generation. Yet connecting a new generator to the U.S. power grid is difficult. The interconnection process can take years, and less than a quarter of the wind and solar projects that begin the process ultimately connect (Rand et al., 2025). Projects waiting to connect can also delay the studies required for projects behind them. We study how congestion and the design of this process affect investment in new generation.

To connect to the transmission grid, a project joins a waitlist known as the interconnection queue. The grid operator then conducts up to three engineering studies, giving priority to earlier entrants. The studies estimate the project's interconnection cost, which it must pay to connect. This cost covers the equipment needed to connect the project to the grid and any upgrades the broader network needs to accommodate it. As each study arrives, the project learns more about its eventual cost and decides whether to keep waiting or withdraw. After its last study, it decides whether to pay the cost and connect.

Historically, developers could preserve the option to connect by paying only small fees and deposits to enter and remain in the queue. These fees and deposits may not have fully reflected the delays projects imposed on others. Grid operators devote substantial resources to processing requests that are ultimately withdrawn. These concerns have motivated recent reforms to speed up study delivery and to raise the cost of entering and remaining in the queue. This paper quantifies the externalities in the queue and uses a dynamic equilibrium model to estimate how such reforms affect completed generation capacity.

Using novel data from PJM, the largest U.S. regional grid operator, we first document long delays and substantial uncertainty. Our data cover all generator interconnection requests from 2008 through 2020: 4,174 requests and their 7,492 engineering studies. PJM expects the third and final interconnection study to take 6 months, but the median wait is 15 months and the 90th percentile is 36 months. Interconnection costs are also highly dispersed: the median second-study cost is $50,000 per megawatt (MW), while the 90th percentile is $420,000 per megawatt, roughly a quarter of the installation cost for wind and solar projects. Observably similar projects can have very different interconnection costs, consistent with developers' reports that these costs are difficult to predict (Caspary et al., 2021).

We next show that high interconnection costs cause projects to withdraw from the queue.

Withdrawal rates rise with cost at each study. However, a developer may pursue a project with high expected interconnection costs only if it is otherwise especially profitable, creating a correlation between cost and unobserved profitability. We therefore instrument for cost with the change in cost across studies, which is difficult to predict. A high second-study cost (above $100,000 per megawatt) increases the probability of withdrawal before the third study by 31 percentage points. The effect of the third-study cost on withdrawal is similar.

We then quantify three potential externalities. First, a congestion externality arises if projects ahead in the queue delay the studies of those behind them. Second, a cost externality arises if projects affect one another's interconnection costs by using or expanding shared transmission infrastructure. Third, a local output externality arises if completed projects reduce local wholesale electricity prices and thus the expected profits of nearby projects still in the queue.

We find that the congestion externality is quantitatively important. We estimate how the number of higher-priority projects affects a project's probability of receiving a new study in a given quarter. Once a project requests a study, a developer cannot control when it arrives, and priority is based only on entry date. A 10% increase in higher-priority projects reduces the probability of receiving the third and final study by 13.2%, on average.

In contrast, we find little evidence that the cost or local output externalities are quantitatively important in our setting. We test whether other projects in the queue or completed interconnections affect a project's interconnection cost and find small, mostly statistically insignificant effects. Using a staggered difference-in-differences design, we likewise find that a new project beginning operation has little effect on wholesale prices at its location relative to other locations in PJM. The latter result is consistent with PJM's deliverability requirement, under which connecting projects must fund transmission upgrades that mitigate local grid congestion. These findings motivate our focus on the congestion externality in the model.

We then develop a dynamic equilibrium model of the queue to quantify the effects of policy reforms. Projects choose whether to enter the queue before knowing their eventual interconnection cost. Once in the queue, they decide whether to keep waiting or withdraw as each study reveals new information, and congestion affects how quickly the next study arrives. Because these decisions in turn determine the size and composition of the queue, we define and solve for a nonstationary equilibrium in which project strategies and queue congestion

are jointly determined, and prove its existence.

We estimate the model in three steps. First, we estimate models of state transitions, including new study arrival rates, interconnection costs, and the additional information revealed by each study, and use these estimates to construct project beliefs. Second, we use observed withdrawal and completion decisions to recover project payoffs and waiting costs. We prove that the distribution of payoff heterogeneity and the waiting costs are identified by within-project changes in interconnection costs across studies. Because these changes are difficult to predict, each new study shifts a project's continuation value. These parameters are identified from how withdrawal and completion decisions respond to these shifts. Third, we use the implied value of entering the queue to estimate entry costs. The key identifying variation is the decline in renewable construction costs over time.

We use the estimated model to quantify the effects of two policy changes: speeding up study delivery and increasing the cost of entering the queue. Both are part of recent reforms. PJM's 2025 Tapestry partnership aims to use AI to speed up engineering studies (PJM Interconnection, 2025*b*), while Federal Energy Regulatory Commission (FERC) Order 2023 penalizes study delays, raises study deposits, and introduces withdrawal penalties (FERC, 2023).

We find that faster study delivery substantially increases completed capacity. For 2012–2020 queue entrants, speeding up study delivery by 50% increases completed capacity by 32%, or 17.8 GW, and reduces average waiting time by 21%. Of the additional capacity, 60% (10.7 GW) is renewable. At a social cost of carbon of $185 per metric ton, this increase in renewable capacity would create $2.3 billion in annual benefits.

Increasing the cost of entering the queue has much smaller effects. We consider a nonrefundable fee and two refundable deposits, including a staged deposit similar to PJM's ongoing reform. At $20,000 per megawatt,[1] the effects on completed capacity range from a 1.2% decrease under the nonrefundable fee to a 0.8% increase under the staged deposit, with little effect on average waiting time. Budget-neutral versions of the deposit and staged deposit, which rebate forfeited deposits to projects that complete, increase completed capacity by 3.6% and 4.2%.

[1]This level is high enough to distinguish the effects of the three fees and deposits and is comparable to the fee PJM adopted in 2026 for a temporary program that addresses short-term reliability concerns (PJM Interconnection, 2026). The level in PJM's broader reforms is $4,000 per megawatt, at which the effects are smaller but in the same direction.

The limited effects of fees and deposits reflect a positive correlation between a project's value and its contribution to congestion. Higher-value projects are willing to remain in the queue longer and therefore contribute more to congestion, but are also less deterred by higher fees and deposits. Higher fees and deposits at entry instead screen out projects that would have contributed little to congestion. The staged deposit is more effective because it screens projects after the first study, when they have learned more about their interconnection costs and payoffs. However, projects that remain at later stages have higher value and are more likely to complete. The same positive correlation between value and congestion therefore limits the gains from raising fees and deposits at later stages.

Related Literature

The interconnection queue is a critical but understudied bottleneck to investment in new electricity generation. While it has received attention in the energy policy literature (Gergen, Cannon Jr and Torgerson, 2008; Alagappan, Orans and Woo, 2011; Gorman et al., 2024), it has been largely absent from formal economic analysis, primarily due to data constraints. We collect novel data on the PJM queue and make them publicly available. In a companion paper (Johnston, Liu and Yang, 2026), we use these data to study the determinants and consequences of interconnection costs. In concurrent work, Gowrisankaran, Langer and Ryan (2026) study cost externalities under a different queue design in California. Another closely related paper is Gonzales, Ito and Reguant (2023), which studies the effect of transmission grid expansion on renewable entry; our paper instead focuses on the interconnection queue as a barrier to entry. We also contribute to a broader literature on how transmission constraints affect competition, emissions, and investment (Wolak, 2015; Ryan, 2021; Davis and Hausman, 2016; Fell, Kaffine and Novan, 2021; LaRiviere and Lyu, 2022; Doshi, 2026), as well as on the institutional barriers to expanding transmission infrastructure (Davis, Hausman and Rose, 2023; Hausman, 2025).

This paper contributes to the energy and environmental economics literature studying investment and industry dynamics (Ryan, 2012; Gowrisankaran, Reynolds and Samano, 2016; Fowlie, Reguant and Ryan, 2016; Blundell, Gowrisankaran and Langer, 2020; Butters, Dorsey and Gowrisankaran, 2021; Elliott, 2022; Gowrisankaran, Langer and Zhang, 2022; Leisten and Vreugdenhil, 2023; Chen, 2025). A distinctive feature of our setting is that projects' entry

and continuation decisions create congestion that slows the necessary engineering studies, thereby affecting other projects' investment decisions. By quantifying how interconnection queue reforms affect renewable entry, we contribute to the literature on public policy and renewable energy investment (Metcalf, 2010; Hitaj, 2013; Johnston, 2019; Aldy, Gerarden and Sweeney, 2023; Deschenes, Malloy and McDonald, 2023).

We also contribute to the empirical literature on dynamic assignment mechanisms (Agarwal et al., 2021; Waldinger, 2021; Verdier and Reeling, 2022; Liu, Wan and Yang, 2024; Huitfeldt, Marone and Waldinger, 2024) by studying a queuing problem in a novel and consequential market. Methodologically, we develop a tractable equilibrium concept for a nonstationary queue with time-varying unobserved heterogeneity. The size and composition of the queue shifted substantially over our sample period, changes that standard oblivious equilibrium (Weintraub, Benkard and Van Roy, 2008) does not accommodate. We build on the nonstationary oblivious equilibrium of Weintraub et al. (2010). Following Igami (2017) and Yang (2020), we use a finite-horizon framework in which projects anticipate aggregate trends, such as declining renewable construction costs and rising queue congestion. We establish equilibrium existence and identification of payoff heterogeneity and waiting costs in this environment.

2 PJM Interconnection Process

The grid operator, PJM, runs the interconnection process, which involves two other parties: developers and transmission owners (TOs). Developers (e.g., NextEra Energy) enter their projects (e.g., 2.5 MW Front Royal Solar Field) in the queue and pay the interconnection cost identified in the studies. These developers can be either independent power producers (more common for renewables) or regulated utilities (more common for natural gas). PJM conducts the interconnection studies in conjunction with the local TO, which supplies the design standards and engineering input and constructs the upgrades (PJM Interconnection, 2020*a*). We assume that the cost developers pay reflects the competitive cost of these upgrades. PJM is a nonprofit grid operator, and TOs are regulated utilities that earn no return on generator-funded upgrades.[2]

[2]PJM also plans transmission separately from the interconnection process, through the Regional Transmission Expansion Plan (RTEP). This investment is driven by reliability needs and funded by consumers through transmission rates. Over our sample period, the RTEP did not consider projects waiting in the queue

2.1 Queuing Rules

Projects can enter the queue at any time, but each year is typically divided into two windows. Projects that enter in the same 6-month window are put in the same cohort and receive up to three studies sequentially. Through these studies, projects learn increasingly accurate information about their interconnection cost. At any point in the process, a project may withdraw from the queue. PJM may require just one or two studies if it determines that a project is not required to make significant network upgrades or share costs with other projects. After its final study, which we call the terminal study, is issued, the project chooses to leave the queue or sign an interconnection service agreement in which it agrees to pay the final interconnection cost, thus completing the interconnection process.

Developers must pay to enter the queue and to advance through it. Before entry, a developer must secure land sufficient to build a project (PJM Interconnection, 2023*b*), typically through exclusive options to lease the land. Developers must also pay a deposit at entry and to advance to each subsequent study. The deposit amount depends on the request size and stage, with larger requests typically requiring higher deposits. For the median request size of 20 MW, the initial deposit was $10,000 to $15,000, while the deposits for the second and third studies were $10,000 and $50,000, respectively. Projects that withdraw have their deposit returned, less a 10% nonrefundable portion and any study costs already incurred (PJM Interconnection, 2023*a*).

The official timeline sets a target duration for each step of the interconnection process. For projects that apply within the same cohort, PJM starts conducting the first studies (feasibility studies) one month after the closing of the time window. Within three months, projects are supposed to receive their first studies. Upon receipt, projects have one month to decide whether to advance to the second study (system impact study). The second study is then scheduled to take four months; upon receipt, projects have one month to decide whether to request the third and final study (facility study). The third study is scheduled to take six months. Finally, projects withdraw or agree with PJM on final details and sign the interconnection service agreement, over a 6.5-month period (PJM Interconnection, 2021*a*).[3]

(PJM Interconnection, 2020*b*).

[3]This agreement also sets a commercial operation date. After signing the interconnection service agreement, the project may suspend the process for up to 3 years (up to 1 year if the suspension has a negative impact on subsequent requests) (PJM Interconnection, 2025*c*), though these suspensions are rare.

2.2 Study Information

Each study reports an interconnection cost estimate along with engineering information. The interconnection cost has two components: the direct cost to connect the project to the grid, and the cost of network upgrades to relieve system overloads the new project triggers.[4] These costs cover both the equipment and its installation. The first study reports preliminary direct costs. The second study reports the network upgrade costs, including the project's share, and updates the direct costs. The third study finalizes engineering specifications and costs.

The engineering tests reported in each study determine which network upgrades the project is responsible for and whether it shares that responsibility with other projects. Three tests—generator deliverability, multiple facility contingency, and short circuit analysis—jointly quantify how much the project will overload the grid; they are conducted in either the first or the second study, depending on project characteristics and local transmission network conditions. Two additional tests—short circuit dynamic analysis and system protection analysis—identify whether the project triggers the same network violation as other projects in the queue, in which case PJM assigns it to a cost-share group jointly responsible for the upgrade.

Studies also report additional information that affects a developer's assessment of the project. They specify the types of equipment required to connect the project, which can affect the costs of permitting and right-of-way access. Studies expressly exclude these costs, which can be substantial (Liu, 2025). Studies may also flag requirements whose cost is not included in the interconnection cost, like bringing facilities up to the transmission owner's published standards.[5]

2.3 Timing of Interconnection within Project Development

Projects apply for interconnection early in project development. The permitting and interconnection processes occur simultaneously, and PJM developers rate the length of the interconnection process as a larger barrier than siting and permitting (Silverman et al., 2024). Projects signing long-term contracts to sell their power, which include nearly all renewable

[4] We combine the two components into a single cost measure for our analysis; see Johnston, Liu and Yang (2026) for more detail and descriptive statistics on each.

[5] For example, the impact study for a 190 MW addition at an existing gas plant (AA2-030) assesses only a $100,000 cost for the transmission owner's oversight, but the project "is required to construct all connection facilities in accordance with the TO published standards. ComEd would review and approve all customer relay protection design drawings and relay settings."

projects, do so while in the queue or after receiving the terminal study. After signing the interconnection service agreement, the project and the required interconnection infrastructure are constructed, typically in less than a year, far shorter than the time spent in the queue.[6]

The slow, unpredictable interconnection process can disrupt other steps in project development. Power buyers are willing to sign long-term contracts only a few years before a project begins operation, but developers must apply for interconnection much earlier. Developers must therefore enter a project into the queue before knowing whether it will secure a contract. A long wait may also cause signed long-term contracts or a developer's land lease option to expire, prompting the developer to renegotiate the lease or abandon the project (Silverman et al., 2024).

Waiting projects also face risks unrelated to interconnection that can cause cancellation. A project may not be selected in a competitive solicitation for a power purchase agreement, fail to secure a necessary permit, or encounter local opposition that blocks siting. Ken Young, CEO of the wind and solar developer Apex Clean Energy, describes the risk of local opposition: "The longer the development takes, even people that want to support projects are likely to wear out. The pressure builds, and it is easier to be against things" (Norton Rose Fulbright, 2026). Each additional quarter in the queue is another quarter of exposure to these shocks, so a slower interconnection process increases cancellations.

2.4 Queue Design Reforms

Recent regulatory reforms aim to address the growing interconnection backlog. In 2023, FERC, the federal regulator overseeing U.S. transmission policy, approved Order No. 2023, which aimed to shift queue priority from "first-come, first-served" to "first-ready, first-served" (Hale and Christian, 2023). The Order has four main components: (a) raising the cost to enter and remain in the queue, (b) imposing penalties for withdrawing from the queue, (c) imposing penalties for delays in completing studies, and (d) transitioning from studying each request serially to cluster studies, under which requests are grouped into large clusters and receive a single study per cluster. The study-delay penalties in (c) fall on the transmission provider, which in PJM's case is PJM itself rather than the transmission owners; the penalties cannot

[6]For wind projects, construction typically takes six months to a year (AWEA, 2019); solar projects follow similar timelines. Network upgrades can take longer, but projects can sometimes begin operating before all upgrades are completed.

be recovered through transmission rates, though PJM may seek FERC approval to recover a penalty from a transmission owner at fault (FERC, 2023). Implementation is ongoing: PJM, for example, opened its first reformed cluster cycle in 2026, with applications due April 27 (PJM Interconnection, 2025*a*).

3 Interconnection Request Data and Descriptive Evidence

Our data cover PJM interconnection requests and the engineering studies those requests received. We first describe the data construction and then document four facts that underpin the structural model: entry to the queue increases over time, waits for later studies are long and uncertain, changes in interconnection costs across studies are difficult to predict, and withdrawals concentrate in the quarter a study arrives. Using cost changes across studies as an instrument, we also show that high interconnection costs cause withdrawal.

3.1 Data Sources and Construction

Our main data cover the 4,174 interconnection requests to PJM from 2008 to 2020 with statuses observed through the first quarter of 2023.[7] Each request is to connect a single project at a specific point on one transmission owner's network. This point is typically a substation but can also be a transmission line tap.[8] A project is either a new plant or a capacity increase at an existing plant; the study process is the same for both. Requests are almost always distinct projects rather than resubmissions: only 6.8% of requests enter at the site of a previously withdrawn request, and 4.1% do so within two years of that withdrawal.[9]

We manually collected data from the PDFs of the 7,492 engineering studies these projects received. From each study we record the issue date, interconnection cost estimate, the engineering tests performed and their results, and whether the project is assigned a network allocation.[10] From PJM, we also know the queue entry date, entry cohort, and withdrawal or operation date. These dates let us reconstruct, for each quarter, the set of requests waiting for each study and each request's priority relative to others, both PJM-wide and within its

[7] Earlier studies use irregular formats that make it difficult to identify the relevant interconnection costs.

[8] We associate each project with a substation: the substation where it connects, or, if it connects through a line tap involving multiple substations, the nearest of those substations.

[9] We treat requests with coordinates within 1 km of each other as the same site.

[10] Network allocation indicates whether a project is explicitly assigned a network upgrade cost in the study; this cost may be zero.

TO. Table 1 reports summary statistics.

Table 1: Summary Statistics

	1st Study		2nd Study		3rd Study	
	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.
Cost per MW of capacity	0.12	0.40	0.17	0.47	0.11	0.21
Wait time (mos.)	4.9	1.5	11.3	8.1	15.1	9.8
Size in MW	97	196	105	190	157	257
Capacity increase at existing plant	0.21	0.41	0.23	0.42	0.15	0.36
Solar	0.59	0.49	0.60	0.49	0.61	0.49
Natural Gas	0.17	0.37	0.16	0.36	0.19	0.39
Wind	0.09	0.29	0.10	0.30	0.12	0.33
Battery	0.10	0.30	0.10	0.29	0.05	0.21
Coal, oil, diesel	0.02	0.14	0.02	0.12	0.01	0.08
Other	0.03	0.18	0.03	0.17	0.02	0.15
Network allocation	0.04	0.20	0.59	0.49	0.62	0.48
Receive engr. tests	0.81	0.39	0.88	0.32	0.04	0.19
Distance to substation (km)	7.91	10.51	8.03	11.34	8.93	13.24
Wholesale electricity price ($/MWh)	35.36	14.54	36.93	13.82	40.74	16.86
Ordinance	0.28	0.45	0.31	0.46	0.32	0.47
Observations	4,168		2,495		829	

The sample is projects that entered the queue in 2008–2020. Cost per MW of capacity is the interconnection cost estimate, in millions of 2020 dollars, divided by the project's size in MW. Wait time for the 1st Study is wait in months for the first study after joining the queue. Wait time for the 2nd Study is wait in months for the second study after receiving the first study. Wait time for the 3rd Study is wait in months for the third study after receiving the second study. Reported wait times are capped at the 90th percentile (9 mos. for the first study and 36 mos. for the second and third studies) to limit the influence of outliers. The corresponding uncapped means are 5.3, 14.9 and 17.9. Capacity increase at existing plant is an indicator for a request to increase maximum generating capacity at an existing power plant; the size for these requests is the incremental capacity, not the total plant capacity. Network allocation is an indicator for receiving an explicit allocation of network upgrade costs, even if the allocated amount is $0; network upgrade costs are typically first reported in the second study. Receive engr. tests is an indicator for receiving any of three engineering tests: generator deliverability, multiple facility contingency, and short circuit analysis. Distance to substation is the distance from project to substation in km. Wholesale electricity price is the project substation's average real-time locational marginal price during peak hours in the quarter the study is issued. Ordinance is an indicator for a local ordinance restricting renewable energy development.

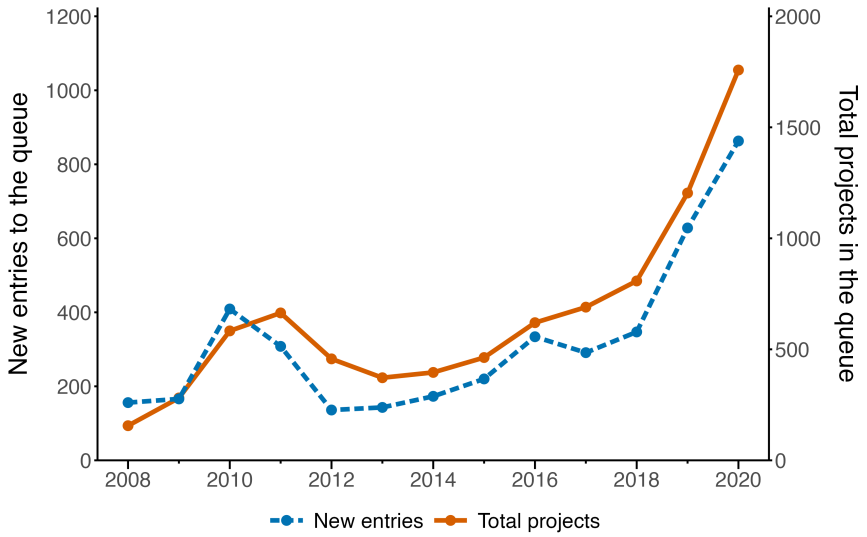
3.2 Queue Composition and Growth

Solar, natural gas, and wind dominate the PJM queue, together making up 85% of requests from 2008 to 2020. Natural gas projects tend to be much larger: they make up 17% of requests but 42% of requested capacity. The composition of requests shifted over our sample, with solar gaining share after 2015 and batteries after 2017. We refer to wind, solar, and battery

projects as "renewable" and all others as "non-renewable".

The number of interconnection requests has increased dramatically over time, driven by falling installation costs for renewable projects, especially solar (Seel et al., 2024; Wiser et al., 2024). Figure 1 plots, by year, new requests (dotted line) and the average number of projects waiting in the queue (solid line). The increase began around 2015 and was concentrated in renewables.[11] Because renewables are smaller on average, the rise in requested capacity was less pronounced than the rise in requests: 42 GW of new requests in 2008 compared to 64 GW in 2020. Consistent with a technology-driven explanation, similar increases in renewable requests occurred across all U.S. regional grid operators (Gorman et al., 2024).

Figure 1: Queue Size Over Time



The figure plots the size of the PJM interconnection queue from 2008 to 2020. The solid orange line is the average number of projects waiting in the queue, computed as the mean across the twelve months of the year of the number of projects that have entered the queue and not yet withdrawn or received their terminal study. The dotted blue line is the total number of new entries each year.

3.3 Developers and Transmission Owners

Developers. We can identify the developer for a subset of projects. From the second study onward, PJM lists each request's interconnection customer, often the limited liability

[11] The spike around 2010 was due to a temporary federal program that gave solar and wind developers the option to claim a cash grant in lieu of the nonrefundable investment tax credit (see Aldy, Gerarden and Sweeney (2023) for program details).

company holding the project rather than the developer that controls it. We match these names to developers using news articles, regulatory documents, and developer websites. We identify the developer for 46.1% of projects overall, 61.4% of those that reach the second study, and 85.2% of those that reach the third.

Although developers can control several projects, treating each project as an independent decision maker, as we do in the model, is a reasonable approximation. First, no developer is large enough to internalize much of the congestion its projects create: the 1,923 matched projects map to 405 unique developers, the largest of which (Invenergy) accounts for 4.3%, and the fifteen largest together account for less than 32% (Appendix Table E.1). Second, a developer has little reason to manage its projects as a portfolio. Projects are small relative to output and input markets, so one project has little direct effect on the profitability of another. A developer should therefore advance every individually profitable project. Financing or staffing constraints could force developers to choose among their projects, but developers can and do sell projects rather than withdraw them.[12]

Transmission Owners. PJM has 22 transmission owners that provide the engineering input the studies rely on.[13] Appendix Table E.2 shows that the average TO has 190 projects and 18.5 GW of queued capacity in our sample.

Projects are individually small relative to their TO. On average, a project accounts for about 0.5% of its TO by both count and capacity. Even at the 25th percentile of TO size, with 33 projects by count or 3.6 GW by capacity, the corresponding share is only about 3%. Developer concentration is also low within TOs: the average TO contains projects from 32 distinct developers.

3.4 Waiting Times

The first study arrives quickly and with little variation, but the later studies arrive more slowly and with more variation. Projects receive the first study after a mean of 5 months,

[12]For example, Savion developed the TWE Myrtle solar project in Virginia (interconnection request AD1-144) and sold it to Dominion Energy in 2019 (Savion, 2022), before the project reached commercial operation in June 2020. Geenex began developing the Sweetleaf solar project in North Carolina (interconnection request AD1-056) in 2017 and sold it to CleanChoice Energy in February 2026 while it was still active in the queue (CleanChoice Energy, 2026).

[13]TOs can own incumbent generation and projects in the queue, but we find no evidence this ownership correlates with interconnection wait times or costs.

with a standard deviation of just 1.5 months. The second and third studies take far longer: against PJM's official targets of 4 and 6 months, the second arrives after a mean of 11 months (a standard deviation of 8 months) and the third after 15 months (a standard deviation of 10 months), where waiting times are capped at the 90th percentile of 36 months before calculating the means. These delays have been attributed to the number of backlogged interconnection requests and a lack of engineering staff capacity (Shoemaker, 2021; Petersen, Siegner and Coequyt, 2026).

While priority is by entry cohort, study arrival is stochastic. Projects in earlier cohorts usually, but not always, receive a given study earlier than projects in later cohorts. For example, two cohorts of projects entered in 2012. The average second-study arrival date was 2013Q2 for the earlier cohort and 2014Q4 for the later cohort, yet 3.5% of the earlier cohort had still not received a second study by 2014Q4.[14]

Time spent in the queue increased over our sample, though not as dramatically as the number of projects. On average, projects that reach the terminal study receive it 27 months after entering the queue. The median wait increased from 13 months for projects entering in 2008-2011 to 19 months in 2012-2015 and 36 months in 2016-2019. Appendix Figure E.2 provides wait time statistics by entry cohort.

3.5 Interconnection Costs

Interconnection costs are low for most projects but very high for some. We use cost per megawatt as our measure of cost.[15] For the second study, 29% of projects have interconnection costs less than 0.01 million per megawatt. The median of the cost distribution is 0.05 million per megawatt, the 75th percentile is 0.14 million per megawatt and the 90th percentile is 0.42 million per megawatt. For comparison, installation costs for wind and solar projects are roughly 1.5 million per megawatt.

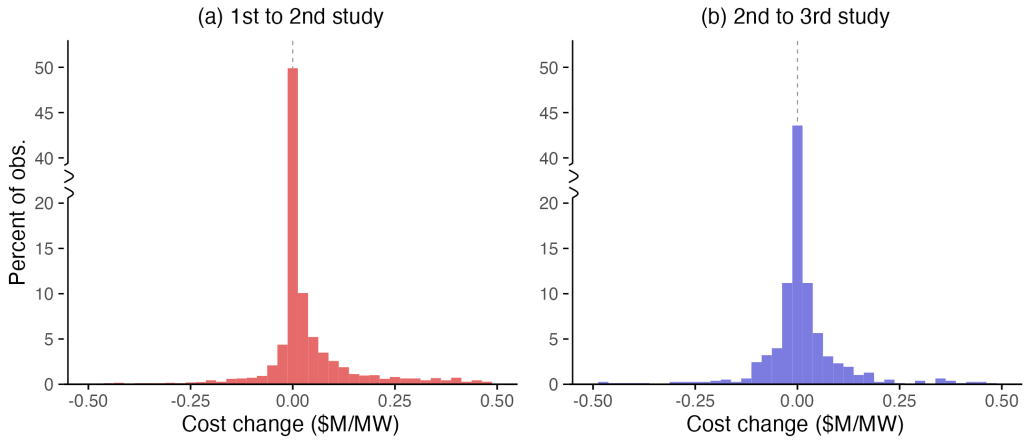
Developers face substantial uncertainty about what their interconnection cost will be. According to an industry expert we interviewed, developers can often use engineering models to predict whether an interconnection will overload the grid, but it is much harder to predict

[14]One reason wait times are variable is that these studies sometimes need to be revised: 22% of projects in our sample had their second study revised. We model the arrival of the final version of each study, not the intervening versions, which are not always posted by PJM.

[15]There do not appear to be economies of scale in interconnection costs for moderately sized projects. For projects from the 10th to 90th percentile in size, a 1 standard deviation increase in capacity is associated with a 0.04 standard deviation decrease in the second-study cost per megawatt.

what remedies a transmission owner will require and what they will cost. Many projects also share network upgrade costs, and a project's cost can rise or fall when another project sharing the same upgrade withdraws. Consistent with these sources of uncertainty, observable project characteristics explain less than half the variation in interconnection costs, and projects in the same geographic area can have very different costs (Appendix Figure E.1; Johnston, Liu and Yang, 2026).

Figure 2: Within-Project Changes in Interconnection Cost Estimates across Studies



Panel (a) shows a histogram of the difference in the first and second study interconnection cost estimates for projects that received the second study. N = 2,495; 151 projects with cost differences outside the plotted range are excluded (126 above and 25 below), as are 4 without a first study. Panel (b) shows a histogram of the difference in the second and third study interconnection cost estimates for projects that received the third study. N = 829; 19 projects with cost differences outside the plotted range are excluded (10 above and 9 below), as are 51 without a second study. The sample is projects queuing from 2008–2020. Y-axis is the percent of observations in each bin. Costs are in millions of 2020 dollars per MW.

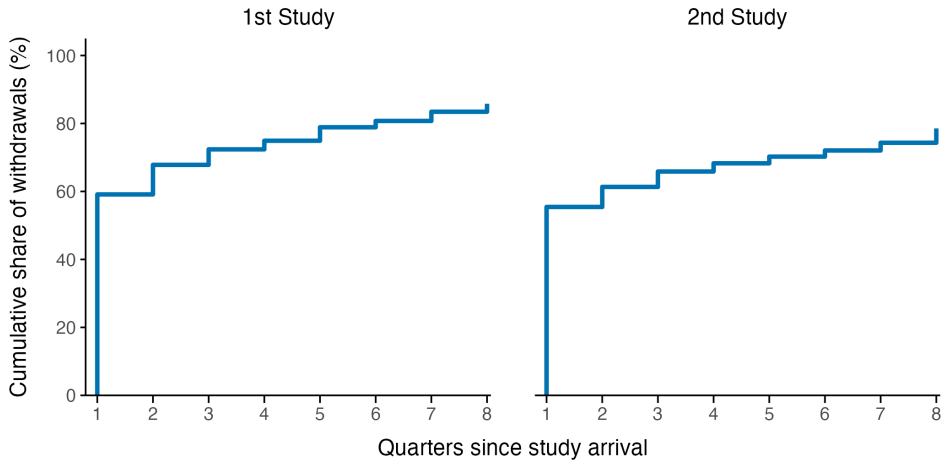
Changes in cost across studies within a project are also hard to predict. Figure 2 plots the change in cost between consecutive studies for projects that waited for the next study. If developers could predict their cost changes, those expecting a decrease would be more likely to wait than those expecting an increase, and this selection would shift the distribution of observed changes downward. Yet the figure shows no systematic decrease in interconnection costs across studies.[16]

[16]Costs on average increase from the first to the second study because the second study includes the contribution to shared network upgrade costs, while the first study does not. For projects that do not receive a network allocation, the distribution of this cost difference is symmetric around zero.

3.6 Withdrawals

Withdrawals concentrate sharply within one quarter of study arrival. Figure 3 plots the cumulative share of projects that have withdrawn from the queue by quarters since the arrival of each study. Each panel is restricted to projects that required a further study to complete the interconnection process but withdrew before receiving the next study. Both curves rise steeply at first and then flatten: following the first study arrival, 59% withdraw within one quarter and 86% within two years. The pattern after the second study is similar, with 55% withdrawing within the first quarter. Because the deposit to request the next study is small, the concentration of withdrawals just after a study arrives likely reflects the information it reveals rather than the cost of continuing.

Figure 3: Withdrawals Relative to Study Arrival



Share of projects that withdrew within t quarters of receiving their first study (left panel) or second study (right panel). For each curve, the sample includes all projects that entered the queue from 2008 to 2020, received the relevant study, and required a further study, but withdrew before receiving it ($N = 1{,}184$ and $N = 662$ respectively). The figure displays only the first eight quarters for visualization purposes. In the left panel, 59.1% withdrew within the first quarter and 14.2% withdrew more than eight quarters after receiving the first study; in the right panel, 55.4% withdrew within the first quarter and 21.5% withdrew more than eight quarters after receiving the study.

We next test whether high interconnection costs explain these withdrawals. Specifically, we regress an indicator for withdrawing from the queue on an indicator for having a high (above 0.1 million per megawatt) interconnection cost. In this regression, we define withdrawing as leaving the queue before the next study arrives, or before beginning operation for projects

that have received their terminal study.

Table 2: Effect of Interconnection Costs on the Probability of Withdrawing from the Queue

	1st Study	2nd Study		3rd Study	
	OLS	OLS	IV	OLS	IV
High cost (>0.1M/MW)	0.115***	0.186***	0.311***	0.155**	0.322***
	(0.019)	(0.027)	(0.042)	(0.061)	(0.122)
Mean of dependent var.	0.32	0.53	0.53	0.57	0.57
Share with high cost	0.25	0.34	0.34	0.29	0.29
First-stage F			218.2		46.3
Observations	4,134	2,259	2,259	663	663

The sample is projects that entered the queue in 2008–2020 and received a first study; projects still active at the end of the sample are excluded, and each column is limited to projects receiving the relevant study. Of the 51 projects that received a third study but no second study, 2 also lack a first study and so fall outside this sample; the third-study columns exclude the remaining 49, so the OLS and IV samples coincide. A further 2 observations drop out as fixed-effect singletons. Standard errors in parentheses; clustered by substation. The dependent variables indicate withdrawal before the next study or, after the terminal study, before beginning operation. High cost indicates that the study's interconnection cost estimate exceeds $0.1 million per MW. All specifications control for project size (three bins), fuel type, whether the project increases capacity at an existing plant, transmission owner FE, and queue year FE; the second- and third-study columns also control for the prior study's high-cost indicator. The IV instruments the high-cost indicator with the within-project change in the cost estimate across studies, in millions of dollars per MW, winsorized at the 1st and 99th percentiles. First-stage F is the Kleibergen-Paap rk Wald statistic. Ninety-five percent Anderson–Rubin confidence sets, robust to weak instruments, are [0.23, 0.39] and [0.08, 0.56] for the second- and third-study IV coefficients. *** p<0.01, ** p<0.05, * p<0.10.

Across all three studies, projects with a high interconnection cost estimate are more likely to withdraw from the queue (Table 2). The OLS estimates imply that a high interconnection cost is associated with an increase in the probability of withdrawal of 12 percentage points after the first study (36% of the mean withdrawal rate), 19 percentage points after the second (35%), and 16 percentage points after the third (27%).[17]

The OLS estimates may not isolate the causal effect of costs. Interconnection costs tend to be higher in densely developed areas (Appendix Figure E.1), where projects may also face more effective local opposition, a cancellation risk unrelated to the assessed cost. If so, the OLS

[17]These results are robust to adding controls for permitting difficulties that may be correlated with interconnection costs: distance from project to its substation and an indicator for local solar and wind ordinances.

estimates would be biased upward. Conversely, if developers plan projects in locations they expect to have high interconnection costs only when those projects are otherwise especially profitable, the resulting positive selection would attenuate the OLS estimates. To address both concerns, we instrument for the high-cost indicator with the change in costs across studies, which is difficult to predict and is the same variation that identifies the payoffs and waiting costs in the structural model (Section 6.2).

The IV estimates are larger: high costs increase the probability of withdrawal by 31 percentage points after the second study and 32 after the third. Both are statistically significant, and the weak-instrument-robust confidence sets in Table 2 exclude zero. The gap between IV and OLS is consistent with positive selection attenuating the OLS estimates.[18] Taken together, the estimates imply that high interconnection costs substantially increase the probability of withdrawal.

4 Externalities in the Queue

Projects in the queue affect one another through three channels: PJM's limited capacity to process engineering studies (congestion externality), the physical transmission infrastructure shared across interconnections (cost externality), and the wholesale electricity market (output externality). We find that the congestion externality is quantitatively important, while the cost and local output externalities are not.

4.1 Congestion Externality

We first investigate how queue length affects how long projects wait for studies. The congestion externality would be mechanical under strict first-come, first-served queuing with a fixed processing rate. In PJM, projects are prioritized by queue position, but studies take different amounts of time to complete, so study arrivals do not strictly follow the queue order. How strongly a project's probability of receiving a study depends on queue position is therefore an empirical question.

[18] A potential concern is that the instrument is correlated with unobserved quality because the change in cost is mechanically related to the prior cost level (a cost of zero can only rise), and projects that continue despite a higher prior cost are likely to have higher unobserved quality. Controlling for prior-study cost more flexibly (eight bins that nest the 0.1 indicator) yields similar estimates: 31 and 34 percentage points for the second and third studies, with standard errors of 4 and 12.

We estimate a discrete-time hazard model, where the conditional probability that the second or third study arrives in a quarter is given by a probit. We focus on the second and third studies because, as noted in Section 3.4, the first study arrives quickly, with little variation in wait times. We measure queue position using the natural log of the number of higher-priority projects (those in the same or earlier entry cohorts that have not yet completed or withdrawn), both PJM-wide and within the project's TO, and estimate how each measure affects the conditional arrival probability. Throughout, we condition flexibly on time already spent in the queue, thus comparing projects with the same waiting time. These controls separate the queue-position effect from pure duration dependence in study processing, i.e., a study becoming more or less likely to arrive simply because a project has already waited longer. We allow the effect of higher-priority projects to vary with whether a project is waiting for the second or third study.

Identification. The queue-position coefficients identify the congestion externality because PJM, not the project, determines study arrival. Projects with higher expected profits are less likely to withdraw, so those that survive to the front of the queue tend to be more profitable. Yet this selection does not bias the estimated queue-position effect. As the PJM manual and industry interviews confirm, study priority depends solely on queue position rather than profitability, and projects cannot act to expedite their studies. The remaining threat is that project characteristics affect both how long PJM takes to complete a study and queue position. We control directly for the main characteristics affecting how long a study takes: project size, fuel type, and the prior study's information (cost estimate, engineering tests, and whether it is assigned a network upgrade allocation). Together, these arguments support the identifying assumption that the number of higher-priority projects is independent of the error term in the probit model.

Results. We find a significant congestion externality, which is largest for the third study. Table 3 reports the average marginal effects of higher-priority projects on own study arrival. Our preferred specification, column (3), implies that a 10% increase in higher-priority projects reduces the second-study arrival probability by 2.9% and the third-study arrival probability by 13.2%.[19] Columns (2) and (5) show that these effects are similar when we omit the prior-study

[19]We assume the PJM-wide and TO counts of higher-priority projects increase by the same percentage, so the effect of a 10% increase is the total queue-position effect in Table 3 multiplied by 0.10, divided by the

controls or add transmission owner and year fixed effects to the model.

Withdrawals by nearby projects can trigger study revisions, slowing delivery of the final version and thus imposing a further congestion externality. Column (4) adds the number of withdrawals in the same TO in the last two quarters. Because larger queues generate more withdrawals, we also control for the total number of projects in the TO's queue. The estimated coefficient on withdrawals is close to zero and statistically insignificant. Measuring withdrawals and queue size in MW rather than project counts yields similar results (Appendix Table E.3, column (5)).

These estimates are robust to flexible time controls, addressing the concern that PJM's study process may have evolved over this period. PJM did not make major changes to its process, but TOs may have learned to process studies more efficiently or hired more engineers in response to rising demand. To account for these effects, we re-estimate our preferred specification (column (3) of Table 3) controlling for linear and quadratic time trends, as well as their interactions with an indicator for renewable generation to allow for technology-specific learning. We also estimate a model with year-by-quarter fixed effects, a specification that identifies the effect of queue position solely from cross-sectional differences in arrival probability for projects at different queue positions. Finally, we estimate a model with transmission owner-by-year fixed effects, which absorb shocks common to a transmission owner in a given year that could drive both queue length and study delays. Appendix Table E.4 shows that we consistently find a negative and statistically significant total effect of the number of higher-priority projects on study arrival for both the second and third studies. The estimates imply that a 10% increase in higher-priority projects reduces study arrival probability by 1.9 to 3.1% for the second study and 9.3 to 13.2% for the third study.

A second concern is that the effect is driven by the end of the sample, when the queue was most congested. Restricting to projects that queued before 2019, which excludes 39.6% of the projects in our sample, we still find statistically significant and negative effects of queue position on study arrival, although the third-study effect is smaller: 3.4% for the second study, 4.3% for the third (Appendix Table E.4, column (6)).

Finally, in support of our claim that PJM's study delivery does not respond to expected project profitability, we re-estimate the arrival model including two proxies for profitability:

study-specific mean arrival rate for that specification's estimation sample.

the local wholesale electricity price and an indicator for binding local wind or solar ordinances. We do so with and without transmission owner and year fixed effects (Appendix Table E.3). The effects of both variables on study arrival are small. The price effect is significant at the 5% level only without the fixed effects. Even in this specification, a one-standard-deviation increase in the wholesale price, $18 per MWh, raises the quarterly probability of second-study arrival by 0.77 percentage points. That is about one seventh of the 5.1 percentage point reduction from raising each of the two queue-position measures by one standard deviation, computed from the same specification. The ordinance effect is indistinguishable from zero and changes sign across specifications. The queue-position effects remain similar.

4.2 Cost Externality

Nearby projects can affect one another's interconnection costs through the network infrastructure they share. Those queued ahead of a given project may consume available local capacity, leaving the project responsible for an upgrade. When several projects trigger a common upgrade, PJM assigns them to share its cost; if one member of a cost-share group withdraws, the others' costs can rise or fall. A completed project that paid for a costly network upgrade can also lower later projects' costs if that upgrade leaves spare capacity behind.

4.2.1 Spillovers from Other Projects in the Queue

Higher-Priority Local Capacity. PJM models higher-priority projects as in service when conducting engineering studies, so these projects can consume available local transmission capacity. More higher-priority local capacity could therefore increase the probability a project triggers costly network upgrades, raising its interconnection cost.

We regress the indicator for a high interconnection cost (above $0.1 million per megawatt) on the amount of higher-priority capacity in a project's local area when its study is issued. We repeat the analysis for the network-upgrade component of cost. We define the local area four ways: within 10 km, within 20 km, within the same transmission owner, and within regions defined by similarity in wholesale prices.[20] We include geographic-region and queue-year fixed effects and control for recent transmission investment by the grid operator. The resulting estimates are nonetheless suggestive rather than causal, since the entry and withdrawal decisions

[20] We use hierarchical clustering to group substations into 50 regions based on location and similarity in locational marginal prices. For more detail, see Johnston, Liu and Yang (2026).

that determine higher-priority capacity are themselves affected by the state of the network.

We find no evidence that more higher-priority capacity raises a project's cost under any of the four definitions, whether we measure total cost or its network-upgrade component alone. The point estimates in Appendix Table E.5 are uniformly small: even the largest in absolute value, which is negative, implies that a doubling of higher-priority capacity reduces the probability of a high cost by about 2.3 percentage points, or roughly 6% of its sample mean of 0.37. This result is robust to alternative definitions of a high interconnection cost and to using the total capacity of active waiting projects rather than higher-priority capacity.

Withdrawals Within Cost-Share Groups. When multiple projects trigger the same network upgrade, PJM assigns them to a cost-share group and allocates the cost in proportion to each project's impact. In the second study, PJM assigns a network upgrade cost allocation to 59% of projects, though the allocated amount is zero for about half of them. We observe complete group membership for 481 projects across 158 groups, covering 23% of projects that are assigned a network upgrade cost in their second studies. Conditional on belonging to a fully observed group, 36% of projects see at least one member withdraw before their own terminal study.

We test whether these withdrawals systematically affect costs for remaining projects. Because both the total group cost and its allocation can change when a project withdraws, the effect on remaining group members is theoretically ambiguous: if PJM still requires the upgrade, the remaining members bear a larger share of its cost; if the withdrawal removes the need for it, their cost falls. We also test whether member withdrawals predict a project's own withdrawal.

The estimated effects of member withdrawals on cost are small and statistically insignificant (Table 4, columns (1) and (2)). A one-standard-deviation increase in the withdrawn share (0.22) implies a \$0.02–\$0.03 million per megawatt reduction in cost between studies, about a tenth of a standard deviation. The corresponding 95% confidence intervals rule out cost increases larger than 6% of a standard deviation of cost changes or decreases larger than one-quarter of a standard deviation. One concern is that projects anticipating cost increases from other members' withdrawals may themselves withdraw before receiving the next study. These cost increases would then go unobserved, potentially masking a positive cost effect. If this selection is important, other members' withdrawals should predict a project's own

withdrawal, but we find no significant relationship (column (3)).

4.2.2 Spillovers from Completed Interconnections

A completed interconnection can lower a later project's cost by leaving spare transmission capacity, but cost-sharing and the project-specific nature of upgrades may limit this spillover.[21] We test for the spillover by regressing an indicator for a high cost estimate (above \$0.1 million per megawatt) on whether the most recently completed interconnection in the same area when the study was issued was costly, defined as a total interconnection cost above \$1 million; 26% of completed interconnections meet this threshold. We exclude prior interconnections that share costs with the focal project, so the regressor captures the effects from earlier completion instead of the effects from being in the same cost-share group.

The estimated effects on a project's subsequent total cost are statistically insignificant across the 10 km, 20 km, and transmission-owner definitions of the local area (Table 5, Panel A). The corresponding confidence intervals exclude effects larger than 16 percentage points against a sample mean of 37%. For the local-area definition based on similarity in wholesale prices, a costly prior interconnection is associated with a statistically significant 3.2 percentage point reduction in the probability of a high total cost estimate, or 9% of the sample mean. Under this definition, however, the effect reverses sign when a costly prior interconnection is defined by cost per megawatt rather than by total cost (Appendix Table E.6). For the network-upgrade component of interconnection cost, where these spillovers are most likely to occur, the estimates under all four area definitions are statistically insignificant (Table 5, Panel B). The network-upgrade estimates remain statistically insignificant under alternative definitions of a costly prior interconnection (Appendix Table E.7).

Discussion. Across these channels, we find little evidence of systematic cost externalities. The null effect of higher-priority capacity is likely explained by PJM's cost-sharing rules, which spread the cost of an upgrade across the projects that jointly trigger it, not the entire active queue. The within-group null effect may reflect two opposing forces: a member's withdrawal raises remaining members' shares when the upgrade is still required but eliminates the cost

[21] In FERC Order 2023, the Commission states that "absent cost sharing provisions among clusters, interconnection customers may significantly benefit from earlier-in-time network upgrades but not share in the cost of those network upgrades in a manner that is roughly commensurate with benefits" (p. 320), suggesting that this externality is more relevant in systems without cost sharing (FERC, 2023).

when it is not. The slightly negative point estimates suggest these forces roughly offset. The largely null effect of completed interconnections, in contrast, may reflect the project-specific nature of upgrades: an upgrade is prescribed to resolve the violations caused by the projects that trigger it, so the spare capacity it leaves behind is incidental.

4.3 Local Output Externality

By lowering local wholesale prices, a completed interconnection could reduce the expected revenues, and hence the entry and completion incentives, of nearby projects. This channel is most likely to be present at locations with binding transmission constraints, where a new project's output cannot flow freely across the grid and instead depresses local prices.

We estimate how local wholesale prices change when a new project begins operation. We first construct a monthly panel with an observation for each substation, which is a potential grid connection point. Our measure of monthly wholesale prices is the average peak-hour locational marginal price at that substation. We use the event-study difference-in-differences estimator from Callaway and Sant'Anna (2021). The comparison group is substation-months where no new projects above a size threshold begin operation; we use thresholds of 20 and 100 MW.

We find little effect of a new project beginning operation on local prices. Figure 4 shows that the estimated price change is small and statistically indistinguishable from zero in most post-period months. The point estimates indicate a transitory decline of roughly \$1–\$2 per MWh, against a mean price of \$35 per MWh.

Local Price Effect at Locations of Transmission Constraints. Because the output externality may operate only at locations where projects face binding transmission constraints, we re-estimate the event study restricting the treatment group to substations where the new project was responsible for paying transmission network upgrade costs. These projects connect to an already-congested local grid, so any local effects should be most detectable in this subsample. We continue to find a null effect (Appendix Figure E.3).

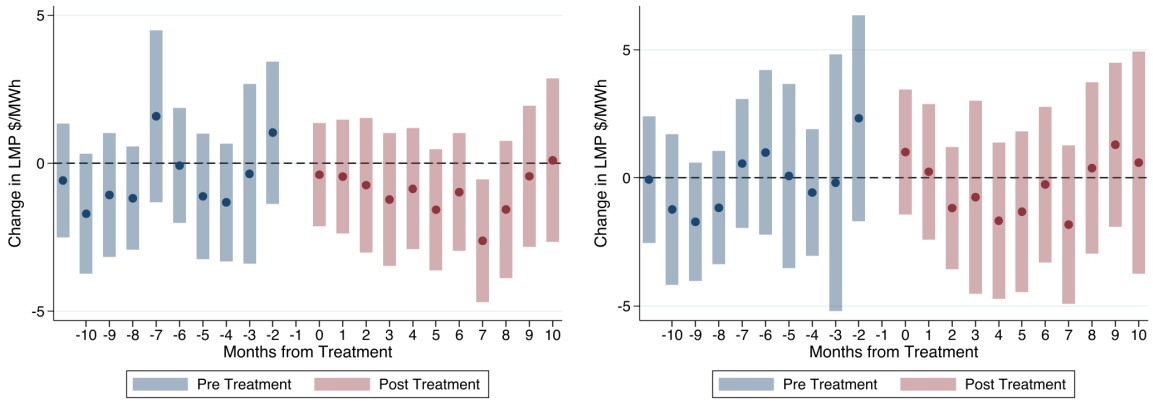
Discussion. This null result is consistent with the institutional details of the interconnection process. PJM's deliverability requirement obligates most new projects to upgrade the transmission infrastructure to the point that they can deliver their power to demand centers

even during peak demand hours. These upgrades relieve local congestion, limiting the effect of added generation on local prices and therefore the expected revenues of future projects in the same location.

As more projects complete and add to supply, PJM wholesale prices should fall. We incorporate this aggregate effect in the model by constructing project revenues from the wholesale-market equilibrium.

Figure 4: Effect of Interconnection on Local Wholesale Prices

(a) Adding 20 MW New Capacity (b) Adding 100 MW New Capacity



Panel (a) plots event-study estimates of the effect of a new project with capacity of at least 20 MW beginning operation at a substation on local wholesale electricity prices. Panel (b) uses a 100 MW threshold. Treated observations are substation-months in which a project above the corresponding capacity threshold begins operation; control observations are substation-months without such entry. The dependent variable is the monthly average peak-hour locational marginal price at the substation level (mean = \$34.89/MWh), constructed from PJM locational marginal price data over 2008-2020. Estimates are obtained using the Callaway and Sant'Anna (2021) difference-in-differences estimator with doubly robust inverse probability weighting. Standard errors are clustered at the substation level.

Table 3: Effect of Queue Position on Study Arrival

	(1) Probit	(2) Probit	(3) Probit	(4) Probit	(5) Probit
ln(# Higher priority)	-0.043***	-0.031***	-0.031***	-0.022***	-0.020
	(0.007)	(0.006)	(0.006)	(0.006)	(0.012)
ln(# Higher priority) x Waiting for third study	-0.026***	-0.017**	-0.016**	-0.014*	-0.011
	(0.007)	(0.008)	(0.008)	(0.007)	(0.007)
ln(# Higher priority in TO)		-0.016***	-0.016***	-0.043***	-0.020***
		(0.004)	(0.003)	(0.007)	(0.007)
ln(# Higher priority in TO) x Waiting for third study		-0.013*	-0.012*	-0.009	-0.011*
		(0.007)	(0.007)	(0.007)	(0.007)
Waiting for third study	-0.209***	-0.218***	-0.216***	-0.231***	-0.243***
	(0.022)	(0.021)	(0.023)	(0.024)	(0.023)
ln(# of withdrawals in queue within TO)				-0.000	
				(0.005)	
ln(# Projects in queue within TO)				0.030***	
				(0.007)	
Total queue-position effect (linear combinations)					
At second study	-0.043***	-0.046***	-0.047***	-0.064***	-0.040***
	(0.007)	(0.006)	(0.006)	(0.008)	(0.012)
At third study	-0.069***	-0.076***	-0.075***	-0.087***	-0.062***
	(0.006)	(0.006)	(0.006)	(0.007)	(0.011)
Prior Study Controls			X	X	X
Transmission Owner FE					X
Year FE					X
Observations	28,917	28,917	28,917	28,917	28,917
Pseudo R^2	0.166	0.173	0.176	0.179	0.195

The sample includes projects queued from 2008–2020, observed over time at the quarterly level as they wait in the queue for study arrival. Average marginal effects from probit regressions are reported. Standard errors in parentheses, clustered by transmission owner by entry cohort (416 clusters). The dependent variable is an indicator for study arrival (overall mean = 0.108; mean for second study is 0.160 and for third study is 0.056). Implied percentage effects reported in the text are computed from unrounded average marginal effects and unrounded mean arrival rates. All specifications control for project size (three bins), fuel type (wind, solar, battery), if the project is a capacity increase at an existing plant, and time spent waiting in the queue (a separate indicator for each quarter since queue entry up to quarter 8 as well as their interactions with an indicator for whether the project is awaiting its third study). Specifications (3)-(5) additionally include controls from the immediately prior study: the project's own cost estimate (e.g., the first study costs for projects awaiting the second study; three bins), an indicator for having ever received one of the three engineering tests and an indicator for whether the project receives a network allocation. Specification (4) further controls for the log number of withdrawn projects waiting for the same or a later study within the same transmission owner over the past two quarters, and the log number of projects in the queue within the same transmission owner. The total-effect rows report linear combinations of the average marginal queue-position effects. *At second study* sums the system-wide and transmission-owner queue effects, while *At third study* additionally includes their interactions with the indicator for awaiting the third study. *** p<0.01, ** p<0.05, * p<0.10.

Table 4: Effects of Cost-Share Group Member Withdrawal on Own Cost and Withdrawal

	Change in cost per MW		Own withdrawal
	Total	Network upgrade	
	(1)	(2)	(3)
Share of group MW withdrawn	-0.132	-0.100	-0.009
	(0.093)	(0.089)	(0.010)
Transmission Owner FE	X	X	X
Queue Year FE	X	X	X
Mean of dep. var. [SD]	-0.017 [0.289]	-0.042 [0.274]	0.023 [0.149]
Mean of indep. var. [SD]	0.077 [0.224]	0.077 [0.224]	0.068 [0.197]
Observations	206	206	8,928

The sample is projects that entered the queue in 2008–2020 that are in a cost-share group whose full membership we observe. Standard errors are in parentheses and clustered by cost-share group. Columns (1)-(2) restrict the sample to projects that have received the third study, since projects typically begin to observe their cost-share group in the second study; the dependent variable is the change in cost per MW from the second to the third study. Column (1) uses total interconnection cost per MW, and column (2) uses network upgrade cost per MW. Column (3) reports linear-probability estimates from a project-quarter discrete-time hazard panel; the dependent variable is an indicator for project withdrawal in the current quarter. The independent variable, Share of group MW withdrawn, is the share of cost-share group peers withdrawn during the relevant exposure window (the inter-study window for columns (1)-(2); since the focal project's most recent study for column (3)). All specifications control for project size (three bins), fuel type (wind, solar, battery), whether the project is a capacity increase at an existing plant, and the log of total cost-share group size in MW. *** $p<0.01$, ** $p<0.05$, * $p<0.10$.

Table 5: Effect of a Costly Prior Interconnection on the Probability of High Cost

	10km (1)	20km (2)	Transmission owner (3)	Hierarchical cluster (4)
Panel A. High Total Cost Estimate (mean = 0.37)				
Costly prior interconnection	0.057	-0.003	-0.011	-0.032**
	[-0.042, 0.157]	[-0.089, 0.082]	[-0.076, 0.055]	[-0.063, -0.000]
Geographic Region FE	X	X	X	X
Queue Year FE	X	X	X	X
Observations	2,328	2,328	2,328	2,328
R^2	0.528	0.525	0.239	0.241
Panel B. High Network Upgrade Cost Estimate (mean = 0.13)				
Costly prior interconnection	0.108	0.058	-0.045	0.037
	[-0.224, 0.474]	[-0.132, 0.267]	[-0.112, 0.078]	[-0.031, 0.113]
Geographic Region FE	X	X	X	X
Queue Year FE	X	X	X	X
Observations	2,328	2,328	2,328	2,328
R^2	0.475	0.481	0.177	0.204

The sample is projects queuing from 2009–2020; the 2008 cohort is excluded because the queue data begin in 2008, leaving it with almost no observable prior interconnections. Observations are project-study pairs for the second and third studies. Wild cluster bootstrap (9,999 replications) clustered at the substation level in the 10 km and 20 km columns, the transmission owner level in the transmission owner column, and the hierarchical cluster level in the hierarchical cluster column; 95% confidence intervals in brackets. In Panel A, the dependent variable is an indicator for whether the total cost estimate in that study is high, defined as exceeding \$0.1 million per MW. In Panel B, the dependent variable is an indicator for whether the network upgrade cost estimate in the second or third study is high, defined as exceeding \$0.1 million per MW. The regressor of interest, Costly prior interconnection, is an indicator defined differently across panels. In Panel A, the indicator equals one if the most recently completed interconnection in the local geographic region, as of the study issue date, had total interconnection costs exceeding \$1 million; completed interconnections that share costs with the focal project are excluded. The local geographic region varies across columns (10 km radius, 20 km radius, transmission owner, or hierarchical cluster). In Panel B, the indicator equals one if that most recently completed interconnection had network upgrade costs exceeding \$1 million. All specifications control for project size (three bins), fuel type (wind, solar, battery), whether the project is a capacity increase at an existing plant; whether PJM Regional Transmission Expansion Plan investment was above the 75th percentile of the RTEP distribution within the same transmission owner, measured over a four-year window spanning three years before and one year after the study issue date; the log of total entrant capacity (MW) within the previous three years in the local geographic region; and the log capacity (MW) of higher-queued projects in the local geographic region. The geographic region fixed effects are substation fixed effects in the 10 km and 20 km columns, transmission owner fixed effects in the transmission owner column, and hierarchical cluster fixed effects in the hierarchical cluster column. *** p<0.01, ** p<0.05, * p<0.10.

5 Model

Each project solves a finite-horizon optimal stopping problem. Time is discrete, indexed by $t = 1, 2, \ldots$ in quarters. Potential projects arrive at the beginning of a period and decide whether to pay an entry cost to join the queue or leave immediately. Projects that join receive up to three studies. Between studies, projects incur waiting costs and can withdraw at any time. Each project also faces a per-period probability of exogenous withdrawal due to events outside the interconnection process. After the terminal study, a project decides whether to complete interconnection or withdraw.

Projects do not directly interact with each other. Each project responds to a vector of variables called the *queue status*, which determines its study arrival probabilities and operating revenues. The transitions of the queue status, aggregated from strategies of atomistic projects, are deterministic. The queue status has both PJM-wide and TO-specific components, since PJM conducts the studies within each TO's territory in conjunction with that TO. This abstraction is reasonable because, even at the TO level, an individual project is small relative to the total queued capacity (Section 3.3).

Below, we first specify the project decision problem. Then we define an equilibrium in which each project's strategy is a function of its beliefs about the path of the queue status, and these strategies together generate the path consistent with those beliefs.

5.1 Decision Problem

We describe a project's decision problem recursively, beginning with the completion decision and working backwards. PJM may require fewer than three studies if the project needs no significant network upgrades and does not share costs. A project's last required study is its *terminal study*, after which it decides whether to complete or withdraw. The first or second study can be terminal, while the third study always is. After a non-terminal first study, the project enters *Stage 1*, waiting for its second study; after a non-terminal second study, it advances to *Stage 2*, waiting for its third study. A project waits at most $T = 32$ quarters after entry; one that reaches T without a terminal study receives its terminal study immediately and makes the completion decision.

Terminal Study Decision. Upon receiving its terminal study in period t, project i chooses to *complete* or to *withdraw* for a payoff of zero. The net payoff from completing, on a per-megawatt basis, is

$$\begin{aligned}\pi_{it} =& r_{it} - \omega_{it} - c_{it} + \xi_i \\ =& \tilde{r}_{it} + \chi_t(x_i)\theta - \omega_{it} - c_{it} + \xi_i \end{aligned} \tag{1}$$

where r_{it} is the present value of expected operating profit, ω_{it} is the expected cost of construction, and c_{it} is the interconnection cost in the terminal study. We decompose r_{it} into $\tilde{r}_{it}$, the present value of revenue from power production, and $\chi_t(x_i)\theta$, which captures other contributions to profit, e.g., operating costs and technology-specific subsidies, through a vector $\chi_t(x_i)$ that varies with project characteristics x_i and completion time t and a parameter vector θ. The time-invariant characteristics x_i include fuel type, capacity, and the TO in which the project is located. This decomposition lets revenues from power production $\tilde{r}_{it}$ respond endogenously to capacity changes in our counterfactual simulations. We describe $\tilde{r}_{it}$ and ω_{it} in more detail in Section 6.2.

The term ξ_i captures payoff information unobserved by the researcher that has accumulated by the terminal study. The component ξ_i^{S1} is revealed at the first study. If that study is terminal, $\xi_i = \xi_i^{\text{S1}}$; otherwise, ξ_i^{S2} is revealed when the second study is received, and $\xi_i = \xi_i^{\text{S1}} + \xi_i^{\text{S2}}$. As described in Section 2.2, studies can signal additional cost-relevant information during the development process. More broadly, ξ^{S1} and ξ^{S2} reflect information a project learns about conditions outside the interconnection process while waiting, such as the contract terms it can expect from power buyers.

Project i also learns a final payoff shock ξ_i^f at the terminal study and completes if $\pi_{it}+\xi_i^f > 0$. Conditional on terminal-study receipt in period t, we assume ξ_i^f is normally distributed with mean 0 and standard deviation $\sigma \cdot (t - e_i)^\kappa$. Here, e_i is the quarter in which the potential project arrives and makes its entry decision; a project that enters joins the queue in e_i+1. The dependence on $t - e_i$ accommodates larger payoff dispersion among projects that have waited longer. The distribution of ξ_i^f thus depends on the arrival time of the terminal study, but for notational simplicity, we suppress the time subscript. We also assume ξ_i^f is conditionally independent of ξ_i given project characteristics x_i and the interconnection cost c_{it}.

Integrating over ξ_i^f, the expected value of receiving the terminal study given the intercon-

nection cost c is

$$\bar{V}_{it}(c, \xi_i) \;=\; E_{\xi_i^f}\Big[\max\big\{\tilde{r}_{it} + \chi_t(x_i)\theta - \omega_{it} - c + \xi_i + \xi_i^f,\; 0\big\}\Big]. \tag{2}$$

Because x_i and e_i do not change over time, we absorb them into the i subscript and condition on them throughout.

Information Assumptions. A project has perfect foresight over the probabilities governing study arrivals and costs in each period t. In the equilibrium defined in Section 5.3, the deterministic transitions of the queue status imply the study arrival probabilities; the cost distributions depend only on a project's own state. At the start of period t, however, projects do not know whether a new study will arrive that period or, if one does, the realization of its cost c' or whether it is terminal. We maintain these assumptions below.

Stage-2 Continuation. After receiving a non-terminal second study, project i decides each period whether to wait or withdraw. The project's state consists of the current time t, the second-study cost c, study information z, and the accumulated unobserved heterogeneity $\xi = \xi^{\mathrm{S1}} + \xi^{\mathrm{S2}}$. The variables (c, z, ξ) are specific to each project i, but for notational simplicity, we suppress their i subscript. We assume that the costs have finite support. A waiting decision is made at the start of period t. The continuation value satisfies

$$\begin{aligned} W_{it}^{\mathrm{S2}}(c, z, \xi) \;=\; \max\Bigg\{ -\gamma^{\mathrm{S2}} \;+\; \beta\delta_i^{\mathrm{S2}}\Bigg[\textstyle\sum_{c'} q_i^{\mathrm{S2}}(c'|t, c, z)\, \bar{V}_{it}(c', \xi) \\ +\; \big(1 - Q_i^{\mathrm{S2}}(t, c, z)\big)\, W_{i,t+1}^{\mathrm{S2}}(c, z, \xi)\Bigg],\; 0\Bigg\}, \end{aligned} \tag{3}$$

where γ^{S2} is the per-megawatt waiting cost,[22] and $\beta = 0.987$ is the quarterly discount factor, implying an annual discount factor of 0.95. The term $1 - \delta_i^{\mathrm{S2}} \equiv 1 - \delta^{\mathrm{S2}}(x_i)$ is the per-period probability of exogenous withdrawal, capturing events outside the interconnection process such as failed permitting or expired land leases and varying with project characteristics x_i. The probability that the terminal study arrives with cost c' during period t is $q_i^{\mathrm{S2}}(c'|t, c, z)$,

[22] To simplify exposition, we omit study request fees from the equations. In the estimated model, a project pays the applicable fees and deposits once when it requests the next study after a non-terminal study, based on the PJM manual (PJM Interconnection, 2023*b*).

and $Q_i^{S2} = \sum_{c'} q_i^{S2}(\cdot)$ is the total probability of study arrival. The study information z is an indicator for whether the engineering tests described in Section 2.2 have been performed, which predicts how fast future studies arrive.[23]

Stage-1 Continuation. After receiving a non-terminal first study, project i decides each period whether to wait or withdraw. The state consists of time t, the first-study cost c, study information z, and unobserved heterogeneity ξ^{S1}. If the project waits and receives its second study during period t, a terminal study leads to the completion decision in t, while a non-terminal study moves the project into Stage 2 in $t+1$. The continuation value satisfies

$$
\begin{aligned}
W_{it}^{S1}(c,z,\xi^{S1}) \;=\; \max\Bigg\{ & -\gamma^{S1} \;+\; \beta\delta_i^{S1}\Big[\textstyle\sum_{c'} q_i^{S1,f}(c'|t,c,z)\, E_{\xi^{S2}}\big[\bar{V}_{it}(c',\xi^{S1}{+}\xi^{S2}) \mid t,c,z\big] \\
& \qquad +\; \textstyle\sum_{c',z'} q_i^{S1}(c',z'|t,c,z)\, E_{\xi^{S2}}\big[W_{i,t+1}^{S2}(c',z',\xi^{S1}{+}\xi^{S2})\big|t,c,z\big] \\
& \qquad +\; \big(1 - Q_i^{S1}(t,c,z)\big)\, W_{i,t+1}^{S1}(c,z,\xi^{S1})\Big],\; 0\Bigg\},
\end{aligned}
\tag{4}
$$

where γ^{S1} is the per megawatt waiting cost, $1-\delta_i^{S1}$ is the per-period probability of exogenous withdrawal, $q_i^{S1,f}(c'|t,c,z)$ is the probability that a terminal second study arrives with cost c' during period t, $q_i^{S1}(c',z'|t,c,z)$ is the probability that a non-terminal second study arrives with updated state (c',z') during period t, and $Q_i^{S1} = \sum_{c'} q_i^{S1,f}(\cdot) + \sum_{c',z'} q_i^{S1}(\cdot)$ is the total probability of study arrival during period t. We assume that, like ξ^f, ξ^{S2} is normally distributed with mean 0 and standard deviation $\sigma^{S2}\cdot(t-e_i)^{\kappa^{S2}}$. We also assume that ξ^{S2} is conditionally independent of ξ^{S1} and of the second study's outcome (c',z') given project characteristics x_i, the current interconnection cost c, and study information z. We again suppress the time subscript on ξ^{S2} for notational simplicity.

Entry. A potential project i arrives in period t and decides whether to enter or leave immediately.[24] We set the arrival time $e_i = t$. If it chooses to enter, it joins the queue and receives

[23] Whether the second study assigns the project a network upgrade cost allocation does not predict the third study's cost or arrival probability conditional on the second-study cost and project characteristics, so we do not include it in z.

[24] A potential entrant thus receives a one-time entry opportunity. This abstracts from the option to delay entry. Delay is costly for two reasons: waiting consumes the option period on the leases used for site control,

its first study in $t+1$. The project's expected value of entry is

$$\begin{aligned} \mathrm{EV}_i = q_i^f E\big[\bar{V}_{i,t+1}(c, \xi^{\mathrm{S1}}) \mid \text{terminal first study}\big] \\ + (1 - q_i^f) E\big[W_{i,t+1}^{\mathrm{S1}}(c, z, \xi^{\mathrm{S1}}) \mid \text{non-terminal first study}\big], \end{aligned} \tag{5}$$

where q_i^f is the probability that the first study is terminal, and each expectation is over the distribution of $(\xi^{\mathrm{S1}}, c, z)$ at the first study in $t+1$ conditional on (x_i, e_i). The entry value is evaluated at the moment of first-study receipt, abstracting from variation in waiting time up to that point, which is small relative to the variation in later study arrivals. We assume that ξ^{S1} is normally distributed with mean 0 and standard deviation σ^{S1}. We also assume that ξ^{S1} is conditionally independent of c and z given project characteristics x_i. Project i pays C_i and enters the queue if $\mathrm{EV}_i \geq C_i$, where the entry cost C_i follows a distribution that depends on a vector of project characteristics x_i^{ent}. Conditional on (x_i, e_i), the entry cost is independent of subsequent study outcomes and payoff shocks. We specify its distribution in Section 6.3.

5.2 State Transitions and Initial State Distribution

Queue congestion affects the probability that a project receives its next study in a given quarter. Conditional on arrival, we model the distribution of the study's outcome: whether it is terminal, the new interconnection cost c, and, for a non-terminal study, the updated study information z. This distribution depends on project characteristics and prior study information, but not on waiting time or queue congestion. The shocks to study arrival and to cost changes are independent of each other and of the unobserved payoff heterogeneity. We specify the distribution of the first-study state conditional on project characteristics. Section 6.1 and Appendix D describe how we parameterize and estimate these probabilities.

5.3 Equilibrium

We assume that in each period t, an exogenous mass of potential projects $N_t(x)$ decides whether to enter the queue. The portion of the mass that enters is determined by the entry rule $EV_i \geq C_i$, and the entering mass then decreases deterministically because of endogenous withdrawals, exogenous withdrawals, and completions.

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and projects entering in the meantime receive higher priority.

Queue Status. Each project's study arrival probabilities and revenues depend on a queue-status vector $\mathcal{K}_t(e, x)$, where the project arrives in period e, and x collects the project's time-invariant characteristics. In the empirical model, $\mathcal{K}_t(e, x)$ consists of three components that vary with t: (1) the number of queued projects with weakly higher priority, i.e., those that arrive in period $\tilde{e} \leq e$, which determines the project's position in the PJM-wide queue in period t; (2) the number of such higher-priority projects located in the same TO; and (3) the capacity of each fuel type in service at the start of period t. The first two components affect study arrival probabilities, and the third affects the present value of revenue through wholesale electricity prices. Projects take the path $\{\mathcal{K}_t(e, x)\}_t$ as given when solving their decision problems.

Optimal Withdrawal Strategy. We characterize the optimal strategies as cutoff rules, in which a project waits if the accumulated unobserved heterogeneity ξ exceeds a function of the observed information from the received studies and project characteristics. Under conditions we provide in Appendix A, endogenous withdrawals occur only immediately after projects receive a study. In periods in which no study arrives, projects do not endogenously withdraw, and only exogenous withdrawals occur with probability $1 - \delta^{(\cdot)}$. This behavior is consistent with Section 3.6, where we document that withdrawals concentrate in a short period after a study arrives. This characterization separates endogenous withdrawals from exogenous ones and simplifies estimation.

Equilibrium Definition. We define a queuing equilibrium as a path of queue statuses $\{\mathcal{K}_t(e, x)\}_t$ for every (e, x), project strategies, mass distributions, and revenues, satisfying three conditions: (i) project entry, waiting, and completion strategies are optimal given beliefs, as defined in Section 5.1; (ii) project beliefs about study arrival probabilities, revenues, and the joint distribution of costs and study information are consistent with those implied by the queue statuses and the prior study's information (Section 5.2 and Appendix C); and (iii) the path of queue statuses is generated by the entry, withdrawal, and completion decisions of projects.

Because we model projects as a continuum, the entry, withdrawal, and completion probabilities, together with exogenous withdrawal, move masses deterministically. The resulting masses in turn imply the path of $\mathcal{K}_t$ that projects take as given. The equilibrium is therefore

a fixed point in which the queue status is consistent with the individual decisions it induces. Appendix A formalizes the definition of the equilibrium and proves its existence.

Our model emphasizes the congestion externality, which we find to be quantitatively important in Section 4. A project's strategy is, in equilibrium, a function of its belief about its queue position, which reflects queue congestion and affects study arrival. Given that we do not find quantitatively important cost or local output externalities in Section 4.2 and Section 4.3, we model projects as responding to aggregate queue conditions rather than to the decisions of other individual projects.

Relationship to Similar Equilibrium Concepts. This equilibrium concept is a large-market rational-expectations equilibrium for a nonstationary environment. It is related to dynamic industry and search models in which agents know equilibrium aggregate objects but still face idiosyncratic uncertainty. For example, firms in Hopenhayn (1992) know the stationary industry environment but not their future productivity shocks, and workers and firms in search models know market tightness and matching rates but not the date of a match. A Markov-perfect equilibrium (Ericson and Pakes, 1995) would require each project to track the individual states of all other projects. Instead, as in oblivious equilibrium (Weintraub, Benkard and Van Roy, 2008) and nonstationary oblivious equilibrium (Weintraub et al., 2010), each project's strategy conditions on its own state and on the equilibrium path of aggregate states.[25]

6 Identification and Estimation

We estimate the model in three steps, discussing identification at each step. First, we estimate state transition and exogenous withdrawal probabilities directly from the data; these serve as beliefs in the project's decision problem. Second, taking these beliefs as given, we use maximum likelihood to recover payoffs, waiting costs, and shock distributions from ob-

[25]Projects in our model correctly anticipate the nonstationary changes in equilibrium objects that affect payoffs and state transition probabilities. In particular, we assume that projects have rational expectations over both technology costs, i.e., the time variation in ω, and interconnection queue congestion. The anticipation of falling renewable costs is consistent with contemporaneous policy and industry forecasts, such as the DOE's 2012 SunShot scenario and IRENA's 2016 cost-reduction projections (DOE, 2012; International Renewable Energy Agency, 2016). Developers also understood that increasing renewable development would congest the queue and slow queue processing, as described in AWEA's petition to FERC and in FERC's subsequent Notice of Proposed Rulemaking on interconnection reform (AWEA, 2015; FERC, 2016).

served withdrawal and completion decisions. Third, we construct a pool of potential entrants, compute their model-implied entry values, and estimate the distribution of entry costs from observed entry decisions. Our estimation sample is projects entering the queue from 2012–2020; we take pre-2012 entrants' observed actions as given. Starting in 2012 avoids an entry spike that coincided with a temporary federal program (Section 3.2). We use project actions through the first quarter of 2023, before PJM began transitioning from serial to cluster studies.

6.1 Transition and Exogenous Withdrawal Probabilities

We discuss identification and give an overview of estimation of the state transitions and exogenous withdrawal probabilities. Detailed specifications are in Appendix D, with estimates in Appendix Tables D.1 and D.2.

We estimate the study arrival process using a hazard model with a probit link. The specification is a simplified version of the hazard models in Section 4.1.[26] It relies on the same identification argument: in PJM's study process, priority for studies is based only on entry date, and PJM does not expedite or delay studies in response to the project profitability that drives selection into the surviving sample. Although the estimation sample conditions on survival, withdrawal depends on payoffs while study arrival does not, so survivor selection does not bias the arrival estimates.

We separately estimate the cost and study-information transitions, assuming that the unobserved shocks to study arrival and cost transitions are independent.[27] We use an ordered probit for the cost transition across four cost bins, and probit models for changes in the study-information indicator z and for whether the second study is terminal. The identifying assumption is that the error terms in these models are independent of the regressors in the estimation sample. A concern is that developers might selectively withdraw based on private information about future cost changes, biasing the survivor sample. We find no empirical

[26] The transition model does not include calendar time, a simplification supported by the finding that queue-position effects are not sensitive to the inclusion of time controls (Appendix Table E.4).

[27] The independence assumption simplifies the characterization of project strategies and the proof of equilibrium existence. When we allow the cost transition to be correlated with the study arrival through unobservable normal shocks in the ordered-probit and probit models, we estimate a correlation coefficient of -0.29 (se=0.04). The estimate implies that a new study with a higher cost estimate arrives more slowly. Allowing for the correlation yields similar payoff and waiting cost estimates and main counterfactual results. The results are available upon request.

or anecdotal evidence suggesting that developers can correctly predict these cost changes (Section 3.5).

For the distribution of first-study states (c, z), we estimate an ordered probit for the cost bin and probits for the study information z and for whether the first study is terminal. We multiply their predicted marginal distributions to obtain the joint distribution, assuming that the three outcomes are independent conditional on project characteristics (see Appendix D.4 for details).

Because endogenous withdrawals only occur immediately after a study arrives (Section 5.3), we treat withdrawals in the quarter a study arrives as endogenous and all other withdrawals as exogenous. Exogenous withdrawals account for less than half of withdrawals, and they are distributed fairly evenly over the wait window (Section 3.6). We therefore model them as a discrete-time hazard that is constant within each study stage,[28] with the stage-specific hazard depending on project characteristics: fuel type, an indicator for a capacity increase at an existing plant, and log capacity. This specification predicts an average per-period exogenous withdrawal probability of 0.029 with little difference between the two stages.

6.2 Project Payoffs and Waiting Costs

Measured Payoff Components. Estimation takes the revenue from power production $\tilde{r}_{it}$ and construction cost ω_{it} as known inputs to the project's payoff. We construct $\tilde{r}_{it}$ from hourly PJM wholesale electricity prices over the sample period, summed over a 25-year operating horizon and discounted at 5% (Appendix C); for batteries, whose operation involves buying and selling power, we set $\tilde{r}_{it}$ to zero. We approximate revenues using observed wholesale prices because we do not observe the long-term contracts through which many projects sell power. This approach lets revenues respond endogenously to entry in the counterfactuals, where we recompute $\tilde{r}_{it}$ under each candidate equilibrium by reconstructing the electricity supply curve with the updated capacity mix. For ω_{it}, we use EIA installation costs adjusted for tax credits (Appendix Table E.8). For solar and battery projects, we multiply installation costs by 0.7 to reflect the investment tax credit. For wind, we reduce installation costs by the value of the production tax credit (PTC), which we compute from predicted production.[29]

[28] The estimation and counterfactual simulation results are robust to including waiting time in the exogenous withdrawal hazards.

[29] We predict the wind production using capacity factors calculated from EIA data. The PTC pays a fixed rate for 10 years. Eligibility for each rate required meeting both a construction-start deadline and an

Identification. Taking the transition and exogenous withdrawal probabilities, present value of revenue from power production $\tilde{r}_{it}$, and construction cost ω_{it} as given, we use the projects' withdrawal and completion decisions to identify the following parameters: (1) the payoff parameters θ; (2) the distributions of the unobserved heterogeneity ξ^{S1}, ξ^{S2}, and ξ^{f}; and (3) the waiting costs γ^{S1} and γ^{S2}. Because interconnection costs enter the payoff in dollars per megawatt, alongside measured revenues and construction costs, all parameters are denominated in dollars and no scale normalization is required.

Appendix B proves identification using continuous variation in interconnection costs.[30] The proof proceeds backwards from the terminal study. Intuitively, changing the third-study cost moves the terminal shock threshold for completion dollar for dollar, so completion probabilities can trace out the shock's distribution. The proof assumes that, conditional on observed information from earlier studies and elapsed waiting time, each new study's cost is independent of the unobserved payoffs.[31] The complication is that projects reaching the third study are selected: those with higher unobserved payoffs are more likely to remain in the queue. The proof therefore must account for this selection before using third-study cost variation to identify the shock distribution.

The proof addresses this selection by using variation in second-study costs to isolate projects at the margin of continuing. Variation in the second-study cost shifts the continuation cutoff, changing which projects withdraw immediately after that study. Comparing changes in immediate and subsequent withdrawal probabilities, accounting for the probability of receiving the third study, reveals how marginal projects respond to the third-study cost. These projects share the same accumulated unobserved heterogeneity, $\xi^{\mathrm{S1}} + \xi^{\mathrm{S2}}$, so their terminal withdrawal probabilities trace out the distribution of ξ^{f} as the third-study cost varies, with its location pinned down by the mean-zero normalization. Applying the same logic to

in-service deadline, and we assume the latter was the binding constraint during our sample period. Assuming a 1-year lag between the terminal study and the in-service date, we assign rates by terminal-study year using the schedule in place before the Inflation Reduction Act: \$25/MWh for 2018 or earlier, \$20/MWh for 2019, \$15/MWh for 2020–2022, and \$0 thereafter.

[30] We assume a finite support for the costs in the existence proof in Appendix A but need continuous variation in costs for identification. Therefore, the identification assumptions are satisfied only approximately with discretized costs in the model.

[31] Some projects may form beliefs about the change in the third-study costs, such as expecting the costs to be revised downwards if the second-study costs are high, but we assume those beliefs depend only on the information we can observe from the first and the second studies. The change in the interconnection costs of the third study is thus conditionally independent of the selection on unobserved payoffs given the information from prior studies.

first-study costs identifies the distribution of ξ^{S2}; first-study withdrawal decisions then identify the distribution of ξ^{S1} and the payoff parameters.

The waiting costs are identified last: a project at the continuation cutoff is indifferent between withdrawing and waiting, and once the payoffs, distributions of unobserved heterogeneity, and beliefs are identified, the waiting cost is the only remaining unknown in this indifference condition.[32] Because the waiting costs are identified from withdrawal behavior, they measure the per-period opportunity cost of remaining an active project, not only direct expenditures.

Estimation. Given the estimates of transition and exogenous withdrawal probabilities in Section 6.1, we use maximum likelihood to estimate the payoff parameters, the scale parameters in the parametric distributions of the shocks, and the waiting costs. For the latent (ξ_i^{S1}, ξ_i^{S2}) in each project's state space, we discretize the support of unobserved heterogeneity to a 30-point grid.[33] Using the first-step estimates of state transitions as beliefs, we then solve the project's optimal decision problem for each combination of a grid value of the unobserved heterogeneity, observed study outcomes, and project characteristics. For each observed project that enters the queue, we integrate over the unobserved heterogeneity to compute the likelihood of its observed withdrawal and completion decisions.[34] For projects still active at the end of the sample, the likelihood contribution is the probability of the observed decisions through the last observed quarter.

Table 6 reports the estimates, with Panels A and C in million dollars per megawatt and

[32] Unobserved heterogeneity could also enter through waiting costs, but it is challenging to separately identify heterogeneity in both payoffs and waiting costs. If waiting costs were heterogeneous, continuation cutoffs would vary with them, so the projects tipped by a marginal increase in a study cost would no longer share the same accumulated heterogeneity, and the ratio argument could not recover the distribution of the terminal shock. Nor does the data pattern that motivates unobserved heterogeneity – later-stage projects are less likely to withdraw than recent entrants – distinguish the two sources. Heterogeneous waiting costs generate it through dynamic selection: later-stage projects are increasingly selected on low waiting costs, which, combined with the growing dispersion of ξ^f, lowers withdrawal at the terminal study. However, payoff heterogeneity alone generates the same pattern: projects with higher payoff realizations are more likely both to remain in the queue and to complete at the terminal study. We therefore allow for heterogeneity only in payoffs.

[33] In the dynamic program, beliefs about future latent values are represented as probability distributions over these grid points. For a project after a non-terminal first study, the belief is over $\xi^{S1} + \xi^{S2}$. Thus, when a project has not yet received a later study realization, its continuation value integrates over the possible future realizations by summing the value function over the grid, weighted by the corresponding probabilities. The grid spans $[-10, 10]$, which is wide relative to the sum of estimated scales of $\xi^{S1} + \xi^{S2}$ in Table 6 even when the waiting time is long. The large support makes the approximation insensitive to boundary points in practice.

[34] For projects receiving a terminal first study, the unobserved heterogeneity in their payoffs is $\xi^{S1} + \xi^f$. For other projects, it is $\xi^{S1} + \xi^{S2} + \xi^f$.

Panel B in thousand dollars per megawatt per quarter. In Panel A, we find that project payoffs vary across technologies. The omitted group is new natural gas generation. In the payoff function, we include the present value of wholesale revenue over the service life for all fuel types except batteries. We also include construction costs. Therefore, the fuel-type coefficients should be interpreted as the payoff of a fuel type relative to gas after netting out these two components. We estimate higher coefficients for wind and solar, likely because these coefficients absorb production costs, which are near zero for these technologies and substantial for gas. The battery coefficient is the highest, likely because it absorbs revenue as well as production costs. The point estimates also imply higher payoffs for larger projects, although they are imprecisely estimated. Renewable payoffs net of measured revenues and construction costs rise after 2018 relative to gas, consistent with the expansion of state renewable portfolio standards and broader wholesale market access for batteries, policies we do not explicitly model.

Panel B shows substantial waiting costs: $2,330 per megawatt per quarter while a project waits for the second study and $1,000 per megawatt per quarter while it waits for the third. We interpret these as the carrying costs of keeping a project in the queue and on schedule. The equivalent annual cost while waiting for the third study, about $4,010 per megawatt, is consistent with a project holding its position: maintaining site control and a minimal development team.[35] The higher cost while waiting for the second study, about $9,310 per megawatt per year, is consistent with the spending of a project in active development, when developers advance permitting, engineering, and the agreements required for the next study. An itemized estimate of development spending for a utility-scale solar project in PJM implies roughly $7,300 per megawatt per year in 2020 dollars, excluding deposits held by PJM (Leyline Renewable Capital, 2022). Our estimate is higher, consistent with waiting costs that include, beyond direct spending, any required return on the developer's capital above the model's 5% discount rate.

In a specification that allows waiting costs to vary by fuel type, with gas as the omitted group, the estimated additional waiting costs for solar, wind, and batteries are statistically indistinguishable from zero for both the second and third studies. The largest of these differ-

[35] Developers typically hold land under option during development, converting to a lease once the project proceeds; industry estimates of development-period site-control spending for a utility-scale solar project in PJM are roughly $3,400 per megawatt per year in 2020 dollars (Leyline Renewable Capital, 2022).

ences in absolute value is for wind projects waiting for the third study, whose waiting cost is $130 per megawatt per quarter lower than for gas.

Panel C shows that unobserved payoff heterogeneity is substantial and that its dispersion grows as a project waits. The standard deviation of ξ^{S1}, the heterogeneity revealed at the first study, is $1.34 million per megawatt, similar to the construction cost of a new project. The dispersions of ξ^{S2} and ξ^{f} grow with waiting time ($\kappa^{S2} = 0.30$ and $\kappa = 0.22$), and both are precisely estimated. Evaluated at 4 quarters since entry, one standard deviation of ξ^{S2} is $0.83 million per megawatt and one standard deviation of ξ^{f} is $0.73 million per megawatt. At 8 quarters, they increase to $1.02 million per megawatt and $0.86 million per megawatt, respectively.

6.3 Entry Costs

We construct a set of potential entrants and estimate entry costs. We group entrants by transmission owner, fuel type, size, and whether the project increases capacity at an existing plant. Within each group, we set the number of potential entrants in each year to the maximum observed annual count, and draw the characteristics of the added non-entrants from the group's observed entrants. We provide details in Appendix D.5.

To ensure that the equilibrium simulation remains tractable, we include a 5% random sample (514 projects) of these non-entrants. The resulting pool implies an average entry probability of 84%. The mass of the potential entrants, $N_t(x)$, is the sum of the observed entrants and the potential non-entrants with characteristics x in each period t. As a robustness check, we also report the estimates using twice as many non-entrants.

Identification. Identification follows from the threshold-crossing structure of entry: a potential entrant enters if its entry value exceeds its entry cost draw. Using the payoff parameters and continuation values from Section 6.2, we compute the entry value in Equation (5) for observed entrants and for potential entrants that do not enter. The entry value is a dollar amount implied by the prior step's estimates, so the distribution of entry costs is identified by how entry decisions respond to variation in the entry value.

The key source of variation underlying the entry values is the decline in renewable construction costs ω over time. Construction costs enter the entry value with a known coefficient through the payoff (Equation (1)) but do not enter the entry cost distribution, which is fixed

over time conditional on project characteristics (x^{ent} contains no calendar-time terms). The decline in construction costs therefore raises the entry values of later cohorts without changing their entry costs.[36]

Because developers in our model anticipate the decline in renewable construction costs, a concern is that they time their decisions to exploit it, either by delaying entry or by entering early and waiting in the queue, which would compress this variation. Delaying entry is not possible in our model because a potential entrant's opportunity to enter arises only once. Nor can a project enter and wait inside the queue for costs to decline: studies arrive on PJM's schedule, and the terminal study forces a completion decision, so early cohorts face lower entry values even though they anticipate the decline. The increase in renewable entry relative to the increase in equilibrium entry values thus identifies the distribution of entry costs.

Unobserved Heterogeneity. Our entry model treats potential projects as ex ante identical up to observed characteristics: the payoff heterogeneity ξ realizes only after entry, and non-entrants differ from entrants only through their entry-cost draws. This approach simplifies estimation and allows us to separately estimate project payoffs and entry costs. The key assumption is that heterogeneity known to developers at entry, such as site quality beyond what observables capture, is limited compared with the post-entry realizations of interconnection costs or unobserved payoff heterogeneity ξ. This assumption is consistent with developers submitting exploratory interconnection requests to learn about interconnection costs (Gorman et al., 2024). Furthermore, if unobserved site quality known to developers were the dominant source of heterogeneity, good sites would attract repeated requests after a withdrawal. Instead, repeated entry is rare (Section 3.1).

Estimation. We estimate the entry cost distribution by maximum likelihood, assuming that the entry cost C is supported on $[0, \bar{C}]$ and that $C/\bar{C}$ has a beta distribution.[37] The entry cost has mean $\exp(x^{\text{ent}}\zeta^{\text{ent}})$, where x^{ent} is the vector of project characteristics reported in Table 7.

[36] Using the prior step's estimates, we find that the capacity-weighted average value of entering the queue for renewables rises from \$0.129 million per megawatt for the 2012 cohort to \$0.336 million per megawatt for the 2020 cohort. For gas, construction costs were roughly flat while entry values fell over this period due to rising queue congestion and falling wholesale electricity prices.

[37] We prefer the beta distribution to a lognormal because the lognormal's right tail puts substantial probability on implausibly large entry costs. The two distributions are similar at the 85th percentile (\$0.214 for lognormal vs. \$0.246 million per megawatt for the beta distribution), but the 99th percentile for the lognormal is \$1.609 million per megawatt, comparable to the construction cost of a plant, against \$0.564 under the beta.

The implied distribution of $C/\bar{C}$ is $\text{Beta}\big((\exp(x^{\text{ent}}\zeta^{\text{ent}})/\bar{C})\psi, (1-\exp(x^{\text{ent}}\zeta^{\text{ent}})/\bar{C})\psi\big)$, determined by the mean and a free precision parameter ψ; the corresponding variance of the entry cost is $\frac{\exp(x^{\text{ent}}\zeta^{\text{ent}})(\bar{C}-\exp(x^{\text{ent}}\zeta^{\text{ent}}))}{\psi+1}$. We set $\bar{C}$ to \$0.65 million per megawatt, slightly above the highest entry value implied by the prior step's estimates (\$0.61 million per megawatt).

Table 7 reports the estimates. Panel A shows that, on a per-megawatt basis, entry costs are lower for renewables than for non-renewables, and larger projects face higher entry costs. Panel B shows that entry costs are low for most projects. The baseline estimates in column (1) imply that entry costs are \$0.030 million per megawatt at the median and \$0.25 million per megawatt at the 85th percentile, relative to a median entry value of \$0.24 million per megawatt (\$0.21 million per megawatt for renewables).

Estimates with twice as many non-entrants (column (2)) imply higher entry costs but a similar relationship between these costs and covariates. The increase is especially large, in proportional terms, at lower quantiles and follows mechanically from the construction: a larger pool implies a lower average entry probability, which the estimation matches by raising entry costs. Columns (1) and (2) both imply 2,708 entrants, close to the 2,709 observed in the data.

Table 6: Estimates of Payoff and Waiting Cost Parameters

Panel A. Payoff Estimates	
Wind	0.663***
	(0.108)
Solar	0.997***
	(0.048)
Battery	1.282***
	(0.064)
Capacity increase at existing plant	0.309***
	(0.080)
ln(size)	0.049
	(0.030)
Renewables × ln(size)	0.029
	(0.041)
After 2018	-0.120
	(0.078)
Renewables × after 2018	0.402***
	(0.090)
Intercept	-1.680***
	(0.086)
Panel B. Waiting Costs ×1000	
Waiting for third study	-1.324***
	(0.191)
Intercept	2.327***
	(0.258)
Panel C. Unobservables: Baseline Scale σ and Waiting Time Effect κs	
Study 1: σ^{S1}	1.338***
	(0.097)
Study 2: σ^{S2}	0.549***
	(0.028)
Study 2: κ^{S2}	0.296***
	(0.017)
Terminal Study: σ	0.541***
	(0.022)
Terminal Study: κ	0.220***
	(0.007)

The sample is projects entering the queue in 2012–2020, with withdrawal and completion decisions observed through the first quarter of 2023 (N = 2,709 projects). Parameters in panels (A) and (C) are in units of $ million per MW. Panel B's units are in $ thousand per MW per quarter. Panel A uses gas projects as the omitted group. The after 2018 indicator is based on the terminal study year. The κ parameters are unitless. Standard errors in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Standard errors account for the statistical errors in the first-stage transition-function estimates using a Murphy–Topel correction (Murphy and Topel, 2002).

Table 7: Entry Cost Distribution Estimates

	(1)	(2)
	Baseline	Larger Pool
Panel A. Entry Cost Distribution Parameters		
Renewables	-0.377**	-0.390***
	(0.177)	(0.145)
Capacity increase at existing plant	0.049	0.094
	(0.215)	(0.112)
ln(size)	0.154***	0.117***
	(0.032)	(0.024)
Intercept	-2.673***	-2.080***
	(0.238)	(0.208)
ψ	1.901***	1.802***
	(0.376)	(0.284)
Panel B. Implied Entry Cost Quantiles ($M/MW)		
50th percentile	0.030	0.087
75th percentile	0.145	0.254
80th percentile	0.189	0.304
85th percentile	0.246	0.362
90th percentile	0.319	0.431
95th percentile	0.424	0.517

The sample is observed projects entering the queue in 2012–2020 (2,709) and simulated potential non-entrants (514 for column (1) and 1,028 for column (2)). The entry cost distribution is estimated as a beta distribution on $[0, \bar{C}]$. Let the mean cost $\mu_i = \exp(x_i^{ent}\zeta^{\text{ent}})$ and let ψ denote the precision parameter. Then $C_i/\bar{C} \sim \text{Beta}(\alpha_i, \beta_i)$ with $\alpha_i = (\mu_i/\bar{C})\psi$ and $\beta_i = (1 - \mu_i/\bar{C})\psi$. Columns (1) and (2) use 514 and 1,028 potential entrants that do not enter, respectively, along with 2,709 entrants observed in the data. Quantile rows report the implied quantiles of the entry cost distribution in $M/MW, based on the estimated mixture over observed entry covariates. Standard errors are in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Standard errors account for the statistical errors in the first-stage transition-function estimates and second-stage model estimates using a Murphy–Topel correction (Murphy and Topel, 2002).

7 Equilibrium Simulation and Model Fit

To simulate the equilibrium in Section 5.3, we start from the observed queue and iteratively update project strategies, queue statuses, project beliefs, and wholesale prices until convergence.[38] We use this procedure for the model-fit analysis below and for all counterfactual simulations.

We assume that, in the no-policy (as-is) equilibrium, projects expect wholesale prices after 2020 to remain at the 2021 level. Specifically, the 2021 prices for each quarter and hour repeat in each subsequent year. As described in Appendix C, we predict prices for 20 representative hours per quarter, so revenues reflect prices during the hours in which each project produces. The generation-weighted average price is \$32.0/MWh in 2021. This assumption reflects the relatively stable prices in PJM over the decade before 2021. It ignores the later surge in demand, driven largely by data centers, and the associated rise in prices, which projects entering in 2020 or earlier had little reason to anticipate.

Wholesale prices adjust endogenously in the counterfactuals. As more projects complete and add to PJM supply, prices can remain stable only if demand grows. We therefore compute the implied demand for 2021–2028 that makes the as-is equilibrium consistent with the stable-price assumption, with 2028 being the last year in which a 2020 entrant can still act under the maximum wait of 32 quarters. In policy counterfactuals, we hold demand fixed — observed for 2012–2020 (Appendix C) and implied for 2021–2028 — and recompute wholesale prices given each counterfactual's completions. These prices enter the wholesale revenues in projects' payoff functions. The new equilibrium is achieved when project strategies, queue statuses, the wholesale prices, and project beliefs are consistent with each other.[39] We assume that wholesale prices after 2028 remain at the 2028 equilibrium level. We also report counterfactual policies' effects on this long-run wholesale price, computed as a generation-weighted average over the representative hours of 2028.

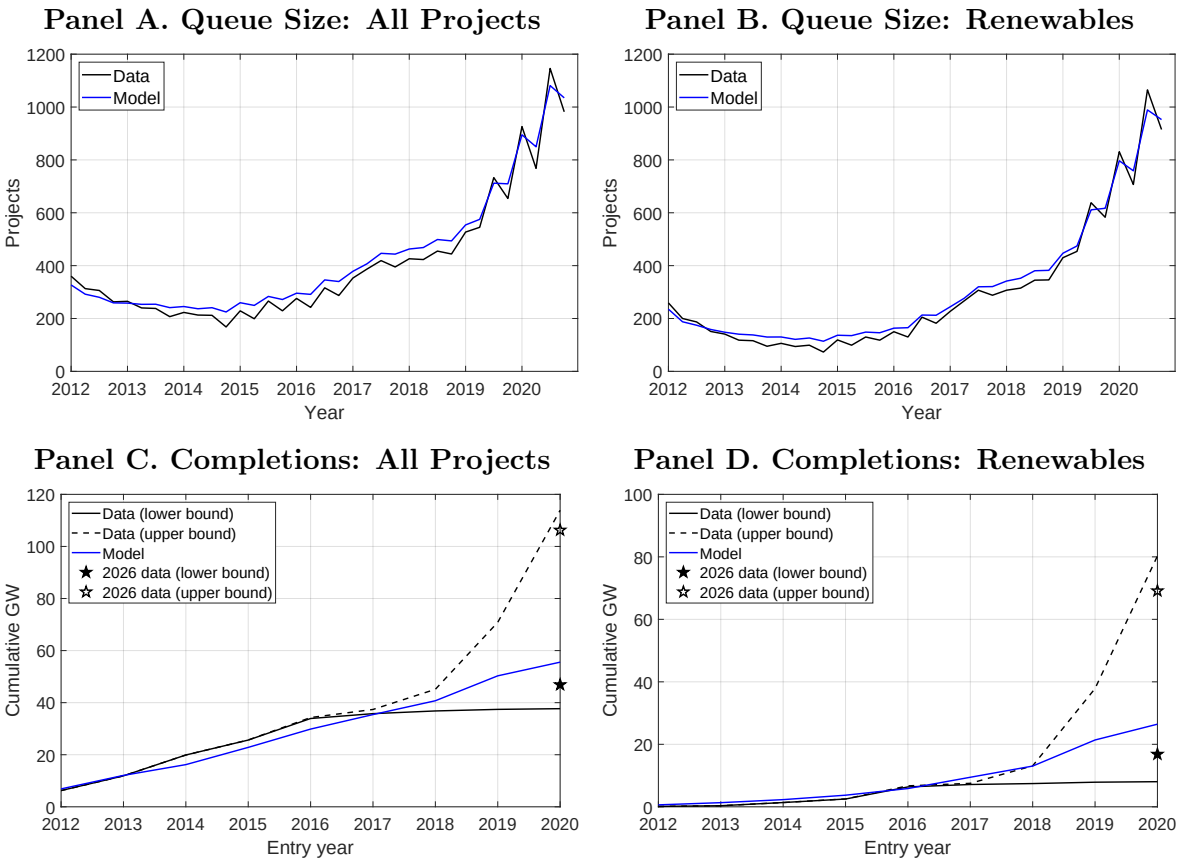
[38] We cannot rule out the presence of multiple equilibria, but starting from queue statuses perturbed around the observed values yields highly similar equilibrium outcomes.

[39] As a simplification, we normalize the present value of batteries' wholesale revenues to 0 and use the battery coefficient in the payoff function to capture the wholesale revenue. Effectively, we assume that battery wholesale revenue, which depends on the price differences across hours during a day, is invariant to the equilibrium price levels that we endogenize. Batteries account for 3.9% of observed entrant capacity from 2012 to 2020.

Model Fit. The model fits the data well in the estimation sample (Figure 5). Panels A and B plot predicted and observed queue sizes, which are the numbers of waiting projects at a given time, for all projects and renewables; both predictions track the data closely. Panels C and D plot cumulative eventual completion by entry year, defined as the sum of eventually completed capacity by projects entering in or before a given year. The blue line is the model's prediction; the solid black line shows capacity completed by 2023 (a lower bound on eventual completion), and the dashed black line above it adds capacity still waiting in 2023 (an upper bound). The model prediction is close to the data in early years and stays within the bounds in later years, for both total and renewable capacity. Our model predicts that 55.6 GW will eventually be in service in PJM from those entering between 2012 and 2020.

The model prediction is also consistent with the completions observed after the estimation sample. We collected 2026 data on the 2012–2020 entry cohorts; by then, more projects had received terminal studies, and more had completed. Our prediction falls within the implied tighter bounds (indicated by stars).

Figure 5: Model Fit: Queue Size and Completions



Panels A–B plot queue counts by calendar quarter (2012–2020) for all projects and renewable projects, respectively. Panels C–D plot cumulative eventual completed capacity by entry year. The solid black line is observed completed capacity by queue entrants from 2012 to 2020 based on their statuses as of 2023; the dashed line adds projects whose status is still unresolved at that date; the blue line is the model's predicted eventual completed capacity. The solid star shows observed completions, as of 2026, of those entering between 2012 and 2020; the hollow star adds projects still in queue.

8 Counterfactuals

Using the simulation procedure in Section 7, we evaluate two reforms: speeding up engineering studies and increasing the cost of entering and remaining in the queue. The counterfactuals allow for equilibrium responses in entry, waiting, completion, and wholesale prices.

8.1 Study Speedup

Recent initiatives aim to accelerate engineering studies: grid operators are beginning to use AI to produce them, and FERC Order 2023 penalizes transmission providers for study delays. In the model, we scale up the study arrival probabilities Q_i^{S1} and Q_i^{S2} in Equations (3) and (4) by 10 to 50%, capping each at one. We scale each of the underlying probabilities $q_i^{\mathrm{S1,f}}(\cdot)$, $q_i^{\mathrm{S1}}(\cdot)$, and $q_i^{\mathrm{S2}}(\cdot)$ by the same factor, so the distribution of study outcomes conditional on arrival is unchanged.

We find that a speedup substantially increases completed capacity. Table 8 reports levels under the as-is equilibrium and changes under each speedup for completion, waiting time (the time between a project's entry and its completion or withdrawal), entry, entry surplus, and the long-run wholesale price. A 50% speedup increases total completion by 32% and reduces waiting time by 21%.[40] Entry rises by only 2.2%, so the increase in completed capacity comes mostly from higher completion among projects that enter the queue. The additional entry is concentrated in projects above 100 MW (Appendix Table F.1), and non-renewable projects account for 5.6 of its 5.7 GW.[41]

The speedup raises entry surplus and lowers the long-run wholesale price. Entry surplus is defined as $\sum_i k_i E \max\{EV_i - C_i, 0\}$ across all potential projects, where k_i is nameplate capacity in MW. It increases with the degree of speedup, but at a diminishing rate.[42] Most of the gain accrues to non-renewable projects ($3.8 of the $4.4 billion). Renewable installation costs fall by about $0.02 million per megawatt for each quarter a project waits, so completing one quarter earlier raises costs by roughly 12% of the average renewable entry surplus ($0.20

[40] The reduction in waiting time is proportionally lower than the increase in study arrival probabilities for two reasons. First, a speedup increases the incentive to wait in the queue, which can lengthen waiting time for projects that eventually withdraw. Second, about 16% of projects receive a terminal first study, so speeding up later studies does not affect them.

[41] The smaller renewable entry response is not driven by a shortage of potential entrants. In 2020 alone, there are over 5 GW of potential renewable entrants that do not enter in the as-is equilibrium. The potential capacity in prior years is even higher.

[42] Surplus gains by project size and fuel type are reported in Appendix Table F.2.

million per megawatt). Gas projects, whose installation costs are flat over this period, face no such offset. These surplus figures cover only project developers. The added capacity under a 50% speedup decreases the long-run wholesale electricity price by 3.8%. The lower wholesale price also benefits consumers, though with inelastic demand this is largely a transfer from incumbent generators rather than an additional welfare gain.

Table 8: Study-Speedup Counterfactuals

	(1)	(2)	(3)	(4)	(5)	(6)
	As-is	10%	20%	30%	40%	50%
Completed (GW)	55.6	+4.2	+8.4	+12.0	+15.1	+17.8
Renewable	26.4	+2.4	+5.0	+7.3	+9.1	+10.7
Non-renewable	29.1	+1.8	+3.4	+4.8	+6.0	+7.1
Waiting time (quarters)	10.4	-0.5	-0.9	-1.4	-1.8	-2.2
Renewable	11.5	-0.5	-1.0	-1.5	-2.0	-2.4
Non-renewable	6.8	-0.3	-0.6	-0.9	-1.1	-1.4
Entry (GW)	262.5	+1.4	+2.7	+3.8	+4.7	+5.7
Renewable	156.1	0.0	+0.1	+0.1	+0.1	+0.1
Non-renewable	106.4	+1.4	+2.6	+3.7	+4.6	+5.6
Entry surplus ($Bn)	59.6	+1.0	+2.0	+2.9	+3.7	+4.4
Renewable	36.3	+0.1	+0.3	+0.4	+0.5	+0.7
Non-renewable	23.3	+0.9	+1.7	+2.4	+3.1	+3.8
Wholesale price ($/MWh)						
Long run	32.0	-0.3	-0.6	-0.9	-1.1	-1.2

Column (1) reports the as-is equilibrium in levels; columns (2)–(6) report changes relative to as-is. Entries and completions are in GW. Waiting time is in quarters. Surplus rows report total entry surplus, defined as $\sum_i k_i E \max\{EV_i - C_i, 0\}$ across all potential projects, in $Bn. Here k_i is nameplate capacity in MW. Long-run price is the 2028 equilibrium wholesale price, averaged across quarters using the sum of wind, solar, and gas generation per MW as weights. Demand in 2021–2028 is chosen so that, given as-is capacity, equilibrium prices remain at the 2021 level (Section 7), and the calculated demand is held fixed across counterfactuals.

Environmental Benefits. The additional renewable capacity under the 50% speedup avoids about 12.5 million metric tons of CO_2 annually. We compute these benefits using EPA's AVoided Emissions and geneRation Tool (AVERT v4.3, April 2024), which estimates region-specific marginal emissions avoided when new renewable generation displaces fossil generation. For PJM, AVERT's capacity factors imply that the counterfactual's additional 3.99 GW of wind and 5.81 GW of solar generate 10.3 and 11.4 TWh per year, respectively; applying AVERT's 2023 PJM marginal CO_2 emissions rates gives the 12.5 million metric tons.[43] At a

[43]The avoided air pollutants, such as $PM_{2.5}$, would increase the benefits, while the 7.1 GW increase in gas capacity has an ambiguous effect, since its emissions may be offset by displacing coal-fired generation.

social cost of carbon of $185 per metric ton (Rennert et al., 2022), these avoided emissions are worth $2.3 billion per year.

Option Value of Waiting. The muted entry response reflects two countervailing effects of the speedup on the value of entering the queue. A waiting project holds an option: it completes only on favorable payoff realizations and withdraws otherwise, so the value of waiting rises with the dispersion of those realizations. Because the standard deviations of ξ^{S2} and ξ^{f} increase with waiting time, a faster queue compresses this dispersion and reduces the option value of remaining in the queue. Offsetting this, faster study arrivals reduce accumulated waiting costs, discounting, and exposure to exogenous exits, raising the value of entering the queue.

To isolate the option-value effect, we compress the dispersion of payoff realizations to reflect the shorter waits under the 50% speedup. Specifically, for each waiting time τ, we replace the standard deviation of ξ^{f}, $\sigma \cdot \tau^{\kappa}$, with $\sigma \cdot (\max\{1, \tau - \Delta t\})^{\kappa}$, where $\Delta t = 2.2$ quarters is the average reduction in waiting time under the speedup. We do the same for ξ^{S2}. We hold study arrival probabilities and queue statuses at their as-is equilibrium values and allow projects to re-optimize once given the smaller standard deviations.

We find that compressing the dispersion of payoff realizations alone would lower completion, entry, and surplus relative to the as-is equilibrium (column (3) of Table 9). Completed capacity falls by 1.40 GW.[44] Projects also withdraw faster, leading to a slight reduction in waiting time.[45] This decomposition provides an additional explanation, beyond the high entry costs of remaining non-entrants, for why the entry response to a 50% speedup is smaller than the completion response. The reduction in option value decreases total entry by 4.1 GW. The offsetting increase in entry value accounts for the remaining 9.7 GW. The net effect is the 5.7 GW increase.

Number of Potential Entrants. The increases in completed capacity from the speedup remain similar when the number of potential non-entrants is doubled. The simulations in Table 8 include 514 potential entrants that do not enter in the as-is equilibrium. We re-

[44] Projects for which the floor of 1 quarter binds account for 8.6% of the lost completion relative to the as-is equilibrium, or 0.12 GW.

[45] Queue statuses and study arrival probabilities are held fixed, so this reduction in waiting time is entirely from faster withdrawals.

simulate the as-is equilibrium and the 50% speedup with twice as many sampled non-entrants (1,028 rather than 514, for 3,737 potential projects), using the entry cost distribution re-estimated with the doubled pool (column (2) of Table 7). Under a 50% speedup, entry increases by 7.4 GW, all of it non-renewable (renewable entry falls slightly), 30% more than in Table 8, while completed capacity increases by 17.6 GW (10.8 GW renewable), 1% less. Thus, the larger entry response does not translate into additional completion, because the additional entry increases congestion, which offsets the benefits of the speedup.

Capacity Market Revenue. The additional capacity from a study speedup would lower prices in PJM's capacity market, a response our model does not capture. PJM's capacity market pays resources for being available to meet peak demand, separately from their energy market earnings. Our specification captures these payments with the time-invariant fuel-type coefficients, so it does not account for how new capacity would lower the capacity price. To provide an upper bound on this price decline, we shift the supply curve of a single auction, the 2021/2022 Base Residual Auction (PJM Interconnection, 2018*a*,*b*), outward by all of the capacity the 50% speedup adds, after scaling each technology's capacity by its Effective Load Carrying Capability. We add this capacity at an offer price of zero. The resulting 9.70 GW shift lowers the system-wide price by 17%, from \$140 to \$116 per megawatt-day.[46] A zero offer gives the largest possible price decline, as does ignoring the retirements that a lower price would induce.

Even this upper-bound 17% price decline has a limited effect on entry values and completion, despite substantial revenue losses for projects that complete. For renewable projects, which supply most of the added capacity and earn little capacity revenue per megawatt, the price decline lowers lifetime revenue by about 20% of their average surplus from entering the queue; for natural gas projects, whose capacity payments are a larger share of revenue, the loss is about 61%. These revenue losses have a more limited effect on the expected value of entering the queue because a project realizes them only if it completes, and most do not. Relative to the 50% speedup result without this price decline, the expected entry values decrease by 7.1% for gas and 2.6% for renewables, reducing the gain in total completion by 3.5 GW

[46]The added capacity is 7.07 GW gas, 5.81 GW solar, 3.99 GW wind, and 0.93 GW battery; class-average Effective Load Carrying Capability factors of 0.92, 0.38, 0.13, and 0.50 (PJM Interconnection, 2021*b*) convert it to 9.70 GW of accredited capacity, which we add to the supply curve and intersect with PJM's demand (Variable Resource Requirement) curve.

(14.3 GW rather than 17.8 GW). The decrease in gas completion accounts for 81% of the total decrease. Our results are thus robust to the capacity market price response.

Table 9: Role of Option Value in 50% Speedup Effects

	(1)	(2)	(3)
	As-is	50% speedup	Option value
Completed (GW)	55.6	+17.8	-1.4
Renewable	26.4	+10.7	-0.6
Non-renewable	29.1	+7.1	-0.8
Waiting time (quarters)	10.4	-2.2	-0.2
Renewable	11.5	-2.4	-0.1
Non-renewable	6.8	-1.4	-0.2
Entry (GW)	262.5	+5.7	-4.1
Renewable	156.1	+0.1	-1.7
Non-renewable	106.4	+5.6	-2.3
Entry surplus ($Bn)	59.6	+4.4	-3.4
Renewable	36.3	+0.7	-1.9
Non-renewable	23.3	+3.8	-1.6

Column (1) reports the as-is equilibrium in levels; columns (2)–(3) report changes relative to as-is. Entries and completions are in GW. Waiting time is in quarters. In columns (1)–(3), surplus rows report total entry surplus, defined as $\sum_i k_i E \max\{EV_i - C_i, 0\}$ across all potential projects, in $Bn. Here k_i is capacity in MW. The Option value column takes the as-is equilibrium queue statuses and rescales the standard deviations of ξ^f to $\sigma \cdot (\max\{1, \tau - \Delta t\})^\kappa$ (and similarly for ξ^{S2}) for each value of waiting time τ, where $\Delta t = 2.2$ quarters is the average reduction in waiting time from the speedup. We then allow the projects to re-optimize once, responding to the as-is equilibrium queue statuses but ξ^{S2} and ξ^f shocks with smaller standard deviations.

8.2 Fees and Deposits

Another key component of the ongoing interconnection queue reforms is increasing the cost of entering the queue to reduce congestion. We focus on per-megawatt fees and deposits collected at queue entry that differ in refundability and timing. We then compare their effects with those of flat fees and deposits.

Per-Megawatt Fees and Deposits. We first consider a nonrefundable fee, which the project pays upon entry and does not recover regardless of the project's outcome. Second, we consider a deposit that is fully at risk: the deposit is returned only if the project completes interconnection, so projects that withdraw forfeit the entire deposit. Third, we consider a staged deposit, which is closest to the PJM reform: projects pay a deposit upon entry and receive a 50% refund if they withdraw after the first study rather than request the second

study. If they continue, they recover the deposit only if they complete interconnection. The deposit's recoverability thus decreases as projects advance in the queue.

We additionally consider budget-neutral versions of the two deposits. In these versions, the deposits forfeited by withdrawing projects are redistributed to completing projects on a per-megawatt basis, so that PJM does not retain revenue. Completing projects therefore receive more than the deposit they paid at entry. Such deposits would be hard to implement in real time, since the rebate depends on future withdrawals and completions, but provide a useful benchmark.[47]

Fees and deposits change completion by at most a few percent, with the direction depending on refundability and staging. Table 10 reports the effects at $20,000 per megawatt. We choose this level because it is high enough to distinguish them and plausible: PJM adopted a comparable fee in 2026 for a temporary program that accelerates some projects to address short-term reliability concerns (PJM Interconnection, 2026). For comparison, the level PJM is adopting in wider reforms is $4,000 per megawatt (PJM Interconnection, 2025*d*). The nonrefundable fee reduces completion by 1.2%, and the deposit reduces it by 0.2%. The staged deposit raises completion by 0.8%, while the two budget-neutral deposits raise it by 3.6% and 4.2%. Refunds strengthen the incentive to complete, and staging additionally encourages earlier withdrawals by projects unlikely to complete.

Under the nonrefundable fee, average waiting time is essentially unchanged, but this masks two offsetting changes. Waiting time falls within most groups of projects defined by fuel type, location, entry time, and size (Appendix Figure F.1). At the same time, the fee disproportionately deters small renewable projects (Appendix Table F.5), and 42.0% of the deterred projects are renewables that enter in 2012-2015, which receive studies faster and are more likely to withdraw. So longer-waiting projects make up a larger share of the queue. This composition shift offsets the within-group declines.

Waiting time rises under the deposit and the budget-neutral deposit but falls under the two staged deposits. Refunds tied to completion induce longer waits, whereas staging works in the opposite direction: projects unlikely to complete withdraw after the first study, before the

[47] Our budget-neutral deposits are realistic in retaining no revenue for PJM but stylized in where forfeitures go. Under the ongoing reform, forfeited deposits fund restudies, with any remainder going to other projects in the cluster, whereas our budget-neutral deposits direct all forfeitures to completing projects. The reform also requires additional deposits tied to network upgrade costs, which we abstract from.

Table 10: Fee and Deposit Counterfactuals at $20,000 per MW

	As-is	Fee	Deposit		Budget neutral	
	(1)	(2)	(3)	(4)	(5)	(6)
Completed (GW)	55.6	-0.7	-0.1	+0.4	+2.0	+2.3
Renewable	26.4	-0.2	0.0	+0.3	+1.0	+1.2
Non-renewable	29.1	-0.5	-0.1	+0.1	+1.0	+1.1
Waiting time (quarters)	10.4	0.0	+0.1	-0.4	+0.3	-0.2
Renewable	11.5	+0.1	+0.1	-0.4	+0.3	-0.2
Non-renewable	6.8	-0.1	0.0	-0.3	+0.1	-0.2
Entry (GW)	262.5	-6.0	-4.8	-3.9	-0.9	-0.4
Renewable	156.1	-3.1	-2.5	-2.1	-0.4	-0.3
Non-renewable	106.4	-2.9	-2.4	-1.8	-0.4	-0.1
Entry surplus ($Bn)	59.6	-4.8	-3.8	-3.2	-0.2	0.0
Renewable	36.3	-3.0	-2.4	-2.1	-0.1	-0.1
Non-renewable	23.3	-1.8	-1.4	-1.1	-0.1	+0.1
Payments ($Bn)						
Collected at entry	0.0	+5.1	+5.2	+5.2	+5.2	+5.2
Returned to projects	0.0	0.0	+1.1	+1.5	+5.2	+5.2
Retained	0.0	+5.1	+4.0	+3.7	0.0	0.0
Wholesale price ($/MWh)	32.0	+0.1	0.0	0.0	-0.2	-0.2
Share of payment forfeited						
If withdraw before requesting the 2nd study		100%	100%	50%	100%	50%
If withdraw later		100%	100%	100%	100%	100%
If complete		100%	0%	0%	0%	0%
Forfeitures rebated to completers		No	No	No	Yes	Yes

Column (1) reports the as-is equilibrium in levels; columns (2)–(6) report changes relative to as-is. All five fees and deposits charge $20,000 per MW of nameplate capacity at queue entry. The bottom rows report the share of that payment forfeited at each outcome. In column (2), the entry fee is not refundable. In column (3), the deposit is refundable when the project completes but forfeited if the project withdraws. In column (4), the deposit is 50% refundable if the project withdraws before requesting the second study; thereafter, the deposit is refunded if the project completes but forfeited if it withdraws. In columns (5) and (6), the forfeiture rules are the same as in columns (3) and (4), respectively, before a project completes, but all forfeited deposits are rebated to completing projects on a per-megawatt basis. Entry surplus is defined as $\sum_i k_i E \max\{EV_i - C_i, 0\}$ across all potential projects and is reported in $Bn. Here k_i is nameplate capacity in MW. Wholesale price is the long-run price.

deposit becomes fully at risk.[48] Under the staged deposit, waiting time falls by 0.4 quarters. The two forces partially offset each other under the budget-neutral staged deposit, where waiting time falls by 0.2 quarters.

All five fees and deposits reduce entry, by 0.4 to 6.0 GW, or at most 2.3%. Entry falls least under the budget-neutral deposits, which return every dollar to the queue, and most under the nonrefundable fee.

Because the completion effects are small, so are the wholesale price effects. The long-run wholesale price falls by \$0.2/MWh (0.6% and 0.7%) under the budget-neutral deposits and is essentially unchanged under the others.

For each fee and deposit, the magnitude of the effect on completed capacity increases with the amount required at entry, while the sign remains unchanged up to \$50,000 per megawatt. Appendix Figure F.2 plots completed capacity, in total and for renewables, across this range. Completion increases with the amount under the staged and budget-neutral deposits and falls under the nonrefundable fee and the unstaged deposit. At still higher levels not shown in the figure, completion eventually declines for all five. Under the staged deposit, for example, completion peaks near \$100,000 per megawatt at 57.6 GW, or 3.7% above as-is. At the reform level of \$4,000 per megawatt, the effects are smaller than at \$20,000 per megawatt but in the same direction.

Flat Fees and Deposits. Flat fees and deposits, which require the same dollar amount regardless of project size, have modest and positive effects on completion.[49] Appendix Table F.3 reports the effects at \$400,000 per project, equivalent to \$20,000 per megawatt for a 20 MW project, which is the median project size. At this level, the nonrefundable fee and the deposit each increase completed capacity by about 0.65%, and renewable capacity by 0.3% and 0.4%, respectively. The staged deposit increases completed capacity slightly more, by 1.2%, and renewable capacity by 1.2%. Appendix Figure F.3 plots completed capacity at levels up to \$400,000; under all three, it increases with the amount. The three fees and deposits have

[48] An alternative staged deposit refunds 50% until projects request the third study, moving the at-risk transition one study later. This deposit adds 0.88 GW more completed capacity than the staged deposit in Table 10 and reduces waiting time by a further 0.39 quarters. Paralleling the staged deposit's effect after the first study, this deposit reduces the number of projects waiting for the third study, where congestion has a larger effect on study arrival (Table 3).

[49] The size-tiered deposits in FERC Order 2023's default procedures (FERC, 2023) fall between the per-megawatt and flat designs.

more similar effects under the flat design than under the per-megawatt design because the flat design mainly deters small projects. For the larger projects that remain, the amount is small per megawatt, so the refund provisions matter little.

The burden of the flat design falls almost entirely on small projects. Under the flat nonrefundable fee, 224 fewer projects enter, or 8.3% of as-is entrants, while entry falls by only 1.4 GW, or 0.5% (Appendix Tables F.3 and F.5). More than 80% of the projects deterred by each flat design are at or below 10 MW, and 97% are at or below 20 MW. Appendix Tables F.4 and F.5 further show that, across all three fees and deposits, the flat design deters about three times as many renewable entrants as the per-megawatt design and reduces surplus per megawatt more for these small projects. Projects above 100 MW are essentially unaffected.

This concentration among small projects also explains the effects on waiting times. The flat nonrefundable fee and deposit lengthen average waiting time, while the staged deposit leaves it essentially unchanged (Appendix Table F.3). In the as-is equilibrium, projects at or below 10 MW tend to exit the queue after fewer studies: 46% receive a terminal first study and a further 16% a terminal second study, compared with 16% and 9% of all entrants. As a result, they leave the queue after 4.1 quarters on average, compared with 10.4 quarters for all projects. By disproportionately deterring such projects, the flat design does little to shorten waits for the projects that remain.

Discussion. Per-megawatt fees and deposits that are nonrefundable or fully at risk upon entry do little to relieve congestion. They deter the lowest-value projects that would have spent little time in the queue. Higher-value projects, which are more willing to wait and more likely to complete, contribute the most to congestion but are not deterred from entering the queue. The flat design screens on size instead, which changes who is deterred but not the result, since the very small projects it removes need few studies and leave the queue quickly.

The per-megawatt staged deposit, in contrast, does more to relieve congestion because it screens on the first study's results rather than on what is known at entry. Relative to the unstaged deposit, more projects enter but fewer remain after the first study, and those that remain are more likely to complete. Even so, staging improves completion only modestly because projects that remain at later stages are positively selected: they have higher value and are more likely to complete. Putting deposits at risk or charging nonrefundable fees at later stages would increasingly burden projects that are likely to complete. The same positive

correlation between project value and congestion limits the gains from raising costs at any stage of the queue.

These policies nonetheless raise revenue. For example, at the $20,000 per-megawatt level, the deposit retains $4.0 billion in forfeited deposits from withdrawing projects. Our speedup counterfactuals abstract from the cost of delivering faster studies, which in practice requires real resources such as additional engineering capacity. A bundled reform could therefore use this revenue to fund study acceleration.

9 Conclusion

We use novel data from PJM, the largest U.S. regional grid operator, to study the interconnection queue. We find a congestion externality: if a project has more projects ahead of it in the queue, the probability of study arrival in a given quarter falls. We also find that interconnection costs are a key factor in projects' decisions to withdraw from the queue.

We next study policy reforms using a dynamic model that accounts for equilibrium effects. We find that speeding up study delivery substantially increases completion: a 50% speedup raises completed capacity by 32%, or 17.8 GW. Most of the added capacity is wind and solar, yielding $2.3 billion per year in environmental benefits from avoided carbon emissions. The added capacity also lowers the long-run wholesale price, benefiting consumers. Faster studies are the goal of reforms underway across the U.S., including AI-assisted automation and penalties on transmission providers for delayed studies.

In contrast, increasing the cost of entering the queue, the aim of stricter deposits in current reforms, has more limited effects on completion and barely changes waiting time. This result reflects a positive correlation between a project's value and its impact on congestion: the low-value projects screened out by higher entry costs would also have waited the least and created the least congestion.

The economic mechanisms we identify extend beyond the serial study process that governed PJM's queue during our sample period. U.S. grid operators are now transitioning from serial to cluster studies, which group interconnection requests and study them together. Faster studies still reduce waiting costs and exposure to exogenous exits, and the projects most willing to wait are still the most valuable, so we expect substantial returns to study acceleration under cluster studies as well. Evaluating cluster designs directly is an important

avenue for future research.

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A Model Details

We first characterize optimal strategies in Section 5.1 as cutoff rules, derive the law of motion for project masses used in the equilibrium definition in Section 5.3, and prove equilibrium existence.

Notation. We focus on a particular project and suppress the project-specific subscript. Let x denote time-invariant project characteristics, e the quarter in which the potential project arrives and makes its entry decision, c the interconnection cost, and z study information. A project that enters joins the queue in $e+1$. Let η denote accumulated payoff information: $\eta = \xi^{\mathrm{S1}}$ in Stage 1 and $\eta = \xi^{\mathrm{S1}} + \xi^{\mathrm{S2}}$ in Stage 2, replacing ξ in the arguments of $\bar{V}$, W^{S2} and W^{S1} in (2), (3) and (4). The exogenous probabilities of exits, $1-\delta^{\mathrm{S1}}$ and $1-\delta^{\mathrm{S2}}$, are based on x, and we suppress their i subscript too. We use the queue-status path $\{\mathcal{K}_t(e,x)\}_t$ defined in Section 5.3. It affects study arrivals and observed study outcomes, but not the distribution of new payoff information ξ^{S2}. Finally, we assume that (x, e, c, z, t) have finite supports, the accumulated payoff state η has an absolutely continuous distribution, and each queue-status vector $\mathcal{K}_t(e,x)$ lies in a bounded subset of Euclidean space.

For projects arriving in e, let $T_e = e+T$ denote their terminal period. Waiting decisions are made for $e+1 \le t < T_e$. At the start of T_e, every remaining project receives its terminal study immediately, without another waiting cost or survival draw. Write $p_s^c(c' \mid x, e, c, z)$ for the cost probabilities conditional on study arrival (p_i^c in Appendix D), with project characteristics made explicit and stage s shown as a subscript. The boundary values are

$$
\begin{aligned}
W_{T_e}^{\mathrm{S2}}(x,e,c,z,\eta) &= \sum_{c'} p_{\mathrm{S2}}^c(c' \mid x,e,c,z)\bar{V}(x,e,T_e,c',\eta),\\
W_{T_e}^{\mathrm{S1}}(x,e,c,z,\eta) &= \sum_{c'} p_{\mathrm{S1}}^c(c' \mid x,e,c,z)E_{\xi_{T_e}^{\mathrm{S2}}}\left[\bar{V}(x,e,T_e,c',\eta+\xi_{T_e}^{\mathrm{S2}})\right].
\end{aligned}
\tag{A.1}
$$

Thus a remaining Stage 1 project receives a terminal second study, while a remaining Stage 2 project receives its third study.

A.1 Cutoff Strategy

As in Section 5, we omit study request fees and deposits from the equations to simplify exposition. For $t < T_e$, define the *waiting surplus* $\mathcal{U}_t^s(x,e,c,z,\eta)$ as the value of waiting in

each stage $s \in \{\text{S1}, \text{S2}\}$ (waiting with the first or second study), so that the continuation values are $W_t^s(x, e, c, z, \eta) = \max\{\mathcal{U}_t^s(x, e, c, z, \eta), 0\}$. The waiting surpluses $\mathcal{U}_t^s$ in the two stages are

$$\mathcal{U}_t^{\text{S2}}(x, e, c, z, \eta) = -\gamma^{\text{S2}} + \beta\delta^{\text{S2}} \Bigg[\sum_{c'} q^{\text{S2}}(c'|x, e, t, c, z)\, \bar{V}(x, e, t, c', \eta) + \left(1 - Q^{\text{S2}}(x, e, t, c, z)\right) W_{t+1}^{\text{S2}}(x, e, c, z, \eta) \Bigg], \tag{A.2}$$

$$\mathcal{U}_t^{\text{S1}}(x, e, c, z, \eta) = -\gamma^{\text{S1}} + \beta\delta^{\text{S1}} \Bigg[\sum_{c'} q^{\text{S1},f}(c'|x, e, t, c, z)\, E_{\xi^{\text{S2}}}\left[\bar{V}(x, e, t, c', \eta + \xi^{\text{S2}})\right] + \sum_{c', z'} q^{\text{S1}}(c', z'|x, e, t, c, z)\, E_{\xi^{\text{S2}}}\left[W_{t+1}^{\text{S2}}(x, e, c', z', \eta + \xi^{\text{S2}})\right] + \left(1 - Q^{\text{S1}}(x, e, t, c, z)\right) W_{t+1}^{\text{S1}}(x, e, c, z, \eta) \Bigg]. \tag{A.3}$$

In these equations, the q^s and Q^s objects give the probabilities of study arrival during period t. If no study arrives, the project carries its state into period $t + 1$.

We characterize the project strategies under the following assumptions:

Assumption 1 (Regularity)**.** Waiting costs satisfy $\gamma^{\text{S1}}, \gamma^{\text{S2}} > 0$. The unobserved heterogeneity ξ^{S1}, ξ^{S2}, and ξ^f have absolutely continuous distributions with full support and finite first absolute moments. The discount factor satisfies $0 < \beta < 1$, and the probabilities of surviving exogenous withdrawals satisfy $0 < \delta^s \leq 1$. For every observed waiting state and every candidate queue-status path in the bounded domain defined below, the probability of receiving a terminal study in the current period is strictly positive: $Q^{\text{S2}} > 0$ and $\sum_{c'} q^{\text{S1},f}(c'|x, e, t, c, z) > 0$.

We assume that waiting costs are positive because projects pay for the costs of site control and project management. We also assume that ξ^{S1}, ξ^{S2}, and ξ^f are normally distributed in the model in Section 5. Empirically, we find that the δ^s values are close to 1. Under these assumptions, we first show that $\mathcal{U}_t^s$ has the following properties:

Lemma 1. For fixed (x, e, t, c, z), $t < T_e$, and s, $\mathcal{U}_t^s$ is continuous and strictly increasing in η. The decision to continue after a non-terminal study has a unique cutoff.

Proof. The terminal payoff is $\bar{V}(x,e,t,c,\eta) = E_{\xi^f}[\max\{a_t(x,e) - c + \eta + \xi^f, 0\}]$, where $a_t(x,e) = \tilde{r}_t(x,e) + \chi_t(x)\theta - \omega_t(x,e)$ is the completion payoff excluding the interconnection cost and payoff shocks. The terminal payoff is finite, continuous and strictly increasing in η: full support of ξ^f gives positive probability that completion is profitable at any finite η. It tends to zero as $\eta \to -\infty$ and to infinity as $\eta \to \infty$.

The terminal-study values at T_e inherit these properties. Backward induction in (A.2)–(A.3) therefore shows that continuation values are continuous, convex and weakly increasing in η. The strictly positive probability of a terminal study in Assumption 1 makes each waiting surplus strictly increasing. Expectations preserve continuity because the $\max\{\cdot\}$ operator inside the expectation is Lipschitz in η, and the shocks have finite first absolute moments. The same induction bounds each value function by $M_0(1+|\eta|)$ for some finite constant M_0. As $\eta \to -\infty$, dominated convergence and backward induction give $W_t^s \to 0$ and $\mathcal{U}_t^s \to -\gamma^s < 0$. For this limit, values at any fixed upper value of η, together with the integrable shocks, provide a dominating bound. As $\eta \to \infty$, the terminal-study term makes $\mathcal{U}_t^s \to \infty$. Continuity and strict monotonicity therefore give a unique zero, which is the cutoff. A fixed request fee raises the cutoff and preserves these properties. □

Define the cutoffs

$$\begin{aligned} \eta^{1\star}(x,e,t,c,z) &= \inf\{\eta : \mathcal{U}_t^{S1}(x,e,c,z,\eta) \geq 0\}, \\ \eta^{2\star}(x,e,t,c,z) &= \inf\{\eta : \mathcal{U}_t^{S2}(x,e,c,z,\eta) \geq 0\}. \end{aligned} \tag{A.4}$$

These cutoffs apply immediately after a non-terminal study. A project continues at equality and withdraws below the cutoff. Between studies, it continues if $\mathcal{U}_t^s \geq 0$. At T_e, set both cutoffs to $-\infty$ and $\mathcal{U}_{T_e}^s = W_{T_e}^s$; every remaining project receives its terminal study and makes the completion decision.

The model permits withdrawal in every quarter. We give conditions under which projects nevertheless wait until their next study. We impose these conditions on the set of queue-status paths used in the existence proof.

Assumption 2. Holding (x,e,c,z) fixed in each stage:
(i) Before T_e, the study-arrival probability is a time-invariant function only of the counts of projects with weakly higher priority in PJM and in the same TO, weakly decreasing in both counts.[50]

[50]The empirical specification also includes duration effects. The no-withdrawal result allows these effects

(ii) Through T_e, the normal shock scales for ξ^f and ξ^{S2} and the net deterministic completion payoff $\tilde{r}_t + \chi_t(x)\theta - \omega_t$ weakly increase in t.[51]

(iii) The conditional cost probabilities p_s^c are time invariant. Before T_e, the joint distribution of (c', z') conditional on a non-terminal second study and the probability that an arriving second study is non-terminal are also time invariant.

Lemma 2. Under Assumptions 1 and 2, if the counts of projects with weakly higher priority in PJM and the same TO weakly decrease after entry, a project that continues after a study does not endogenously withdraw before its next study.

Proof. Write Q_t^s for $Q^s(x, e, t, c, z)$. By Assumption 2(i) and the decreasing counts, $Q_{t+1}^s \geq Q_t^s$ for $t + 1 < T_e$. We first show that the Stage-2 continuation value W_t^{S2} weakly increases in t, holding (x, e, c, z, η) fixed. Write

$$h_t = \sum_{c'} p_{\mathrm{S2}}^c(c' \mid x, e, c, z)\bar{V}(x, e, t, c', \eta).$$

The positive-part payoff is convex, so increasing the mean or the normal shock scale raises h_t. Thus Assumption 2(ii)–(iii) gives $h_{t+1} \geq h_t$, and, since $q^{\mathrm{S2}} = Q^{\mathrm{S2}} p_{\mathrm{S2}}^c$,

$$W_t^{\mathrm{S2}} = \max\{-\gamma^{\mathrm{S2}} + \beta\delta^{\mathrm{S2}}[Q_t^{\mathrm{S2}} h_t + (1 - Q_t^{\mathrm{S2}})W_{t+1}^{\mathrm{S2}}], 0\}.$$

Since $h_{T_e-1} \leq h_{T_e} = W_{T_e}^{\mathrm{S2}}$, the boundary gives

$$W_{T_e-1}^{\mathrm{S2}} \leq \max\{-\gamma^{\mathrm{S2}} + \beta\delta^{\mathrm{S2}} h_{T_e}, 0\} \leq h_{T_e} = W_{T_e}^{\mathrm{S2}}.$$

For the induction step, suppose $W_{t+2}^{\mathrm{S2}} \geq W_{t+1}^{\mathrm{S2}}$ for some $t + 2 \leq T_e$. If $h_t \leq W_{t+1}^{\mathrm{S2}}$, the bracket above is at most W_{t+1}^{S2}, so $W_t^{\mathrm{S2}} \leq W_{t+1}^{\mathrm{S2}}$. Otherwise, raising Q_t^{S2} to Q_{t+1}^{S2} raises the bracket; then raising h_t to h_{t+1} and W_{t+1}^{S2} to W_{t+2}^{S2} gives the recursion for W_{t+1}^{S2}. This proves the ordering.[52]

if, for example, for a fixed arrival time e and study state, the overall arrival probability still weakly increases over time along the equilibrium path.

[51] For existence, the decrease in installation costs plus the change in the other deterministic payoff terms must offset the maximal decline in wholesale revenues over the bounded candidate paths in Appendix A.5. This assumption holds for renewable projects but not for gas projects. However, we have numerically verified that Lemma 2, the key result using this assumption, still holds for gas projects in equilibrium, because their values of waiting increase in waiting time, driven by the increasing study arrival probabilities and the dispersions of shocks.

[52] Subtracting the request fee and taking the maximum with zero preserves it, as does omitting the fee at

For Stage 1, compare W^{S2}_{t+1} and W^{S2}_{t+2} at each updated state (c', z'). The ordering just proved, convexity in η and the increasing dispersion of ξ^{S2} imply that the expected non-terminal study value weakly increases. Part (iii) keeps fixed the distribution of (c', z') after a non-terminal second study and, because p^c_{S1} is also fixed, the distribution of c' after a terminal one. The terminal study value increases by the same convexity argument for $\bar{V}$. Since the conditional probability that an arriving second study is terminal is constant, and Q^{S1}_t increases, the two nonnegative study-arrival terms in (A.3) together weakly increase for $t < T_e - 1$. The study-arrival term in (A.2) also increases over these dates.

Suppose withdrawal is strictly optimal in an intervening period $t+1 < T_e$, so $\mathcal{U}^s_{t+1} < 0$ and $W^s_{t+1} = 0$. In (A.2) or (A.3), the no-arrival term is then zero at t and nonnegative at $t+1$, while the study-arrival terms weakly increase. Thus $\mathcal{U}^s_t \leq \mathcal{U}^s_{t+1} < 0$: waiting in t could not have been optimal, even without a request fee. At T_e, the terminal study arrives immediately. Continuing at indifference completes the proof. □

A.2 State Transitions

In the above, we have described the project strategies given their beliefs. Under Lemma 2, projects endogenously withdraw after the arrival of a new study. Between studies, withdrawals occur exogenously at a constant rate that depends on project characteristics and the study stage (Appendix D.6). This section describes a model of how PJM processes studies, which determines a project's state transition probabilities. These probabilities must be consistent with project beliefs in an equilibrium.

A.2.1 Transitions of η

We first describe how the distribution of accumulated payoff information, η, changes over time. To keep the notation tractable, we explicitly use the parametric assumptions on ξ^{S1} and ξ^{S2} in Section 5. The innovation in payoff information is independent of queue statuses and of the observed study-state transition. The only transition in accumulated payoff information occurs when $\eta = \xi^{S1}$ in Stage 1 to $\eta' = \xi^{S1} + \xi^{S2}$ upon the second study's arrival. We assume ξ^{S1} has standard deviation σ^{S1}, and ξ^{S2} has standard deviation $\sigma^{S2} \cdot (t-e)^{\kappa^{S2}}$, where $\kappa^{S2} > 0$.

After a non-terminal first study in period t, the improper density of $\eta \equiv \xi^{S1}$ among

T_e.

projects that do not withdraw is

$$g_t^{\mathrm{S1}}(\eta \mid x, e, c, z) = \frac{1}{\sigma^{\mathrm{S1}}}\,\phi\Big(\frac{\eta}{\sigma^{\mathrm{S1}}}\Big)\,\mathbf{1}\{\eta \geq \eta^{1\star}(x, e, t, c, z)\}, \tag{A.5}$$

where ϕ is the standard normal density. After a non-terminal second study in period t, the improper density of $\eta' = \eta + \xi^{\mathrm{S2}}$ among projects that continue into Stage 2 in $t+1$ and have waited $t-e$ periods is

$$g_t^{\mathrm{S2}}(\eta' \mid \eta, x, e, c', z') = \frac{1}{\sigma^{\mathrm{S2}} \cdot (t-e)^{\kappa^{\mathrm{S2}}}}\,\phi\left(\frac{\eta' - \eta}{\sigma^{\mathrm{S2}} \cdot (t-e)^{\kappa^{\mathrm{S2}}}}\right)\mathbf{1}\{\eta' \geq \eta^{2\star}(x, e, t+1, c', z')\}. \tag{A.6}$$

The normal density component in (A.6) does not depend on (c, z), (c', z'), or $\mathcal{K}_t(e, x)$; the new study state (c', z') and the queue-status path enter (A.6) only through the cutoff.

A.2.2 Transitions of c and z and Study Arrival Probabilities

We assume that the probabilities of (1) receiving a new terminal second study with cost c' in Stage 1, (2) receiving a new non-terminal second study with cost c' and updated study information z' in Stage 1, and (3) receiving the third study with cost information c' in Stage 2 are functions of $(x, e, t, c, z, \mathcal{K}_t(e, x))$, and are continuous in $\mathcal{K}_t(e, x)$. Thus Q^{S1} and Q^{S2} depend on the observed state and queue status, but not on η. In practice, we use an ordered-probit model for the transition of costs, and probit models for the transitions of other states (Appendix D).

A.3 Transitions of the Queue

We now describe how the composition of the queue changes over time given project strategies and how PJM processes studies. Let $m_t^s(x, e, c, z, \eta)$ denote the mass density of projects in stage $s \in \{\mathrm{S1}, \mathrm{S2}\}$ at the start of quarter t, after the most recent withdrawal decision, where its integral with respect to η produces the mass of projects given (x, e, c, z). Let $N_e(x)$ denote the exogenous mass of potential entrants in quarter e, and $q_0(c, z \mid x, e)$ the probability of a non-terminal first study with state (c, z), so that $\sum_{c,z} q_0 = 1 - q^f$. The first-study distribution and the probability that the study is terminal, q^f, are exogenously determined by (x, e). The entry cost is independent of the first-study outcomes and payoff shocks conditional on (x, e). The entry value in (5) follows the same convention of omitting request fees. The entry probability

is

$$p^{\text{ent}}(x,e) = \Pr(EV(x,e) > C(x,e)), \tag{A.7}$$

where C has the scaled beta distribution on $[0,\bar{C}]$ specified in Section 6.3.

Now we describe the transition of the mass of projects that arrive in e and decide to enter. The mass in Stage 1 equals the mass of potential projects multiplied by the entry probability, the probability of a non-terminal first-study state and the density of η for not withdrawing. A project arriving and choosing to enter in e joins the queue and receives its first study in $e+1$. Thus, the mass of projects that arrive in e and are in Stage 1 in period $e+1$ is

$$m^{\text{S1}}_{e+1}(x,e,c,z,\eta) = N_e(x)p^{\text{ent}}(x,e)q_0(c,z \mid x,e)g^{\text{S1}}_{e+1}(\eta \mid x,e,c,z). \tag{A.8}$$

To describe the changes of the queue mass over time, we require notation on the transition probabilities of (c,z) defined in Appendix A.2.2. The q and Q functions below are the processing probabilities evaluated at $\mathcal{K}_t(e,x)$; in equilibrium they equal the probabilities in the project's Bellman equations.

For $e+1 \leq t < T_e$, the Stage-1 mass transition is:

$$\begin{aligned} m^{\text{S1}}_{t+1}(x,e,c,z,\eta) = \delta^{\text{S1}}\left[1 - Q^{\text{S1}}(x,e,t,c,z;\mathcal{K}_t(e,x))\right] \\ \times m^{\text{S1}}_t(x,e,c,z,\eta). \end{aligned} \tag{A.9}$$

The projects leaving the queue include transitions to Stage 2 and exogenous withdrawals at a per-period probability of $1-\delta^{\text{S1}}$. For the projects receiving a terminal second study, the completed capacity in a period is the mass in Stage 1 receiving the terminal second study times the respective capacity and the probability of completion.

The Stage-2 mass consists of continuing Stage-2 projects and Stage-1 projects receiving a

non-terminal second study. We start with $m^{\mathrm{S2}}_{e+1} = 0$. Then,

$$\begin{aligned} m^{\mathrm{S2}}_{t+1}(x,e,c',z',\eta') = \delta^{\mathrm{S2}}\big[1 - Q^{\mathrm{S2}}(x,e,t,c',z';\mathcal{K}_t(e,x))\big] \\ \times m^{\mathrm{S2}}_t(x,e,c',z',\eta')\mathbf{1}\{\mathcal{U}^{\mathrm{S2}}_{t+1}(x,e,c',z',\eta') \geq 0\} \\ + \sum_{c,z} \int \delta^{\mathrm{S1}}\, q^{\mathrm{S1}}(c',z' \mid x,e,t,c,z;\mathcal{K}_t(e,x)) \\ \times g^{\mathrm{S2}}_t(\eta' \mid \eta,x,e,c',z')\, m^{\mathrm{S1}}_t(x,e,c,z,\eta)\, d\eta. \end{aligned} \tag{A.10}$$

The leaving mass includes projects receiving the third study and exogenous exits at a per-period probability of $1-\delta^{\mathrm{S2}}$. For the projects receiving the third study, the completed capacity in a period is the mass in Stage 2 receiving the third study times the respective capacity and the probability of completion. Under Lemma 2, projects that do not receive a new study continue. At T_e, the mass is measured just before the terminal study.

The masses for projects with arrival time e are zero for $t \leq e$ and $t > T_e$. The initial normal density, convolution with the second-study normal innovation, and multiplication by withdrawal indicators preserve absolute continuity of all these mass distributions.

Completed Capacity. Let $\Delta K^{0,\mathrm{comp}}_t$ and $\Delta K^{s,\mathrm{comp}}_t$ denote completions at the first study and from waiting stage s, in MW. Let $k(x)$ be project capacity and let the completion probability be $p^{\mathrm{comp}}_t(c,\eta) = \Pr(a_t(x,e) - c + \eta + \xi^f_t > 0)$, and set $p^{\mathrm{comp,S1}}_t(c,\eta) = E_{\xi^{\mathrm{S2}}_t}[p^{\mathrm{comp}}_t(c,\eta+\xi^{\mathrm{S2}}_t)]$, $p^{\mathrm{comp,S2}}_t = p^{\mathrm{comp}}_t$. The suppressed (x,e) arguments are held fixed. If $q^f_0(c,z \mid x,e)$ is the joint probability of a terminal first study and its state, with $\sum_{c,z} q^f_0 = q^f$, first-study completions are

$$\Delta K^{0,\mathrm{comp}}_{e+1}(x,e) = k(x)N_e(x)p^{\mathrm{ent}}(x,e)\sum_{c,z} q^f_0(c,z \mid x,e) E_{\xi^{\mathrm{S1}}}[p^{\mathrm{comp}}_{e+1}(c,\xi^{\mathrm{S1}})]. \tag{A.11}$$

For $e+1 \leq t < T_e$, let $q^{\mathrm{S2},f} = q^{\mathrm{S2}}$. Completions from waiting stages and from the terminal boundary are, respectively,

$$\Delta K_t^{s,\mathrm{comp}}(x,e) = k(x)\delta^s \sum_{c,z,c'} \int m_t^s(x,e,c,z,\eta)\, q^{s,f}(c' \mid x,e,t,c,z;\mathcal{K}_t(e,x)) p_t^{\mathrm{comp},s}(c',\eta)\, d\eta, \tag{A.12}$$

$$\Delta K_{T_e}^{s,\mathrm{comp}}(x,e) = k(x) \sum_{c,z,c'} \int m_{T_e}^s(x,e,c,z,\eta)\, p_s^c(c' \mid x,e,c,z) p_{T_e}^{\mathrm{comp},s}(c',\eta)\, d\eta. \tag{A.13}$$

Set these flows to zero outside their stated dates. Survival through the last waiting period is already included in $m_{T_e}^s$ and is not applied again at T_e.

A.4 Queue Status and Operating Revenues

We now construct queue status variables based on the mass density functions defined above. The queue position in period t for projects that arrived in period e and joined the queue in $e+1$ is the number of queued projects with higher-or-equal priority, i.e., those that arrive in period $\tilde{e} \leq e$:

$$\sum_s \sum_{\tilde{x}} \sum_{\tilde{e} \leq e} \sum_{c,z} \int m_t^s(\tilde{x},\tilde{e},c,z,\eta)\, d\eta. \tag{A.14}$$

Other queue statuses used in our empirical model defined in Section 5.3 are computed similarly. For example, the same-TO queue position restricts the sum above to projects in the same TO zone. Writing $\mathrm{g}(x)$ for fuel type and $K_t^{\mathrm{comp,g}}$ for capacity in service at the start of quarter t, we have

$$K_{t+4}^{\mathrm{comp,g}} - K_{t+3}^{\mathrm{comp,g}} = \sum_{x:\mathrm{g}(x)=\mathrm{g}} \sum_e \left[\Delta K_t^{0,\mathrm{comp}}(x,e) + \sum_s \Delta K_t^{s,\mathrm{comp}}(x,e) \right].$$

Quarter-t revenues use K_t^{comp}; completions during t enter K_{t+4}^{comp}, one year later.

The wholesale price in t is determined by the capacity of each fuel type in service at the start of t. This in-service capacity adds to the supply curve in Appendix C, and its crossing with the exogenous demand yields the new wholesale price. A project's revenue, $\tilde{r}$, is computed as the discounted sum of energy market revenue per megawatt over a 25-year service life. We assume that the resulting $\tilde{r}$ is continuous in the queue status. In the actual implementation in Appendix C, we construct a step function based on the fuel types. For

equilibrium existence purposes, the function can be smoothed at the break points to ensure existence. We use the step function directly in the equilibrium calculation through an iterative algorithm, and we do not find non-existence in practice.

A.5 Equilibrium Existence

A **queuing equilibrium** consists of cutoff strategies, entry probabilities, mass densities $\{m_t^s\}$, queue-status paths $\{\mathcal{K}_t(e,x)\}_t$ for every (e,x), and revenues. Decisions solve the Bellman equations with boundary (A.1); entry satisfies (A.7); masses and completion flows satisfy (A.8)–(A.13); and queue status and revenue are constructed as in Appendix A.4. The study probabilities used in decisions equal the processing probabilities evaluated at that same queue-status path. For every fixed arrival time e and observed study state, study-arrival probabilities and net deterministic completion payoff weakly increase with time, and study information satisfies Assumption 2(iii).

We prove existence under Assumptions 1 and 2. For each candidate queue-status path, we solve the project problems and use the resulting mass transitions and completion flows to construct a new path. Denote this map by Γ. We apply Brouwer's fixed point theorem.

We next construct a compact and convex set of paths for the function. Total project count is bounded by $\bar{N} = \sum_{e,x} N_e(x)$. Total capacity is bounded by $\bar{M} = \sum_{e,x} k(x) N_e(x)$. Let $K_t^{\mathrm{PJM}}(e,x)$ and $K_t^{\mathrm{TO}}(e,x)$ denote the two queue-position components of $\mathcal{K}_t(e,x)$. We construct the following set of paths $\{\mathcal{K}_t(e,x)\}_t$: for each (e,x), the count components are in $[0,\bar{N}]$, capacity components in $[0,\bar{M}]$, and

$$\begin{aligned} K_{t+1}^{\mathrm{PJM}}(e,x) &\le K_t^{\mathrm{PJM}}(e,x), \\ K_{t+1}^{\mathrm{TO}}(e,x) &\le K_t^{\mathrm{TO}}(e,x), \\ K_{t+1}^{\mathrm{comp,g}} &\ge K_t^{\mathrm{comp,g}}. \end{aligned} \tag{A.15}$$

The first two inequalities apply for $t \ge e+1$, and completed-capacity coordinates are common across projects with the same arrival time e. Since all indices have finite support, these bounds and linear restrictions define a nonempty, compact and convex set of paths.

The first two inequalities in (A.15) hold for paths generated by Γ, because all projects with priority weakly above those that arrive in e have joined by $e+1$ and can only lose mass thereafter. Moving between stages adds no projects. Completed capacity cannot decrease,

and the count and capacity cannot exceed the potential-project bounds. Thus Γ maps the set into itself. Assumption 2 and the path inequalities give the conditions of Lemma 2 throughout this set.

For continuity, nearby queue-status paths must generate nearby paths through Γ. Consider any sequence of candidate queue-status paths for which every component converges at each date to that of a limiting candidate path. By the assumed continuity of study transitions and revenues in queue status (Appendices A.2.2 and A.4), the implied study probabilities and revenues converge to those at the limiting path. Over the finite horizon, absolute payoffs are bounded by a fixed constant plus a fixed multiple of the sum of absolute payoff shocks. This bound applies to all candidate paths and has finite expectation. Dominated convergence and backward induction therefore give convergence of value functions and entry values. The continuous entry-cost distribution gives convergence of entry probabilities, and strict monotonicity in Lemma 1 gives convergence of withdrawal thresholds for $t < T_e$.

As the candidate queue-status path supplied to Γ converges, the induced decisions converge to those under the limiting path for each study history, except at payoff realizations exactly at a limiting cutoff. Such realizations have probability zero because payoff information has a continuous distribution. Decision indicators are bounded by one, and total project mass and capacity are bounded. Together with convergence of the study probabilities, dominated convergence therefore gives convergence of aggregate waiting masses and completion flows. The output of Γ thus converges to its output at the limiting input path, establishing continuity. We choose continuation at indifference, which affects zero mass.

Therefore, Γ is continuous on this nonempty, compact and convex set of paths and maps the set into itself. Brouwer's theorem gives an equilibrium with increasing study arrival probability, with Lemma 2 characterizing project strategies.

B Identification of Payoffs and Waiting Costs

Fix project characteristics x and arrival quarter e, when the potential project makes its entry decision. We show below how to identify the payoff specific to each combination of (x, e). We therefore suppress (x, e) from the arguments of all functions in this section, though e continues to appear where we write total elapsed time since arrival.

Notation. We use $T_e = e + T$ and omit study request fees and deposits from the equations to simplify exposition, as in Appendix A. Study probabilities below are densities in their continuous cost arguments. After a non-terminal first study with state (t, c, z), let $\Lambda^{\mathrm{S1}}(c', z', \tau_1 \mid t, c, z)$ denote the probability that a project that waits receives a non-terminal second study with state (c', z') after τ_1 no-arrival quarters, in period $t + \tau_1$:

$$\Lambda^{\mathrm{S1}}(c', z', \tau_1 \mid t, c, z) = \left[\prod_{h=0}^{\tau_1 - 1} \delta^{\mathrm{S1}}\big(1 - Q^{\mathrm{S1}}(t + h, c, z)\big)\right] \delta^{\mathrm{S1}} q^{\mathrm{S1}}(c', z' \mid t + \tau_1, c, z). \tag{B.1}$$

Here τ_1 is the elapsed time from this non-terminal first-study state to the second study, whether that study is terminal or non-terminal. The analogous probability that the second study is terminal with interconnection cost c' is

$$\Lambda^{\mathrm{S1},f}(c', \tau_1 \mid t, c, z) = \left[\prod_{h=0}^{\tau_1 - 1} \delta^{\mathrm{S1}}\big(1 - Q^{\mathrm{S1}}(t + h, c, z)\big)\right] \delta^{\mathrm{S1}} q^{\mathrm{S1},f}(c' \mid t + \tau_1, c, z). \tag{B.2}$$

After a non-terminal second study arrives at $\tilde{t} = t + \tau_1$, the project enters Stage 2 in $\tilde{t} + 1$ with state $(\tilde{t} + 1, c', z')$. Define the probability that a project that waits receives a terminal study with interconnection cost c'' after τ_2 additional no-arrival quarters, in period $\tilde{t} + 1 + \tau_2$:

$$\Lambda^{\mathrm{S2}}(c'', \tau_2 \mid \tilde{t}, c', z') = \left[\prod_{h=0}^{\tau_2 - 1} \delta^{\mathrm{S2}}\big(1 - Q^{\mathrm{S2}}(\tilde{t} + 1 + h, c', z')\big)\right] \delta^{\mathrm{S2}} q^{\mathrm{S2}}(c'' \mid \tilde{t} + 1 + \tau_2, c', z'). \tag{B.3}$$

The formulas above apply to study dates before T_e. At T_e, $\Lambda^{\mathrm{S1}} = 0$, while the final factor $\delta^{\mathrm{S1}} q^{\mathrm{S1},f}$ in $\Lambda^{\mathrm{S1},f}$ or $\delta^{\mathrm{S2}} q^{\mathrm{S2}}$ in Λ^{S2} is replaced by the boundary cost kernel p^c_{S1} or p^c_{S2} in (A.1); the preceding survival product is unchanged. Empty products equal one.

We use F^{S1}, F^{S2} and G for the distributions of ξ^{S1}, ξ^{S2} and ξ^f, and condition the latter two on total elapsed time since the project's arrival. Thus a second study at $\tilde{t} = t + \tau_1$ draws ξ^{S2} from $F^{\mathrm{S2}}(\cdot \mid \tilde{t} - e)$ and, if terminal, ξ^f from $G(\cdot \mid \tilde{t} - e)$; a terminal third study in $\tilde{t} + 1 + \tau_2$ draws ξ^f from $G(\cdot \mid \tilde{t} + 1 + \tau_2 - e)$. We impose the normalizations $E[\xi^{\mathrm{S1}}] = 0$, $E[\xi^{\mathrm{S2}} \mid \text{total elapsed time}] = 0$, and $E[\xi^f \mid \text{total elapsed time}] = 0$. We use a_t to denote the present value of the profit from starting to operate the plant in t less the construction cost, which corresponds to $r_{it} - \omega_{it}$ in Equation (1).

Theorem 1. Under the conditions in Appendix A that yield cutoff strategies and ensure

that a project continuing after a study does not withdraw before the next study (Lemmas 1 and 2), and the assumptions that

(1) the probabilities defined in (B.1), (B.2), and (B.3), as well as the component transition objects Q^s, δ^s, and the discount factor β, are known and positive on their respective supports;

(2) F^{S1} and the elapsed-time-indexed families F^{S2} and G are absolutely continuous with positive densities, have full support on $\mathbb{R}$, and have finite first absolute moments;

(3) conditional on the lagged observed state and total elapsed time since the project's arrival, the new study cost and information are independent of the shocks ξ^{S1}, ξ^{S2}, and ξ^f. The shocks are mutually independent, and each new study's cost has full support on $\mathbb{R}$;[53]

(4) the current study cost c affects the waiting value only through the study-arrival probability and the distribution of the next study's cost, and the study-arrival probability is continuous in c. For the cost c' of each remaining study, whether the next one or a later one, the distribution of c' conditional on c, other observed states, and the study date is continuous in c, and so is its first absolute moment;

(5) for each such c' and every fixed k, $\Pr(c' \leq k \mid c, \cdot) \to 0$ as $c \to +\infty$ and $\Pr(c' \leq k \mid c, \cdot) \to 1$ as $c \to -\infty$, together with $E[\max\{-c', 0\} \mid c, \cdot] \to 0$ as $c \to +\infty$; and

(6) a non-terminal second study arrives with positive probability in each period before T_e,

the distributions of unobserved heterogeneity F^{S1}, F^{S2}, and G, the payoffs a_t, and the waiting costs γ^{S1} and γ^{S2} are identified. The payoff parameters θ are then identified whenever the matrix collecting $\chi_t(x)$ over dates and project characteristics has full column rank.

Proof. All probabilities below condition on the lagged observed state and total elapsed time since arrival. By Assumption (3), varying a new study's cost changes the continuation cutoff but not the distribution of the payoff shock at that decision node. Because a project that continues after a study does not withdraw before the next one, the Λ's give the entire probability of surviving the wait and receiving the next study, and they do not depend on the payoff shocks. Since the Λ's are known and positive by Assumption (1), we work with probabilities normalized by the relevant Λ; the corresponding unnormalized probabilities are observed in the data.

[53] This argument uses continuous study-cost variation with full support on $\mathbb{R}$. In the empirical model, the costs are nonnegative and discretized, as described in Appendix D.

Cutoff Regularity. Fix unobserved heterogeneity η. The terminal payoff $\bar{V}$ in (2) is bounded by a constant plus the absolute value of the terminal-study cost, and the Stage-2 continuation value is bounded by a constant plus the expected absolute value of the third-study cost given the second-study cost. All continuation values are therefore finite.[54] By Assumption (4), the conditional distribution of the next study's cost and the expectations of these bounds are continuous in the current cost, and therefore so are the expected payoffs after the next study. Backward induction then makes each waiting surplus $\mathcal{U}_t^s$, $t < T_e$, continuous in the current cost. The tail conditions in Assumption (5) make $\mathcal{U}_t^s$ tend to $-\gamma^s < 0$ as cost tends to $+\infty$. They also make $\mathcal{U}_t^s$ tend to infinity as cost tends to $-\infty$, because a project that waits receives a terminal study by T_e unless it exits exogenously. Together with strict monotonicity in η (Lemma 1), this makes each pre-deadline cutoff continuous in cost, with range $\mathbb{R}$. Because cost also affects the study-arrival probability, the cutoffs need not be monotone in cost. Steps 1–3 use only their continuity and range.

Step 1: Terminal Shock. Consider projects that have received a non-terminal second study in $\tilde{t}$, with $\tilde{t}+1 < T_e$ (an event with positive probability by Assumption (6)), and enter Stage 2 with state $(\tilde{t}+1, c', z')$. Let $\eta = \xi^{\mathrm{S1}} + \xi^{\mathrm{S2}}$, and let H^{S2} denote the distribution of η at this decision node before the withdrawal decision, which by Assumption (3) does not vary with (c', z'). By Lemma 1, the Stage-2 decision is characterized by a cutoff $\eta^{2\star}(\tilde{t}+1, c', z')$, where projects continue if $\eta \geq \eta^{2\star}(\tilde{t}+1, c', z')$.

The observed withdrawal probability upon entering Stage 2 after this non-terminal second study is

$$D^{\mathrm{S2}}(\tilde{t}+1, c', z') = H^{\mathrm{S2}}\big(\eta^{2\star}(\tilde{t}+1, c', z')\big)\,.$$

Next consider the observed joint probability, from the same non-terminal second-study state, of continuing into Stage 2, receiving a terminal third study in period $\tilde{t}+1+\tau_2$ with cost c'', and withdrawing at the terminal study. This probability is

$$\Omega^{\mathrm{S2}}(\tilde{t}, c', z', c'', \tau_2) = \Lambda^{\mathrm{S2}}(c'', \tau_2 \mid \tilde{t}, c', z') M^{\mathrm{S2}}(\tilde{t}, c', z', c'', \tau_2).$$

Since Ω^{S2} is observed and Λ^{S2} is known and positive, $M^{\mathrm{S2}} = \Omega^{\mathrm{S2}}/\Lambda^{\mathrm{S2}}$ is known from data. It

[54] The Stage-1 continuation value is bounded in the same way. The constants are finite because the shocks have finite first absolute moments by Assumption (2), and the expected absolute values of the study costs are finite by Assumption (4).

equals

$$M^{\mathrm{S2}}(\tilde{t}, c', z', c'', \tau_2) = \int_{\eta^{2\star}(\tilde{t}+1,c',z')}^{\infty} G\big(c'' - a_{\tilde{t}+1+\tau_2} - \tilde{\eta} \mid \tilde{t}+1+\tau_2-e\big)\, dH^{\mathrm{S2}}(\tilde{\eta}).$$

As c' varies, cutoff regularity and the positive density of H^{S2}, inherited from F^{S2} under Assumption (2), make D^{S2} continuous, with range $(0,1)$. Holding the remaining arguments fixed, c' enters M^{S2} only through this cutoff, so we may index M^{S2} by the observed probability D^{S2}. Differentiating with respect to this probability gives

$$-\frac{dM^{\mathrm{S2}}(\tilde{t}, c', z', c'', \tau_2)}{dD^{\mathrm{S2}}(\tilde{t}+1, c', z')} = G\big(c'' - a_{\tilde{t}+1+\tau_2} - \eta^{2\star}(\tilde{t}+1, c', z') \mid \tilde{t}+1+\tau_2-e\big). \tag{B.4}$$

The left-hand side is known from data. As c'' varies over its support, the right-hand side traces $G(\cdot \mid \tilde{t}+1+\tau_2-e)$ with a horizontal shift. The mean-zero normalization of G identifies both this distribution and the shift

$$b^{\mathrm{S2}}_{\tilde{t}+1+\tau_2}(\tilde{t}+1, c', z') \equiv a_{\tilde{t}+1+\tau_2} + \eta^{2\star}(\tilde{t}+1, c', z').$$

Repeating this argument across third-study arrival dates, including the terminal date T_e, identifies G.

Step 2: Second-Study Payoff Information. Now consider projects that have reached a non-terminal first study with state (t, c, z). Let $\eta^{1\star}(t, c, z)$ denote the Stage-1 cutoff. Entry precedes the realization of ξ^{S1} in (5), and ξ^{S1} is independent of the first-study state by Assumption (3), so its distribution at this node is F^{S1} and the immediate withdrawal probability is

$$D^{\mathrm{S1}}(t, c, z) = F^{\mathrm{S1}}\big(\eta^{1\star}(t, c, z)\big).$$

Suppose a project that continues receives a non-terminal second study at $\tilde{t} = t + \tau_1$, with $\tilde{t}+1 < T_e$, and enters Stage 2 in the following period. The normalized joint probability of continuing after the first study and withdrawing upon entering Stage 2 is

$$M^{\mathrm{S1}}(t, c, z, c', z', \tau_1) = \int_{\eta^{1\star}(t,c,z)}^{\infty} F^{\mathrm{S2}}\Big(\eta^{2\star}(\tilde{t}+1, c', z') - \tilde{\xi}^{\mathrm{S1}} \mid \tilde{t}-e\Big)\, dF^{\mathrm{S1}}(\tilde{\xi}^{\mathrm{S1}}).$$

As in Step 1, cutoff regularity lets us index M^{S1} by the observed probability D^{S1}. Differentiating with respect to this probability gives

$$-\frac{dM^{\mathrm{S1}}(t,c,z,c',z',\tau_1)}{dD^{\mathrm{S1}}(t,c,z)} = F^{\mathrm{S2}}\big(\eta^{2\star}(\tilde{t}+1,c',z') - \eta^{1\star}(t,c,z) \mid \tilde{t}-e\big). \tag{B.5}$$

For any fixed τ_2 reached with positive probability, rewrite the argument on the right-hand side as

$$\eta^{2\star}(\tilde{t}+1,c',z') - \eta^{1\star}(t,c,z) = b^{\mathrm{S2}}_{\tilde{t}+1+\tau_2}(\tilde{t}+1,c',z') - b^{\mathrm{S1}}_{\tilde{t}+1+\tau_2}(t,c,z),$$

where

$$b^{\mathrm{S1}}_{\tilde{t}+1+\tau_2}(t,c,z) \equiv a_{\tilde{t}+1+\tau_2} + \eta^{1\star}(t,c,z).$$

Since $b^{\mathrm{S2}}_{\tilde{t}+1+\tau_2}$ is identified from Step 1 and ranges over $\mathbb{R}$ through variation in c', (B.5) identifies $F^{\mathrm{S2}}(\cdot \mid \tilde{t}-e)$ and the shift $b^{\mathrm{S1}}_{\tilde{t}+1+\tau_2}(t,c,z)$, using the mean-zero normalization of $F^{\mathrm{S2}}(\cdot \mid \tilde{t}-e)$.

Step 3: Payoff Levels and Cutoffs. Using the identified object $b^{\mathrm{S1}}_{\tilde{t}+1+\tau_2}$, the first-study withdrawal probability can be written as

$$D^{\mathrm{S1}}(t,c,z) = F^{\mathrm{S1}}\Big(b^{\mathrm{S1}}_{\tilde{t}+1+\tau_2}(t,c,z) - a_{\tilde{t}+1+\tau_2}\Big). \tag{B.6}$$

Variation in the first-study cost makes $b^{\mathrm{S1}}_{\tilde{t}+1+\tau_2}$ range over $\mathbb{R}$ and thus traces out the CDF F^{S1} with a horizontal shift. Its mean-zero normalization identifies $a_{\tilde{t}+1+\tau_2}$ and F^{S1}. Repeating the argument over third-study dates identifies a_t. The identified a_t then separately identifies the cutoffs $\eta^{1\star}(t,c,z)$ and $\eta^{2\star}(\tilde{t}+1,c',z')$ from the identified shifts b^{S1} and b^{S2}. For terminal second studies at $t+\tau_1 \le T_e$, let $M^{\mathrm{S1},f}$ be the joint probability of continuing after the first study and withdrawing at that terminal study, divided by $\Lambda^{\mathrm{S1},f}$. The same differentiation gives

$$-\frac{dM^{\mathrm{S1},f}}{dD^{\mathrm{S1}}} = \Pr\Big(a_{t+\tau_1} + \eta^{1\star}(t,c,z) + \xi^{\mathrm{S2}} + \xi^{f} \le c'\Big).$$

Varying c' identifies this distribution. Its mean identifies $a_{t+\tau_1} + \eta^{1\star}(t,c,z)$, and hence $a_{t+\tau_1}$, including at T_e. It also identifies the expected terminal payoff at the Stage-1 cutoff. For a non-terminal second study in T_e-1 followed by a third study in T_e, the same argument, using variation in c'', identifies the expected Stage-2 boundary payoff at the Stage-1 cutoff.

Finally, Equation (1) implies that $a_t - \tilde{r}_t + \omega_t = \chi_t(x)\theta$. Since the left-hand side is identified

for each (x,t), θ is identified when the resulting matrix of $\chi_t(x)$ has full column rank.

Step 4: Waiting Costs. At a Stage-2 cutoff in $\tilde{t}+1 < T_e$, the project is indifferent between continuing and withdrawing. By Lemma 2, a project that continues does not withdraw before the next study. Waiting costs are paid only before T_e. Payoffs at T_e are discounted through the preceding waiting periods, with no additional discount at the boundary. Let

$$S^{\mathrm{S2}}(\tau_2 \mid \tilde{t}+1, c', z') = \prod_{h=0}^{\tau_2-1} \delta^{\mathrm{S2}}\big(1 - Q^{\mathrm{S2}}(\tilde{t}+1+h, c', z')\big)$$

denote the probability of still waiting through τ_2 Stage-2 periods before the next terminal-study arrival. The indifference condition at the cutoff equates the expected discounted terminal payoff from continuing, A^{S2}, to the expected discounted waiting costs, $\gamma^{\mathrm{S2}} B^{\mathrm{S2}}$:

$$0 = A^{\mathrm{S2}}(\tilde{t}+1, c', z') - \gamma^{\mathrm{S2}} B^{\mathrm{S2}}(\tilde{t}+1, c', z'),$$

where

$$A^{\mathrm{S2}}(\tilde{t}+1, c', z') = \sum_{\tau_2=0}^{T_e-\tilde{t}-1} \int \beta^{\min\{\tau_2+1, T_e-\tilde{t}-1\}} \Lambda^{\mathrm{S2}}(c'', \tau_2 \mid \tilde{t}, c', z') \times \bar{V}(\tilde{t}+1+\tau_2, c'', \eta^{2\star}(\tilde{t}+1, c', z'))\, dc''$$

and

$$B^{\mathrm{S2}}(\tilde{t}+1, c', z') = \sum_{\tau_2=0}^{T_e-\tilde{t}-2} \beta^{\tau_2} S^{\mathrm{S2}}(\tau_2 \mid \tilde{t}+1, c', z').$$

In the above, $\bar{V}$ is the expected value of receiving the terminal study defined in (2); it is identified from Steps 1–3. Both A^{S2} and B^{S2} are identified, and $B^{\mathrm{S2}} > 0$, so $\gamma^{\mathrm{S2}} = A^{\mathrm{S2}}/B^{\mathrm{S2}}$ is identified.

Similarly, define

$$S^{\mathrm{S1}}(\tau_1 \mid t, c, z) = \prod_{h=0}^{\tau_1-1} \delta^{\mathrm{S1}}\big(1 - Q^{\mathrm{S1}}(t+h, c, z)\big), \qquad B^{\mathrm{S1}}(t, c, z) = \sum_{\tau_1=0}^{T_e-t-1} \beta^{\tau_1} S^{\mathrm{S1}}(\tau_1 \mid t, c, z).$$

The indifference condition at the Stage-1 cutoff $\eta^{1\star}(t,c,z)$, $t < T_e$, implies

$$0 = A^{\mathrm{S1}}(t,c,z) - \gamma^{\mathrm{S1}} B^{\mathrm{S1}}(t,c,z),$$

where

$$\begin{aligned} A^{\mathrm{S1}}(t,c,z) = \sum_{\tau_1=0}^{T_e-t} \int \beta^{\min\{\tau_1+1,T_e-t\}} \Lambda^{\mathrm{S1},f}(c',\tau_1 \mid t,c,z) \\ \times \int \bar{V}(t+\tau_1,c',\eta^{1\star}(t,c,z)+u)\, dF^{\mathrm{S2}}(u \mid t+\tau_1-e)\, dc' \\ + \sum_{z'} \sum_{\tau_1=0}^{T_e-t-1} \int \beta^{\tau_1+1} \Lambda^{\mathrm{S1}}(c',z',\tau_1 \mid t,c,z) \\ \times \int W^{\mathrm{S2}}_{t+\tau_1+1}(c',z',\eta^{1\star}(t,c,z)+u)\, dF^{\mathrm{S2}}(u \mid t+\tau_1-e)\, dc'. \end{aligned}$$

The expected terminal payoffs in the first term are identified in Step 3. In the second term, W^{S2} follows from the identified Stage-2 problem, and Step 2 identifies the distribution of ξ^{S2} whenever $t+\tau_1+1 < T_e$. At $t+\tau_1+1 = T_e$, Step 3 identifies the integrated boundary payoff directly. Hence A^{S1} and B^{S1} are identified, and $B^{\mathrm{S1}} > 0$, giving $\gamma^{\mathrm{S1}} = A^{\mathrm{S1}}/B^{\mathrm{S1}}$. With request fees and deposits, the amounts are known and enter the indifference conditions as constants, so γ^{S1} and γ^{S2} remain identified. □

C Construction of Revenue from Power Production

We construct the present value of each project's revenue from power production as the discounted sum of the project's hourly revenues, i.e., the wholesale price times generation.[55] Prices come from crossing a merit-order supply curve with PJM demand, and generation comes from project capacity and fuel-specific hourly capacity factors. We build the supply curve from data on PJM generating units for 2011 through 2021. Beyond this horizon, we repeat the 2021 prices and capacity factors.

In counterfactual simulations, we recompute wholesale prices because the mix of in-service capacity changes. We build each counterfactual supply curve from the existing generators,

[55] Revenue from power production excludes capacity market payments and tax credits. The fuel-type coefficients in the project payoff functions absorb capacity market payments in estimation, and Section 8.1 bounds the decline in capacity market prices that counterfactual capacity additions would induce. The production and investment tax credits enter through the construction cost ω_{it} (Section 6.2).

with their observed entries and exits held fixed, and from the projects that complete the queuing process. Because demand is perfectly inelastic, evaluating the supply curve at the quantity demanded yields the equilibrium price; when a policy changes the timing or amount of completions, the curve shifts and the price changes.

C.1 Supply Curve

We construct hourly supply curves for the PJM Interconnection from 2011 through 2021. The construction follows Cicala (2022) and Hausman (2025), with modifications for PJM-specific data availability. Throughout, a "unit" refers to an individual generating unit represented separately in the supply curve and generally corresponds to a generator reported on EIA forms. A "plant" refers to a physical site that may contain multiple generating units.

The supply curve is a merit-order representation of competitive offers: we assume each unit offers at marginal cost, rank all units in operation in a given month by marginal cost, and construct the hourly supply curve from their available capacity. Building the curve for each hour requires two unit-level inputs, a marginal cost and an available capacity; the next two subsections describe how we construct each, and the subsequent subsections cover demand, market clearing, and the use of the curve in counterfactuals.

The primary data sources are EIA Forms 860 and 923, which together provide generator-level capacity and sub-fuel,[56] net generation, heat input, and fuel prices; EIA operating-expense data on fuel-specific variable operation and maintenance (O&M) costs; EIA Henry Hub and WTI spot prices; EPA Continuous Emissions Monitoring System (CEMS) data and Power Sector Programs Progress Reports on hourly unit-level emissions and heat input; PJM Data Miner data on hourly load and generation; NRC Daily Power Status reports on daily nuclear reactor operating status; S&P Global emissions allowance prices; NREL solar radiation data; PRISM county-level weather records; and U.S. Census geocode files.

We combine these sources by harmonizing generator, plant, and boiler identifiers across EIA and EPA records, including mapping fossil-plant boilers to generators using EIA linkages and maintaining these mappings over time. We also map plants to counties and time zones and convert hourly observations to a common time standard before merging them with PJM operations data.

[56] Sub-fuels are EIA's detailed categories within each fossil fuel, such as bituminous, subbituminous, lignite, waste, and refined coal, or distillate and residual fuel oil.

C.1.1 Unit-level Marginal Cost

We estimate an hourly marginal cost for each fossil unit. Let j index dispatchable generating units and h index hours. Throughout this appendix, t indexes quarters. This marginal cost is the sum of fuel costs, emissions allowance costs, and variable O&M costs:

$$\mathrm{MC}_{j\mathrm{h}} = \mathrm{FuelCost}_{j\mathrm{h}} + \mathrm{EmissionsCost}_{j\mathrm{h}} + \mathrm{VOM}_{j\mathrm{h}}, \tag{C.1}$$

where $\mathrm{FuelCost}_{j\mathrm{h}}$ is the fuel cost of producing one MWh, $\mathrm{EmissionsCost}_{j\mathrm{h}}$ is the emissions allowance cost per MWh, and $\mathrm{VOM}_{j\mathrm{h}}$ is the variable O&M cost per MWh.

Each unit's fuel cost is determined by its heat rate, sub-fuel mix, and corresponding sub-fuel prices:

$$\mathrm{FuelCost}_{j\mathrm{h}} = \mathrm{HR}_{j\mathrm{h}} \sum_{k} \mathrm{Share}^{k}_{j\mathrm{h}}\, \mathrm{FuelPrice}^{k}_{j\mathrm{h}}. \tag{C.2}$$

Here k indexes sub-fuel categories, $\mathrm{Share}^{k}_{j\mathrm{h}}$ is the heat-input share of sub-fuel k, $\mathrm{FuelPrice}^{k}_{j\mathrm{h}}$ is the corresponding sub-fuel price, and $\mathrm{HR}_{j\mathrm{h}}$ is the unit's heat rate. For natural gas and oil units, we use daily Henry Hub and WTI spot prices, adjusted by plant-year markups estimated from EIA fuel receipt data, to capture daily sub-fuel price variation not observed in the monthly EIA series. Missing plant-level fuel prices are imputed using state-, division-, and national-level averages.

The heat rate $\mathrm{HR}_{j\mathrm{h}}$ measures the amount of fuel energy required to produce one MWh of electricity. Multiplying the heat rate by the unit's weighted-average sub-fuel price gives its fuel cost per MWh. For units monitored by the CEMS, we measure the monthly heat rate as reported heat input divided by net generation and treat values outside a physically plausible range (6 to 100 MMBtu/MWh) as outliers. Because a unit must carry a marginal cost even in months when it does not run, we replace missing and outlier monthly values with predictions from unit-level regressions of the heat rate on operating characteristics such as age and ramping, falling back to vintage- and size-specific averages where a unit has too little data of its own. Heat rates for units outside the CEMS are built from EIA-923 fuel consumption and generation by the same logic.

Emissions costs equal the unit's emissions rate per MWh multiplied by the relevant allowance price. Units are assigned to SO_2 and NO_x programs based on location and program rules. For seasonal NO_x markets, emissions costs apply only during the ozone-season window

for the unit's county.

Variable O&M costs capture non-fuel and non-emissions expenses that vary with generation, including routine maintenance and other production-related materials and expenses. We assign each unit the variable O&M cost for its fuel type and year, in dollars per MWh.

As a final step, we winsorize fossil units' marginal costs at the 1st and 99th percentiles within each year to reduce the influence of extreme outliers.

For non-fossil units, we assign a constant marginal cost for each fuel type. Renewable units, including wind, solar, and hydro, are assigned zero marginal cost. Nuclear units are assigned a marginal cost of \$24.90/MWh, based on the EIA average operating expense for nuclear generation (U.S. Energy Information Administration, 2022).

C.1.2 Available Capacity for Generation

We next estimate the capacity available for generation from each unit in each hour. For fossil units, available capacity is the unit's nameplate capacity, capped at its maximum hourly net generation on that day. The cap prevents the merit order from dispatching capacity beyond what the unit produced that day. Constructing this cap requires hourly net generation. For units monitored by the CEMS, we calculate hourly net generation as reported gross load multiplied by the unit's gross-to-net conversion factor, defined as the monthly ratio of net generation to reported gross load. When this ratio is missing or implausible, we replace it with the unit's average ratio over non-outlier months in the same time period. For units without CEMS coverage, we allocate monthly net generation from EIA-923 across hours of the month in proportion to PJM-wide load.

For non-fossil units, we set available capacity equal to actual generation; we use hourly generation data where possible and otherwise infer hourly generation from daily or monthly data. For wind, we use PJM's reported hourly generation. For solar, we use PJM's reported hourly generation from 2019 onward. For earlier years, when PJM reports no hourly solar generation, we infer hourly PJM-wide generation using plant-level monthly generation and irradiance data. For each plant, we allocate monthly generation across hours in proportion to irradiance at its location, then sum across plants to obtain PJM-wide hourly generation. For nuclear, we distribute each reactor's EIA-923 monthly net generation across days in proportion to its daily power level from NRC status reports and hold generation constant within each

day. For reactor-months with missing power data, we impute generation from a regression on reactor fixed effects, reactor-specific seasonal cycles and trends, and county temperature from PRISM. Hydroelectric, battery, biomass, and other minor fuel types are represented by a synthetic zero-cost unit with capacity equal to PJM's reported hourly generation from these sources.

C.2 Demand

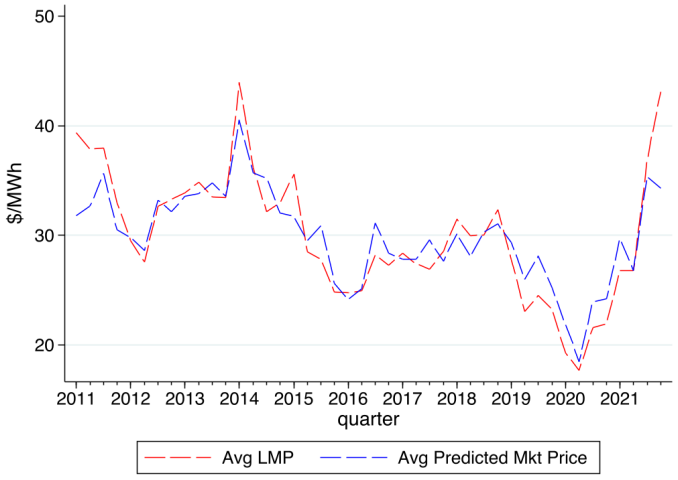
Hourly electricity demand is measured using PJM-wide metered load from 2011 through 2021. In the market-clearing exercise, demand is treated as perfectly inelastic within each hour.

C.3 Market-Clearing Price and Validation

For each hour, we sort available dispatchable units in ascending order of marginal cost and accumulate available capacity along the merit-order curve. The predicted market-clearing price, $\widehat{P}_{\mathrm{h}}$, is the marginal cost of the first unit for which cumulative available capacity equals or exceeds realized PJM-wide metered load.

Figure C.1 compares the quarterly average of $\widehat{P}_{\mathrm{h}}$ with the quarterly average of PJM's real-time hourly LMP from 2011 through 2021.[57] The predicted series broadly tracks the actual LMP series.

Figure C.1: Predicted and Actual PJM Real-Time LMPs



[57] We use the LMP at the PJM-RTO aggregate pricing node, which PJM computes as the load-weighted average of nodal LMPs in its footprint.

C.4 Using the Supply Curve in the Structural Model

The structural model uses the constructed supply curve to predict how market prices respond to counterfactual capacity additions. To make these calculations computationally feasible, we replace each quarter's hours with weighted representative hours, following the clustering approach of Reguant (2019). Within each quarter, we use k-means clustering to divide hours into 20 net-demand clusters and select the median-net-demand hour from each cluster as its representative.[58] Each representative hour h receives a weight w_h, equal to the share of the quarter's actual hours assigned to its cluster, so that the 20 weighted representative hours approximate the quarter's full net-demand distribution. We write $\mathcal{C}_t$ for the set of representative hours in quarter t. Repeating this procedure for every quarter from 2011 through 2021 produces 880 representative hours. Hereafter, h indexes these representative hours, each of which identifies a unique quarter and clock hour; quarter-specific quantities therefore inherit their quarter from h and carry no separate t subscript.

The structural model determines the timing and amount of capacity completing the queuing process at the quarterly level. After a one-year construction lag measured from the completion of the queuing process, this capacity enters service and is added to the representative-hour supply curves, with renewable and non-renewable additions treated differently.

Wind and solar additions are assigned zero marginal cost, but their output depends on resource availability. Let g index fuel types. A wind or solar addition has available capacity equal to its nameplate capacity times $CF_{h,g} \in [0, 1]$, the capacity factor of fuel type g in representative hour h. We estimate $CF_{h,g}$ from PJM data. For each quarter and each hour of the day, we average PJM-wide generation from fuel g across all days in the quarter at that clock hour and divide by the installed nameplate capacity of g in that year. As a result, wind and solar additions shift the supply curve, and depress the clearing price, by more in representative hours when expected output from that fuel type is high.

Non-renewable additions are treated as new natural-gas capacity. To allow new gas units to differ in efficiency, we divide the heat-rate distribution of natural-gas generators operating in PJM since 2011 into ten equal-width bins and weight each bin by its share of post-2011 new gas nameplate capacity. A counterfactual gas addition is then allocated across these bins according to those weights. This creates ten distinct additions to the supply curve, each with

[58] Net demand is defined as metered load minus wind and solar generation. Because wind and solar have zero marginal cost, the clearing price is where net demand meets the supply curve of the remaining units.

its own capacity and marginal cost. The marginal cost of each bin is computed using its mean heat rate, the prevailing annual natural-gas price, emissions cost, and variable O&M cost, following the same marginal-cost definition used for existing units. New gas capacity enters the supply curve at full nameplate capacity, while its dispatch is determined by the merit order. Its effective capacity factor is therefore endogenous.

The capacity in this constructed supply curve may differ from observed PJM capacity because of retirements and other unmodeled capacity changes. To account for these differences, we compute a quarterly fuel-specific residual capacity wedge equal to the observed change in PJM nameplate capacity minus model-predicted completions. Adding this wedge to the merit order reconciles the change in model-implied capacity with the observed change in PJM capacity from one quarter to the next. In counterfactual exercises, we hold the wedge fixed, so the changes in predicted prices are driven only by changes in model-predicted capacity additions.

For each counterfactual capacity vector, we insert the resulting capacity additions and residual capacity wedge into the existing merit-order supply curve, re-sort units by marginal cost, and determine the clearing price $\widehat{P}_{\mathrm{h}}$ in each representative hour by crossing the resulting supply curve with PJM demand. Each quarter therefore yields 20 representative-hour clearing prices.

We also compute a single PJM-wide quarterly price, $\widehat{P}_t = \sum_{\mathrm{h}\in\mathcal{C}_t} \mathrm{w}_{\mathrm{h}}\widehat{P}_{\mathrm{h}}$, which weights representative hours by their frequency. This price does not enter project revenue. Instead, it serves as the convergence criterion when we solve for the queuing equilibrium: we iterate until the PJM-wide price path implied by projects' queue entry decisions reproduces the price path against which those decisions were made.

C.5 From Hourly Prices to Project-Level Revenue

We construct $\tilde{r}_{it}$, the present value of revenue per megawatt from power production, in two steps. First, for each fuel type and quarter, we compute revenue per megawatt of nameplate capacity from the representative-hour clearing prices and the fuel type's capacity factors. Second, we sum this quarterly revenue over a 25-year operating horizon beginning in the project's in-service quarter, discounting at an annual rate of 5%. Projects therefore differ in $\tilde{r}_{it}$ only through their fuel type and in-service quarter.

Consider first a single quarter, and let H_t denote the number of hours in quarter t. The revenue of a project of fuel g per megawatt of capacity in quarter t is the sum, over the quarter's hours, of its generation in each hour multiplied by that hour's clearing price. In terms of the representative hours, this revenue is

$$\mathrm{Rev}_{\mathrm{g},t} = \mathrm{H}_t \sum_{\mathrm{h} \in \mathcal{C}_t} \mathrm{w}_{\mathrm{h}} \, \mathrm{CF}_{\mathrm{h,g}} \, \widehat{P}_{\mathrm{h}}. \tag{C.3}$$

In the above, the term $\mathrm{H}_t \mathrm{w}_{\mathrm{h}}$ is the number of type h hours in the quarter, and the weight is common to all fuels. The capacity factor $\mathrm{CF}_{\mathrm{h,g}}$ is how much energy a project of fuel g produces in this hour. For wind and solar, $\mathrm{CF}_{\mathrm{h,g}}$ is the data-implied capacity factor; for new natural gas, it is the share of the hour's gas capacity that the merit order dispatches, which market clearing determines endogenously. Because wind, solar, and natural gas have different hourly capacity factors, they earn different revenue per megawatt in an hour.

A project of fuel g that goes into service in quarter t_0 earns this quarterly revenue over a 25-year life. Its discounted revenue per megawatt is

$$\sum_{n=0}^{99} \frac{\mathrm{Rev}_{\mathrm{g},t_0+n}}{(1.05)^{n/4}}, \tag{C.4}$$

rescaled to millions of dollars per megawatt. For quarters up to 2021, the first year of operation for projects completing in 2020, $\mathrm{Rev}_{\mathrm{g},t}$ is based on that quarter's representative-hour prices and capacity factors in the data. For quarters after 2021, we carry forward the 2021 quarterly revenue for the corresponding calendar quarter, thereby preserving seasonal variation in prices and capacity factors.

D State Transitions, First-Study State Distribution, and Potential Entrants

D.1 Details on Project States in the Empirical Model

We first describe a project's post-entry state transitions. Interconnection costs are measured in millions of dollars per megawatt. To model cost transitions and simplify state spaces,

we discretize the costs into four bins: $[0, 0.01], (0.01, 0.05], (0.05, 0.2],$ and $(0.2, \infty)$.[59] Let $\ell \in \{1, 2, 3, 4\}$ denote the cost bin containing the most recent study's cost. To calculate the expected value of receiving the terminal study with a cost in bin ℓ, we assume that its actual cost is drawn from the empirical distribution of terminal-study costs in that bin.[60] We can express the states as (s, τ, ℓ, z), where $s \in \{1, 2\}$ denotes which study has been received, $\tau = t - e$ denotes the waiting time since the potential project's arrival and entry decision, ℓ denotes the cost bin, and z indicates whether a set of engineering tests has been performed. Because a project arriving in e joins the queue in $e + 1$, $\tau = 1$ in its first queue period.

The state transitions are built from four sets of probabilities that are functions of a project's current state and characteristics. We use p^A for the probability that a study arrives during the current period t, which depends on the queue status of a project, $\mathcal{K}_t(e, x)$. Conditional on study arrival and the prior state, we use p^c, p^{test}, and p^f for the probabilities of a new cost bin, of the engineering tests being performed conditional on $z = 0$, and of the second study being terminal. The state z is absorbing: once it becomes 1, it remains 1 in subsequent studies.

In the estimated specification, only the study-arrival probability depends on waiting time and queue status. The ordered-probit index for the cost transition has no duration or queue-status regressors, and the terminal-second-study probability and the engineering-test probability depend only on fixed project characteristics. We therefore suppress time and queue status in p_i^c, p_i^f, and the engineering-test probability p_i^{test} below.

D.2 Transition Equations

We now express the transition probabilities in Section 5.1 using the primitive probabilities above.

For a project in Stage 1, the probability that the second study arrives and is terminal

[59] Differences in mean costs across bins account for 37.5% of the total variance of interconnection costs across projects and studies; the remaining 62.5% is within bins, mostly in the open top bin. Few completing projects (6%) have costs above \$0.2 million per megawatt. If we instead restrict the sample to the 88% of the study-project observations with interconnection costs at or below \$0.2 million per megawatt, the cross-bin share rises to 77.4%. The transition, payoff, and waiting cost parameter estimates are robust to using five bins: $[0, 0.01]$, $(0.01, 0.05]$, $(0.05, 0.1]$, $(0.1, 0.2]$, and $(0.2, \infty)$.

[60] In estimation, we use the observed cost when forming the likelihood of completion when a project receives a terminal study.

with new cost bin ℓ' is

$$q_i^{\mathrm{S1},f}(\ell' \mid \ell, t, z; \mathcal{K}_t(e,x)) = p_i^A(s=1,\tau,\ell,z;\mathcal{K}_t(e,x))\, p_i^c(\ell' \mid s=1,\ell,z)\, p_i^f. \tag{D.1}$$

The probability that a non-terminal second study arrives with updated state (ℓ', z') is

$$q_i^{\mathrm{S1}}(\ell', z' \mid \ell, t, z; \mathcal{K}_t(e,x)) = p_i^A(s=1,\tau,\ell,z;\mathcal{K}_t(e,x))\, p_i^c(\ell' \mid s=1,\ell,z)\left(1-p_i^f\right) h(z' \mid z), \tag{D.2}$$

where (suppressing fixed project characteristics)

$$h(1 \mid 1) = 1, \quad h(1 \mid 0) = p^{\mathrm{test}}, \quad h(0 \mid 1) = 0, \quad h(0 \mid 0) = 1 - p^{\mathrm{test}}. \tag{D.3}$$

Therefore, the total probability of study arrival in Stage 1 is

$$\begin{aligned} &Q_i^{\mathrm{S1}}(\ell, t, z; \mathcal{K}_t(e,x)) \\ &\quad = \sum_{\ell'} q_i^{\mathrm{S1},f}(\ell' \mid \ell, t, z; \mathcal{K}_t(e,x)) \\ &\qquad + \sum_{\ell', z'} q_i^{\mathrm{S1}}(\ell', z' \mid \ell, t, z; \mathcal{K}_t(e,x)) \\ &\quad = p_i^A(s=1,\tau,\ell,z;\mathcal{K}_t(e,x)). \end{aligned} \tag{D.4}$$

For a project in Stage 2, every arriving study is terminal. The probability that the third study arrives with new cost bin ℓ' and the total probability of study arrival are therefore

$$\begin{aligned} q_i^{\mathrm{S2}}(\ell' \mid \ell, t, z; \mathcal{K}_t(e,x)) &= p_i^A(s=2,\tau,\ell,z;\mathcal{K}_t(e,x))\, p_i^c(\ell' \mid s=2,\ell,z), \\ Q_i^{\mathrm{S2}}(\ell, t, z; \mathcal{K}_t(e,x)) &= p_i^A(s=2,\tau,\ell,z;\mathcal{K}_t(e,x)). \end{aligned} \tag{D.5}$$

These equations apply before $t = e+T$, where $T = 32$. At the start of $e+T$, every remaining project receives its terminal study immediately, with cost kernel p_i^c for its current stage, and chooses whether to complete or withdraw. No additional waiting cost, request fees and deposits, or exogenous withdrawal draw applies at this boundary.

D.3 Transition Estimates

Table D.1 reports the parameter estimates of these transition probabilities. The cost column reports parameters from the ordered-probit index. The other columns are from probit indices. We also report observed and fitted means. Receiving engineering tests is correlated with slower study arrival.

D.4 First-Study State Distribution

The first-study state is constructed separately from the later transition probabilities and corresponds to period $e + 1$ in the model. Paralleling the later-study transition models, we estimate an ordered probit for the first-study cost bin and probits for engineering-test status and whether the first study is terminal, each with fuel type, size, and whether the project increases capacity at an existing plant as covariates. Assuming these outcomes are independent conditional on project characteristics, we multiply their predicted marginal distributions to obtain the joint distribution. The sample is projects that entered the queue from 2012 to 2020, the same as in Table 6.

D.5 Potential Entrants

We construct the potential non-entrants as a pool of synthetic projects. We group observed entrants by transmission owner, fuel (wind, solar, battery, or gas), whether the project increases capacity at an existing plant, and size class ($\leq$ 20, 20–100, or $>$ 100 MW). For each group, we find the maximum observed entrant count for that group in any year. We then draw synthetic projects from that group in each year so that the total number of observed and synthetic projects is equal to the maximum. Synthetic projects' characteristics thus follow the empirical distribution of the observed entrants. For each year, we assign first-study quarters to the synthetic projects based on the empirical distribution of all observed entrants' first-study quarters in the same year.

The full construction creates 10,275 synthetic projects over 2012–2020. To keep the equilibrium simulation tractable, the baseline estimation uses a random 5% subsample of the synthetic pool (514 projects) as the potential non-entrants. The robustness analysis in Section 6.3 uses a random 10% subsample.

Table D.1: Transition Function Estimates

	Ordered-probit cost state	Study arrival	Receive engr. tests	Second study is terminal
Cutoff 2	0.801***			
	(0.039)			
Cutoff 3	1.162***			
	(0.106)			
Prior cost: $(0.01, 0.05]$	0.550***	0.068*		
	(0.083)	(0.039)		
Prior cost: $(0.05, 0.2]$	1.135***	0.035		
	(0.081)	(0.041)		
Prior cost: > 0.2	2.052***	-0.103*		
	(0.096)	(0.057)		
Waiting for third study		-0.404*		
		(0.209)		
Wind	0.588***	0.023	0.909**	-0.736***
	(0.111)	(0.052)	(0.365)	(0.249)
Solar	0.536***	0.129***	1.757***	-0.628***
	(0.074)	(0.043)	(0.216)	(0.111)
Battery	0.522***	-0.062	1.878***	-1.258***
	(0.095)	(0.056)	(0.340)	(0.154)
Capacity increase at existing plant	-0.288***	-0.037	0.019	0.097
	(0.064)	(0.038)	(0.213)	(0.112)
10–20 MW		0.036	0.746***	-0.588***
		(0.049)	(0.234)	(0.111)
20–100 MW		-0.013	0.845***	-1.481***
		(0.048)	(0.229)	(0.130)
> 100 MW		-0.183***	1.209***	-2.352***
		(0.052)	(0.314)	(0.162)
Receive engr. tests	0.013	-0.170***		
	(0.070)	(0.039)		
ln(# Higher priority)		-0.094***		
		(0.026)		
ln(# Higher priority) × Waiting for third study		-0.185***		
		(0.035)		
ln(# Higher priority in TO)		-0.127***		
		(0.016)		
ln(# Higher priority in TO)× Waiting for third study		-0.036		
		(0.026)		
Observed mean	0.506	0.112	0.744	0.192
Fitted mean	0.491	0.111	0.745	0.190
Observations	2,648	23,627	394	2,002

The sample includes projects that entered the queue from 2012 to 2020 and received non-terminal first studies. The cost-state column uses the 2,648 observed second- and third-study arrivals within this sample; its dependent variable is the cost category reported in the arriving study. The study-arrival column uses 23,627 project-quarter observations while projects await their second or third study. The engineering-test column uses 394 second-study arrivals among projects that had not previously received engineering tests; its dependent variable indicates whether the second study reports such tests. The terminal-study column uses 2,002 second-study arrivals; its dependent variable indicates whether that study is terminal. Projects whose first study is terminal do not contribute to these columns. The ordered-probit cost-state column uses four cost categories: $[0, 0.01]$, $(0.01, 0.05]$, $(0.05, 0.2]$, and > 0.2 million dollars per MW. The first cutoff is normalized to zero; Cutoff 2 and Cutoff 3 are the estimated higher cutoffs. Prior cost rows indicate the project's most recent prior study cost, in million dollars per MW. For the cost-state column, observed and fitted means are shares with new cost above \$0.05 million/MW among study arrivals. Other means compare average binary outcomes with predicted probabilities. The study-arrival probit also controls for queue-duration indicators and their interactions with waiting for the third study. Coefficients are reported with standard errors in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

D.6 Exogenous Withdrawal

Under the conditions of Lemma 2, projects do not endogenously withdraw between studies. We maintain this behavioral restriction when estimating the exogenous withdrawal process and treat observed intervening withdrawals as exogenous. We estimate their probability with a probit model. The estimation sample is a project-quarter panel of projects waiting for their next study. A project enters the sample in the quarter after a non-terminal first or second study. Its last observation is the quarter before its next study arrives or, if it withdraws first, the quarter in which it withdraws. Table D.2 reports parameters from the probit index. The implied average probability of an exogenous withdrawal is 0.029 for both stages.

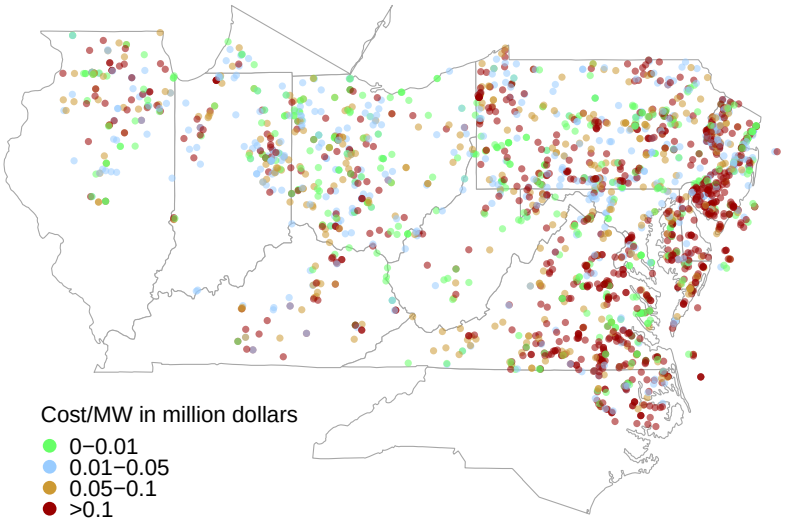
Table D.2: Exogenous Withdrawal Probability

Waiting for third study	-0.058
	(0.040)
Wind	-0.374***
	(0.081)
Solar	-0.622***
	(0.057)
Battery	-0.524***
	(0.069)
Capacity increase at existing plant	-0.391***
	(0.050)
ln(MW)	-0.189***
	(0.015)
Constant	-0.546***
	(0.089)
Observations	18,618
Fitted Mean	0.029

The sample consists of projects that entered from 2012 to 2020. An observation is a project-quarter combination, where a project waits for the second study after a non-terminal first study, or a project waits for the third study after a non-terminal second study. Only quarters with no new study arrival and an observed wait or withdrawal decision are included. The dependent variable equals one for withdrawal and zero for continued waiting. Waiting for third study indicates the latter waiting stage. Coefficient estimates are reported. Standard errors are in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

E Additional Figures and Tables

Figure E.1: Costs by Location



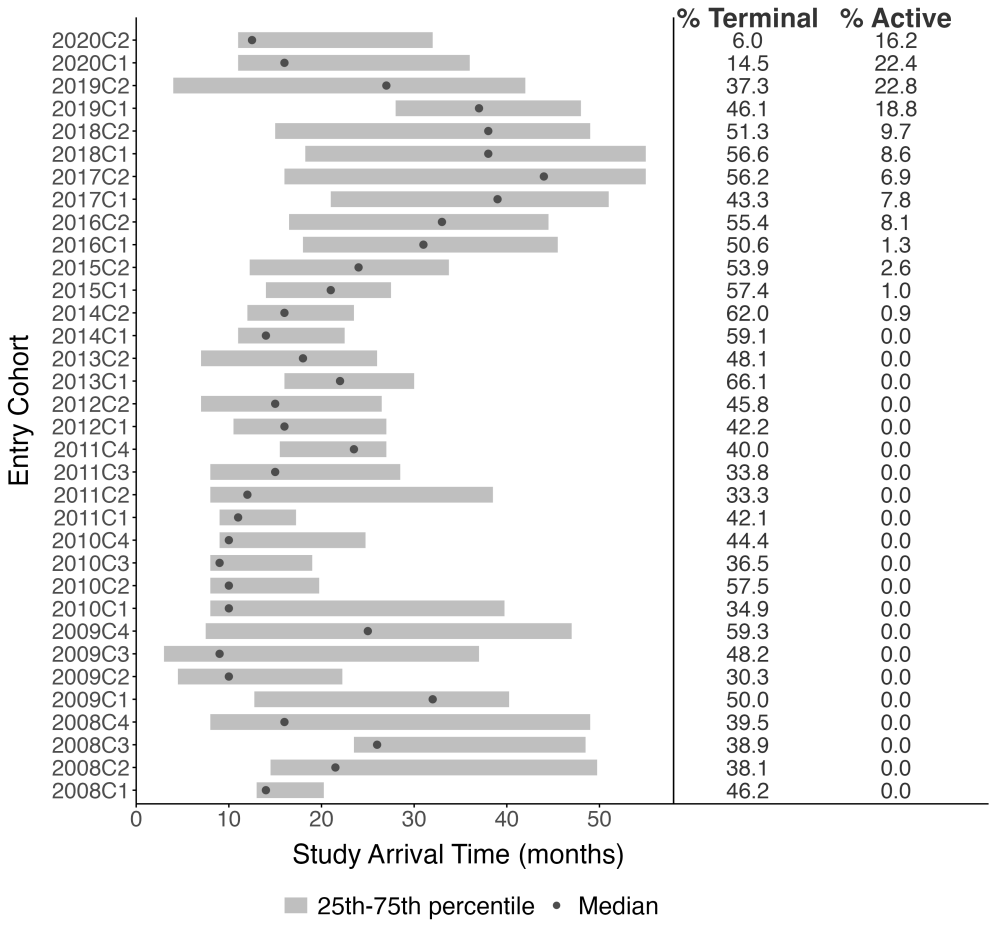
Second study interconnection cost estimates by location. Costs in millions of 2020 dollars per MW.

Table E.1: Top 15 Developers

Developer	No. of projects	Share of projects (%)	Share of capacity (%)	Cohorts with a project	Avg. proj. per cohort
Invenergy	83	4.3	8.1	20	4.2
Community Energy Solar	68	3.5	1.0	19	3.6
Dominion	63	3.3	5.1	21	3.0
PSEG	56	2.9	1.8	18	3.1
LS	47	2.4	3.8	22	2.1
NextEra	38	2.0	2.0	12	3.2
Apex	30	1.6	3.5	14	2.1
EDF	30	1.6	3.2	12	2.5
SunEnergy1	30	1.6	1.5	10	3.0
EffiSolar	29	1.5	0.2	5	5.8
IMG	29	1.5	0.3	10	2.9
EDP	28	1.5	1.8	13	2.2
Exelon	27	1.4	1.7	12	2.3
Urban Grid	27	1.4	1.2	6	4.5
AEP	25	1.3	0.3	12	2.1

Summary statistics for the top 15 developers in our sample by number of projects. Share of projects is the developer's project count as a percentage of the 1,923 queue projects for which a developer could be identified, out of 4,174 interconnection requests in the data. Share of capacity is the developer's total nameplate capacity as a percentage of the 190.3 GW of capacity in those same projects. Cohorts with a project is the number of queue entry cohorts (out of 34) in which the developer has at least one project.

Figure E.2: Study Wait Time (months) by Date of Queue Entry



This figure shows the time to receive the terminal study, measured in months, by entry cohort. The horizontal bar represents the interquartile range (25th-75th percentiles), and the dot indicates the median. "% Terminal" is the share of projects that have received the terminal study among projects entering in a given cohort. "% Active" is the share of projects that are still active among projects in that cohort.

Table E.2: Summary Statistics by Transmission Owner

	Mean	Std. Dev.	25%	75%
Panel A: Transmission owner-level totals and shares				
No. of projects	190	191	33	292
Total capacity (GW)	18.5	22.7	3.6	21.1
Avg. capacity per project (MW)	101.9	54.9	60.4	139.1
No. of withdrawals	102	96	20	168
Total withdrawn capacity (GW)	10.20	10.63	2.12	14.56
No. of completions	33	34	4	55
Completed capacity (GW)	2.34	3.13	0.07	3.42
Share renewables	0.69	0.25	0.63	0.88
No. of unique developers	32	27	10	50
Panel B: Within-transmission owner averages				
New projects per year	14.8	14.5	2.5	22.5
Waiting time (mos.)	26.8	4.7	23.9	29.4
First study cost ($M/MW)	0.14	0.15	0.07	0.16
Time to first study (mos.)	5.3	0.8	4.6	5.4
Second study cost ($M/MW)	0.20	0.17	0.10	0.20
Time to second study (mos.)	14.7	4.4	12.4	15.7
Third study cost ($M/MW)	0.11	0.09	0.05	0.15
Time to third study (mos.)	16.9	4.1	13.6	18.8
Transmission owners		22		

Panel A reports the distribution across transmission owners of aggregate statistics computed within each transmission owner. Unique developers counts distinct developers within each transmission owner; we are unable to match 2,251 of the 4,174 interconnection requests in the data (53.9%) to a developer that controls the project. Panel B first averages within each transmission owner across its projects, then reports the distribution of those transmission owner-level averages. New projects per year is the total number of projects in a transmission owner divided by the number of years spanned by its queue entries. Waiting time is months from queue entry to completion or withdrawal, excluding projects whose last known status is neither. Study costs (in millions of dollars per MW) and times (in months) are conditional on receiving the respective study. Renewables is solar, wind, and batteries.

Table E.3: New Study Arrival Probit Models With Additional Controls

	(1) Probit	(2) Probit	(3) Probit	(4) Probit	(5) Probit
ln(# Higher priority)	-0.031***	-0.032***	-0.020	-0.021*	-0.028***
	(0.006)	(0.006)	(0.012)	(0.012)	(0.007)
ln(# Higher priority) x Waiting for third study	-0.016**	-0.016**	-0.011	-0.011	-0.015**
	(0.008)	(0.008)	(0.007)	(0.007)	(0.008)
ln(# Higher priority in TO)	-0.016***	-0.014***	-0.020***	-0.020***	-0.023***
	(0.003)	(0.004)	(0.007)	(0.007)	(0.006)
ln(# Higher priority in TO) x Waiting for third study	-0.012*	-0.013*	-0.011*	-0.011*	-0.011
	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)
Waiting for third study	-0.216***	-0.215***	-0.243***	-0.243***	-0.221***
	(0.023)	(0.023)	(0.023)	(0.023)	(0.023)
Receive engr. tests	-0.027**	-0.028**	-0.030***	-0.030***	-0.028**
	(0.011)	(0.012)	(0.010)	(0.010)	(0.011)
Network allocation	-0.002	-0.003	0.000	-0.000	-0.003
	(0.012)	(0.012)	(0.012)	(0.012)	(0.012)
ln(Size of withdrawals in queue within TO)					-0.000
					(0.002)
ln(Size of projects in queue within TO)					0.008*
					(0.005)
Wholesale electricity price ($100/MWh)		0.042**		0.032	
		(0.020)		(0.024)	
Ordinance		-0.007		0.004	
		(0.005)		(0.005)	
Prior Study Controls	X	X	X	X	X
Transmission Owner FE			X	X	
Year FE			X	X	
Observations	28,917	28,917	28,917	28,917	28,917
Pseudo R^2	0.176	0.177	0.195	0.196	0.176

The sample includes projects queued from 2008–2020, observed over time at the quarterly level as they wait in the queue for study arrival. Marginal effects from probit regressions are reported. Standard errors in parentheses, clustered by transmission owner - entry cohort (416 clusters). The dependent variable is an indicator for study arrival (overall mean = 0.108; mean for second study is 0.160 and for third study is 0.056). Implied percentage effects reported in the text are computed from unrounded marginal effects and unrounded mean arrival rates. All specifications control for project size (three bins), fuel type (wind, solar, battery), if the project is a capacity increase at an existing plant, time spent waiting in the queue (controlled for flexibly with a separate indicator for each quarter since queue entry), as well as their interactions with an indicator for whether the project is awaiting its third study, and prior study characteristics, including the project's own cost estimate (three bins) from the prior study (e.g., the first study costs for projects awaiting the second study), indicators for having ever received each of the three engineering tests and an indicator for whether the project receives an explicit network upgrade cost. Specifications (2) and (4) control for wholesale electricity prices, defined as the average real-time locational marginal price over peak hours in that quarter, taken at whichever of the project's connected substations has the highest such price, in nominal $/MWh, and ordinance, an indicator for a local ordinance restricting renewable energy development. Specification (5) further controls for the log size of withdrawn projects waiting for the same or a later study within the same transmission owner over the past two quarters, and the log size of projects in the queue within the same transmission owner. *** p<0.01, ** p<0.05, * p<0.10.

Table E.4: New Study Arrival Probit Models With Different Time Controls

	(1) None	(2) Linear	(3) Quadratic	(4) YearxQtr FE	(5) TOxYear FE	(6) None
ln(# Higher priority)	-0.031***	-0.027***	-0.027**	-0.033***	-0.005	-0.033***
	(0.006)	(0.009)	(0.011)	(0.010)	(0.013)	(0.012)
ln(# Higher priority) x Waiting for third study	-0.016**	-0.017**	-0.017**	-0.012	-0.012*	0.014
	(0.008)	(0.008)	(0.008)	(0.008)	(0.007)	(0.012)
ln(# Higher priority in TO)	-0.016***	-0.016***	-0.016***	-0.016***	-0.026**	-0.025***
	(0.003)	(0.003)	(0.004)	(0.003)	(0.010)	(0.005)
ln(# Higher priority in TO) x Waiting for third study	-0.012*	-0.012*	-0.012*	-0.012*	-0.010	-0.005
	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)	(0.011)
Total queue-position effect (linear combinations)						
At second study	-0.047***	-0.043***	-0.043***	-0.049***	-0.031***	-0.058***
	(0.006)	(0.009)	(0.011)	(0.011)	(0.011)	(0.012)
At third study	-0.075***	-0.071***	-0.072***	-0.074***	-0.052***	-0.050***
	(0.006)	(0.008)	(0.010)	(0.010)	(0.010)	(0.013)
Sample Cohorts	2008–2020	2008–2020	2008–2020	2008–2020	2008–2020	2008–2018
Observations	28,917	28,917	28,917	28,295	28,105	14,109
Pseudo R^2	0.176	0.176	0.176	0.210	0.233	0.077

The sample includes projects queued from 2008–2020, observed over time at the quarterly level as they wait in the queue for study arrival. Marginal effects from probit regressions are reported. Standard errors are in parentheses and clustered by transmission owner - entry cohort (416 clusters). The dependent variable is an indicator for study arrival (overall mean = 0.108; mean for second study is 0.160 and for third study is 0.056). Implied percentage effects reported in the text are computed from unrounded marginal effects and unrounded mean arrival rates. All specifications control for project size (three bins), fuel type (wind, solar, battery), if the project is a capacity increase at an existing plant, time spent waiting in the queue (controlled for flexibly with a separate indicator for each quarter since queue entry), as well as their interactions with an indicator for whether the project is awaiting its third study, and prior study characteristics, including the project's own cost estimate (three bins) from the prior study (e.g., the first study costs for projects awaiting the second study), indicators for having ever received each of the three engineering tests and an indicator for whether the project receives an explicit network upgrade cost. Columns (1)-(4) differ in their time controls: column (1) includes none; column (2) includes a linear year trend; column (3) includes linear and quadratic year trends; and column (4) includes year-by-quarter fixed effects. In columns (2) and (3), the trend terms are also interacted with an indicator for renewable generation (wind, solar, or battery). Column (5) instead includes transmission-owner-by-year fixed effects. Column (6) uses the specification in column (1) but restricts the sample to projects entering before 2019; for this sample, the mean arrival rate is 0.173 for the second study and 0.115 for the third, and percentage effects are scaled accordingly. The total-effect rows report linear combinations of the marginal queue-position effects. *At second study* sums the system-wide and transmission-owner queue effects, while *At third study* additionally includes their interactions with the indicator for awaiting the third study. *** p<0.01, ** p<0.05, * p<0.10.

Table E.5: Effect of Active Higher-Queued Capacity on Probability of High Interconnection Cost

	10km (1)	20km (2)	Transmission owner (3)	Hierarchical cluster (4)
Panel A. High Total Cost Estimate (mean = 0.37)				
Log active higher-priority MW	-0.033*	-0.023	-0.010	0.003
	[-0.071, 0.006]	[-0.057, 0.011]	[-0.036, 0.018]	[-0.023, 0.028]
Geographic Region FE	X	X	X	X
Queue Year FE	X	X	X	X
Observations	2,328	2,328	2,328	2,328
R^2	0.520	0.519	0.231	0.230
Panel B. High Network Upgrade Cost Estimate (mean = 0.13)				
Log active higher-priority MW	0.006	0.009	0.004	0.009
	[-0.025, 0.036]	[-0.015, 0.034]	[-0.025, 0.028]	[-0.016, 0.033]
Geographic Region FE	X	X	X	X
Queue Year FE	X	X	X	X
Observations	2,328	2,328	2,328	2,328
R^2	0.477	0.477	0.177	0.204

Projects queuing from 2009–2020; the 2008 cohort is excluded because the queue data begin in 2008, leaving it with almost no observable prior interconnections. Wild cluster bootstrap (9,999 replications) clustered at the substation level in the 10 km and 20 km columns, the transmission owner level in the transmission owner column, and the hierarchical cluster level in the hierarchical cluster column; 95% confidence intervals in brackets. In Panel A, the dependent variable is an indicator for whether the total cost estimate in the second or third study is high, defined as exceeding \$0.1 million per MW. In Panel B, the dependent variable is an indicator for whether the network upgrade cost estimate in the second or third study is high, defined as exceeding \$0.1 million per MW. The regressor of interest is the log of total MW from the focal project and other active projects in the same or earlier interconnection cohort in the local geographic region in the quarter when the focal study is issued. An earlier cohort code indicates higher queue priority. A project is active if it is still waiting in the queue, meaning it has not completed or withdrawn, in the focal study's issue quarter. Columns vary the geographic definition: 10 km, 20 km, transmission owner, and hierarchical cluster. All specifications control for project size (three bins), fuel type (wind, solar, battery), whether the project is a capacity increase at an existing plant; whether PJM Regional Transmission Expansion Plan investment was above the 75th percentile of the RTEP distribution within the same transmission owner, measured over a four-year window spanning three years before and one year after the study issue date. The geographic region fixed effects are substation fixed effects in the 10 km and 20 km columns, transmission owner fixed effects in the transmission owner column, and hierarchical cluster fixed effects in the hierarchical cluster column. *** p<0.01, ** p<0.05, * p<0.10.

Table E.6: Robustness for Effect of a Costly Prior Interconnection on the Probability of High Total Cost

	Definition of nearby prior interconnection							
	10km		20km		Transmission owner		Hierarchical cluster	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A. Total cost > \$1m								
Costly prior	0.065	0.057	-0.003	-0.003	-0.010	-0.011	-0.037**	-0.032**
	[-0.038, 0.169]	[-0.042, 0.157]	[-0.091, 0.083]	[-0.089, 0.082]	[-0.075, 0.057]	[-0.076, 0.055]	[-0.067, -0.007]	[-0.063, -0.000]
Panel B. Total cost > 90th percentile								
Costly prior	0.089	0.060	-0.064	-0.073	-0.038	-0.012	-0.050**	-0.038*
	[-0.072, 0.244]	[-0.082, 0.203]	[-0.177, 0.051]	[-0.194, 0.060]	[-0.168, 0.196]	[-0.164, 0.233]	[-0.094, -0.002]	[-0.080, 0.006]
Panel C. Cost/MW > \$0.1m								
Costly prior	0.009	0.018	-0.013	-0.011	-0.030	-0.018	0.044	0.045
	[-0.119, 0.133]	[-0.115, 0.147]	[-0.130, 0.102]	[-0.135, 0.113]	[-0.112, 0.052]	[-0.082, 0.039]	[-0.012, 0.097]	[-0.011, 0.100]
Panel D. Cost/MW > 90th percentile								
Costly prior	-0.063	-0.028	-0.021	-0.006	-0.027	-0.037	0.068*	0.071*
	[-0.313, 0.162]	[-0.251, 0.180]	[-0.162, 0.122]	[-0.147, 0.137]	[-0.092, 0.102]	[-0.084, 0.135]	[-0.010, 0.144]	[-0.008, 0.148]
Geographic Region FE	X	X	X	X	X	X	X	X
Queue Year FE		X		X		X		X

Projects queuing from 2009–2020; the 2008 cohort is excluded because the queue data begin in 2008, leaving it with almost no observable prior interconnections. Wild cluster bootstrap (9,999 replications) clustered at the substation level in the 10 km and 20 km columns, the transmission owner level in the transmission owner column, and the hierarchical cluster level in the hierarchical cluster column; 95% confidence intervals in brackets. The dependent variable is an indicator for whether the cost estimate in the second or third study is high, defined as exceeding \$0.1 million per MW (mean = 0.37). The 90th percentile for the cost of prior interconnection is 9.17, 9.37, 7.54, and 7.64 million dollars for total cost and 0.27, 0.19, 0.16, and 0.15 million dollars for cost per MW, for the 10km, 20km, transmission owner, and hierarchical cluster definitions, respectively. The mean of the key independent variable, Costly prior interconnection, is 0.10, 0.02, 0.06, and 0.03 for the 10km specification across Panels A–D; 0.16, 0.04, 0.09, and 0.04 for the 20km specification; 0.29, 0.08, 0.15, and 0.07 for the transmission owner level specification; and 0.22, 0.07, 0.11, and 0.07 for the hierarchical cluster level specification. Each estimate corresponds to a separate regression, with rows corresponding to different thresholds for costly prior interconnection and columns corresponding to different definitions of nearby. All specifications control for project size (three bins), fuel type (wind, solar, battery), whether the project is a capacity increase at an existing plant; whether PJM Regional Transmission Expansion Plan investment was above the 75th percentile of the RTEP distribution within the same transmission owner, measured over a four-year window spanning three years before and one year after the study issue date; the log of total entrant capacity (MW) within the previous three years in the local geographic region; and the log capacity (MW) of higher-queued projects in the local geographic region. The geographic region fixed effects are substation fixed effects in the 10 km and 20 km columns, transmission owner fixed effects in the transmission owner column, and hierarchical cluster fixed effects in the hierarchical cluster column. *** p<0.01, ** p<0.05, * p<0.10.

Table E.7: Robustness for Effect of a Costly Prior Interconnection on the Probability of High Network Upgrade Cost

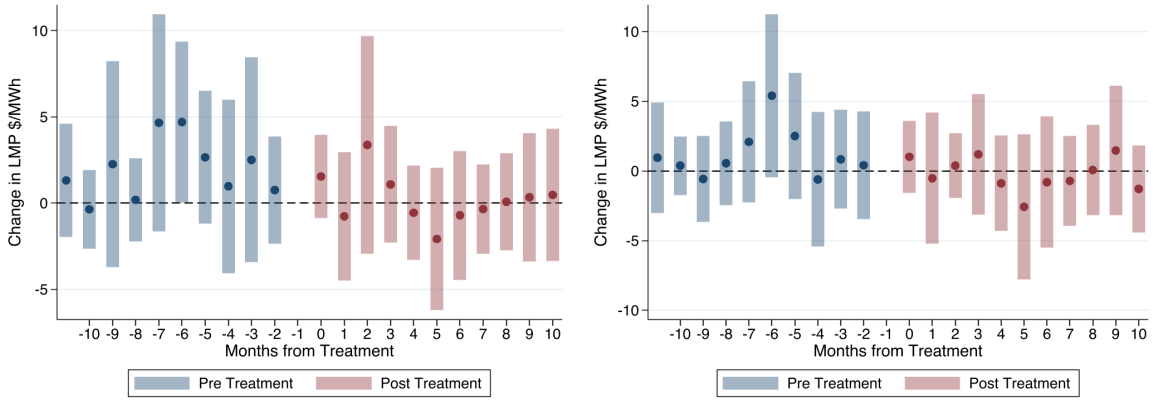
	Definition of nearby prior interconnection							
	10km		20km		Transmission owner		Hierarchical cluster	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A. Network upgrade cost > \$1m								
Costly prior	0.145	0.108	0.036	0.058	-0.060	-0.045	0.012	0.037
	[-0.233, 0.561]	[-0.224, 0.474]	[-0.155, 0.265]	[-0.132, 0.267]	[-0.124, 0.036]	[-0.112, 0.078]	[-0.060, 0.090]	[-0.031, 0.113]
Panel B. Network upgrade cost > 90th percentile								
Costly prior	0.067	0.021	-0.019	0.004	-0.055	-0.042	0.010	0.034
	[-0.334, 0.547]	[-0.293, 0.431]	[-0.225, 0.256]	[-0.218, 0.246]	[-0.113, 0.070]	[-0.091, 0.057]	[-0.061, 0.089]	[-0.037, 0.113]
Panel C. Network upgrade cost/MW > \$0.1m								
Costly prior	0.167	0.163	-0.083	-0.012	-0.084	-0.095	-0.019	-0.067
	[-2.782, 21.244]	[-3.994, 2.926]	[-7.776, 1.810]	[-3.464, 3.136]	[-1.380, 2.007]	[-0.740, 0.673]	[-3.136, 2.626]	[-2.264, 4.060]
Panel D. Network upgrade cost/MW > 90th percentile								
Costly prior	0.207	0.179	0.095	0.112	-0.051	-0.041	-0.008	0.007
	[-0.377, 0.755]	[-0.220, 0.579]	[-0.138, 0.366]	[-0.090, 0.332]	[-0.104, 0.070]	[-0.085, 0.057]	[-0.112, 0.067]	[-0.068, 0.081]
Geographic Region FE	X	X	X	X	X	X	X	X
Queue Year FE		X		X		X		X

Projects queuing from 2009–2020; the 2008 cohort is excluded because the queue data begin in 2008, leaving it with almost no observable prior interconnections. Wild cluster bootstrap (9,999 replications) clustered at the substation level in the 10 km and 20 km columns, the transmission owner level in the transmission owner column, and the hierarchical cluster level in the hierarchical cluster column; 95% confidence intervals in brackets. The dependent variable is an indicator for whether the network upgrade cost estimate in the second or third study is high, defined as exceeding \$0.1 million per MW (mean = 0.13). In this table the costly-prior indicators are defined using the prior interconnection's network upgrade cost. In Panel C, a prior interconnection is classified as costly if its network upgrade cost per MW exceeds \$0.1 million. The 90th percentile for the network upgrade cost of prior interconnection is 2.03, 2.03, 0.07, and 0.61 million dollars for network upgrade cost and 0.04, 0.02, 0.00, and 0.01 million dollars for network upgrade cost per MW, for the 10km, 20km, transmission owner, and hierarchical cluster definitions, respectively. The mean of the key independent variable, Costly prior interconnection, is 0.01, 0.01, 0.00, and 0.01 for the 10km specification across Panels A–D; 0.02, 0.02, 0.00, and 0.02 for the 20km specification; 0.02, 0.05, 0.01, and 0.05 for the transmission owner level specification; and 0.03, 0.03, 0.00, and 0.03 for the hierarchical cluster level specification. Each estimate corresponds to a separate regression, with rows corresponding to different thresholds for costly prior interconnection and columns corresponding to different definitions of nearby. All specifications control for project size (three bins), fuel type (wind, solar, battery), whether the project is a capacity increase at an existing plant; whether PJM Regional Transmission Expansion Plan investment was above the 75th percentile of the RTEP distribution within the same transmission owner, measured over a four-year window spanning three years before and one year after the study issue date; the log of total entrant capacity (MW) within the previous three years in the local geographic region; and the log capacity (MW) of higher-queued projects in the local geographic region. The geographic region fixed effects are substation fixed effects in the 10 km and 20 km columns, transmission owner fixed effects in the transmission owner column, and hierarchical cluster fixed effects in the hierarchical cluster column. The wild cluster bootstrap is numerically unstable for the costly-prior estimates in Panel C, columns (1)–(5), (7), and (8), because very few observations and clusters have a costly prior interconnection under this definition. *** p<0.01, ** p<0.05, * p<0.10.

Figure E.3: Effect of Interconnections with Network Upgrade Costs on Local Wholesale Prices

(a) Adding 20 MW New Capacity that Pays Network Upgrade Costs

(b) Adding 100 MW New Capacity that Pays Network Upgrade Costs



Panel (a) plots event-study estimates of the effect of a new project with capacity of at least 20 MW beginning operation at a substation on local wholesale electricity prices, focusing on substations where the entering project is responsible for transmission network upgrade costs as part of the interconnection process. Panel (b) uses a 100 MW threshold. Treated observations are substation-months in which a project above the corresponding capacity threshold begins operation and is responsible for network upgrade costs; control observations are substation-months without such entry and include substation-months with entry of smaller or non-network-upgrade-paying projects. The dependent variable is the monthly average peak-hour locational marginal price at the substation level (mean = $34.89/MWh), constructed from PJM locational marginal price data over 2008-2020. Estimates are obtained using the Callaway and Sant'Anna (2021) difference-in-differences estimator with doubly robust inverse probability weighting. Standard errors are clustered at the substation level.

Table E.8: Construction Cost Inputs by Fuel Type

	2012	2013	2014	2015	2016	2017	2018	2019	2020
Gas (combined cycle)	1.017	0.917	0.917	0.917	0.978	0.978	0.978	0.958	0.958
Wind (onshore)	–	2.105	1.918	1.814	1.758	1.739	1.424	1.408	1.498
Solar PV	–	2.408	2.270	1.899	1.391	1.523	1.201	1.167	1.076
Battery (4-hour battery)	1.750	1.489	1.331	1.026	0.849	0.701	0.621	0.572	0.521

The costs are in $ million/MW. Gas uses EIA natural-gas combined-cycle installation costs. Wind uses EIA onshore-wind construction costs. Solar uses EIA Form 860/923 utility-scale PV installed costs in $/kWAC, converted to $ million/MW. Battery uses BloombergNEF battery pack prices plus NREL ATB-based balance-of-system costs for a 4-hour utility-scale battery. Solar and battery costs are multiplied by 0.7 to reflect the investment tax credit only; no depreciation adjustment is included. "–" denotes no observation for that year. We extrapolate the missing values by fitting an exponential function.

F Additional Counterfactual Tables and Figures

Table F.1: By-Size Entrant Count Changes from Study Speedups

	(1)	(2)	(3)	(4)	(5)	(6)
	As-is	10%	20%	30%	40%	50%
Renewable	2,063	+1	+1	+1	+1	0
≤20 MW	1,008	0	0	0	0	0
20–100 MW	618	0	+1	+1	0	0
>100 MW	436	0	0	0	0	0
Non-renewable	644	+4	+8	+11	+13	+16
≤20 MW	335	+1	+1	+1	+2	+2
20–100 MW	128	+1	+2	+3	+4	+4
>100 MW	181	+2	+5	+6	+8	+10

Column (1) reports the as-is equilibrium in levels; columns (2)–(6) report changes in the number of entrants relative to as-is under study speedups of 10%, 20%, 30%, 40%, and 50%, respectively.

Table F.2: By-Size Per-MW Entry Surplus Changes from Study Speedups

	(1)	(2)	(3)	(4)	(5)	(6)
	As-is	10%	20%	30%	40%	50%
Renewable	200.8	+0.6	+1.4	+2.4	+3.0	+3.7
≤20 MW	117.5	+1.0	+2.0	+2.9	+3.4	+3.9
20–100 MW	185.4	+0.9	+1.8	+2.5	+2.8	+3.0
>100 MW	219.2	+0.4	+1.2	+2.3	+3.1	+3.9
Non-renewable	163.3	+6.4	+12.2	+17.1	+21.9	+26.5
≤20 MW	253.9	+3.8	+6.8	+8.9	+11.5	+14.1
20–100 MW	221.1	+7.8	+14.7	+20.5	+26.1	+31.4
>100 MW	156.9	+6.4	+12.2	+17.1	+22.0	+26.6

Column (1) reports the as-is equilibrium in levels; columns (2)–(6) report changes relative to as-is under study speedups of 10%, 20%, 30%, 40%, and 50%, respectively. Entry surplus is defined as $\sum_i k_i E \max\{EV_i - C_i, 0\}$ across all potential projects in the indicated category. Here k_i is nameplate capacity in MW. The per-MW average divides this total by the nameplate capacity of all potential projects in the indicated category, including non-entrants, and is reported in $1,000 per MW.

Table F.3: Fee and Deposit Counterfactuals at $400,000 per Project

	As-is	Fee	Deposit		Budget neutral	
	(1)	(2)	(3)	(4)	(5)	(6)
Completed (GW)	55.6	+0.4	+0.4	+0.7	+0.3	+0.6
Renewable	26.4	+0.1	+0.1	+0.3	+0.2	+0.4
Non-renewable	29.1	+0.3	+0.3	+0.4	+0.1	+0.3
Waiting time (quarters)	10.4	+0.5	+0.5	-0.1	+0.3	-0.1
Renewable	11.5	+0.6	+0.6	0.0	+0.4	-0.1
Non-renewable	6.8	+0.2	+0.2	0.0	0.0	-0.2
Entry (GW)	262.5	-1.4	-1.2	-0.7	-0.7	-0.4
Renewable	156.1	-1.5	-1.2	-0.9	-0.6	-0.4
Non-renewable	106.4	+0.1	0.0	+0.2	-0.1	0.0
Entry surplus ($Bn)	59.6	-0.8	-0.6	-0.5	-0.2	0.0
Renewable	36.3	-0.8	-0.6	-0.5	-0.2	-0.1
Non-renewable	23.3	0.0	0.0	+0.1	0.0	+0.1
Payments ($Bn)						
Collected at entry	0.0	+1.0	+1.0	+1.0	+1.1	+1.1
Returned to projects	0.0	0.0	+0.2	+0.3	+1.1	+1.1
Retained	0.0	+1.0	+0.8	+0.7	0.0	0.0
Wholesale price ($/MWh)	32.0	0.0	0.0	-0.1	-0.1	-0.1
Share of payment forfeited						
If withdraw before requesting the 2nd study		100%	100%	50%	100%	50%
If withdraw later		100%	100%	100%	100%	100%
If complete		100%	0%	0%	0%	0%
Forfeitures rebated to completers		No	No	No	Yes	Yes

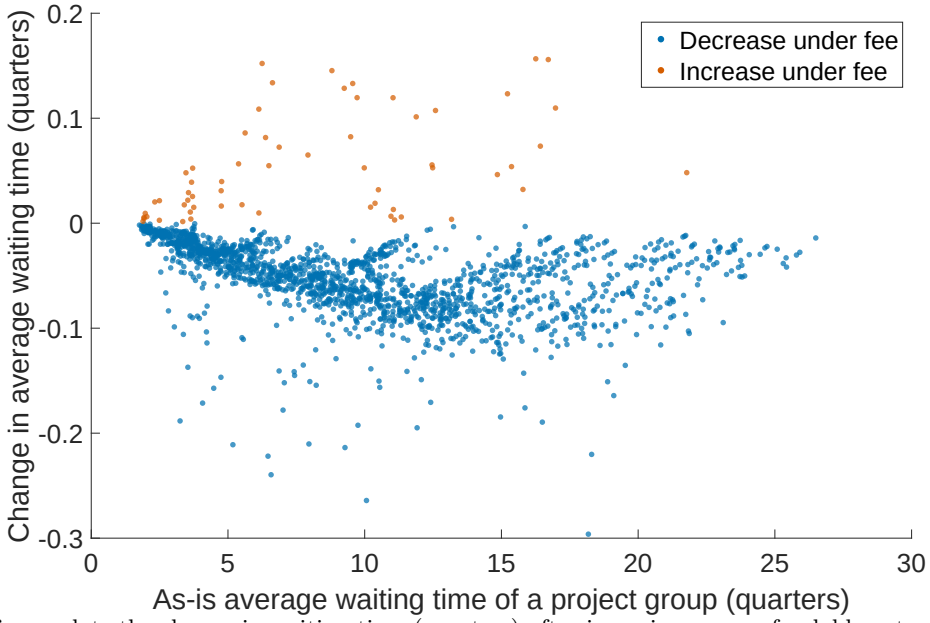
Column (1) reports the as-is equilibrium in levels; columns (2)–(6) report changes relative to as-is. All five fees and deposits charge $400,000 per project at queue entry, regardless of capacity. The bottom rows report the share of that payment forfeited at each outcome. In column (2), the entry fee is not refundable. In column (3), the deposit is refundable when the project completes but forfeited if the project withdraws. In column (4), the deposit is 50% refundable if the project withdraws before requesting the second study; thereafter, the deposit is refunded if the project completes but forfeited if it withdraws. In columns (5) and (6), the forfeiture rules are the same as in columns (3) and (4), respectively, before a project completes, but all forfeited deposits are rebated to completing projects in an equal amount per project. Entry surplus is defined as $\sum_i k_i E \max\{EV_i - C_i, 0\}$ across all potential projects and is reported in $Bn. Here k_i is nameplate capacity in MW. Wholesale price is the long-run price.

Table F.4: By-Size Per-MW Entry Surplus Changes: $20,000/MW and $400,000 Flat

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	As-is		Per-MW			Flat	
Renewable	200.8	-16.7	-13.2	-11.8	-4.2	-3.5	-2.9
≤20 MW	117.5	-15.6	-13.3	-10.4	-21.2	-17.9	-14.0
20–100 MW	185.4	-16.7	-13.5	-12.1	-5.2	-4.2	-3.7
>100 MW	219.2	-16.8	-13.0	-12.0	-1.3	-1.1	-1.0
Non-renewable	163.3	-12.6	-10.0	-7.8	-0.2	-0.1	+0.5
≤20 MW	253.9	-15.7	-11.6	-8.0	-28.0	-19.2	-14.1
20–100 MW	221.1	-14.1	-11.1	-8.1	-4.3	-3.2	-2.1
>100 MW	156.9	-12.4	-9.9	-7.7	+0.9	+0.7	+1.1
Share of payment forfeited							
If withdraw before requesting the 2nd study		100%	100%	50%	100%	100%	50%
If withdraw later		100%	100%	100%	100%	100%	100%
If complete		100%	0%	0%	100%	0%	0%

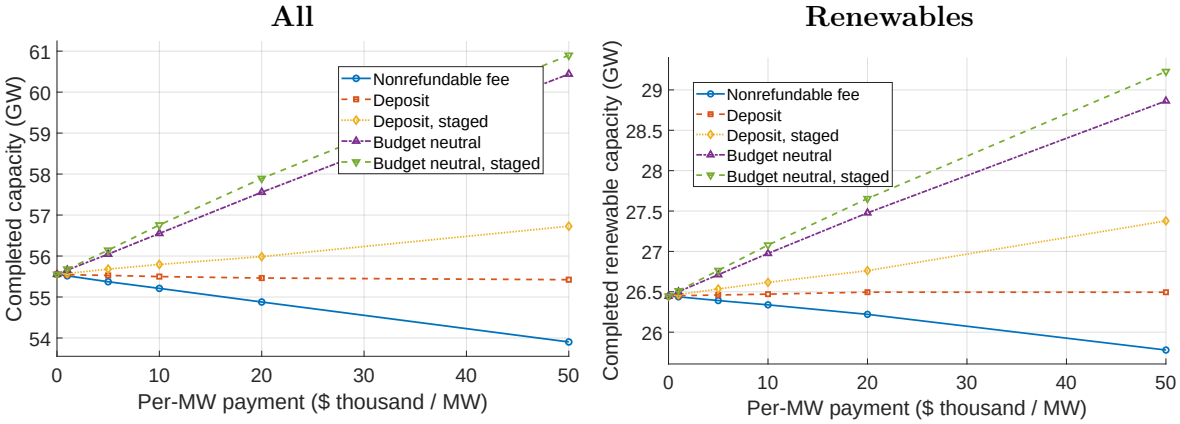
Surplus entries are in $1,000 per MW. Column (1) reports the as-is equilibrium in levels; columns (2)–(4) and (5)–(7) report changes relative to as-is under the entry cost designs of $20,000 per MW of nameplate capacity (per-MW design) and $400,000 per project (flat design), respectively, at queue entry. The bottom rows report the share of that payment forfeited at each outcome. In column (2), the entry fee is not refundable. In column (3), the deposit is refundable when the project completes but forfeited if the project withdraws. In column (4), the deposit is 50% refundable if the project withdraws before requesting the second study; thereafter, the deposit is refunded if the project completes but forfeited if it withdraws. Columns (5)–(7) report the same outcomes under the flat fee. Entry surplus is defined as $\sum_i k_i E \max\{EV_i - C_i, 0\}$ across all potential projects in the indicated category. Here k_i is nameplate capacity in MW. The per-MW average divides this total by the nameplate capacity of all potential projects in the indicated category, including non-entrants.

Figure F.1: Change in Waiting Time by Project Group under a $20,000/MW Nonrefundable Fee



The figure plots the change in waiting time (quarters) after imposing a nonrefundable entry fee of $20,000/MW, relative to the as-is equilibrium. Each group is defined as a fuel type, TO, arrival quarter, and size-bin combination. We group projects into eight size bins with cutoffs at 10, 20, 50, 100, 200, 500, and 1000 MW. There are 1,845 project groups. For each group, we plot the change in waiting time. Waiting time decreases for 1,784 groups, representing about 2,634 projects and 253 GW, and increases for 61 groups, representing about 73 projects and 9 GW.

Figure F.2: Completed GW under Each Per-MW Fee and Deposit



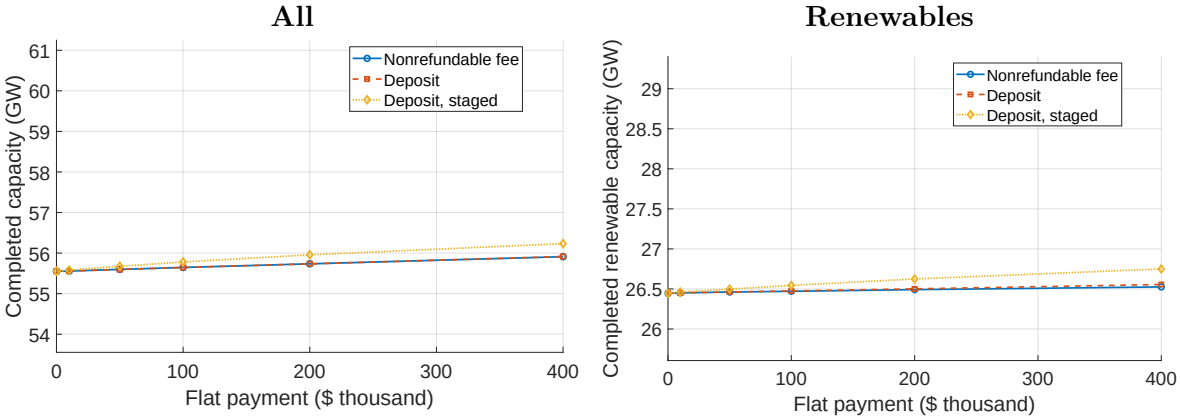
Each point is an equilibrium simulation. Five series: nonrefundable fee (solid), deposit (dashed, square), staged deposit (dotted, diamond), where projects receive 50% of the deposit if they withdraw before requesting the second study, deposit with budget-neutral rebate (dotted, triangle), and staged deposit with budget neutral rebates (dashed, inverted triangle). The as-is equilibrium is when fees are 0.

Table F.5: By-Size Entrant Count Changes: $20,000/MW and $400,000 Flat

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	As-is	Per-MW			Flat		
Renewable	2,063	-60	-52	-37	-183	-158	-102
≤20 MW	1,008	-41	-36	-23	-179	-154	-98
20–100 MW	618	-12	-10	-8	-4	-3	-3
>100 MW	436	-8	-6	-5	-1	-1	-1
Non-renewable	644	-11	-9	-6	-40	-32	-25
≤20 MW	335	-4	-3	-2	-40	-32	-25
20–100 MW	128	-2	-2	-1	-1	-1	0
>100 MW	181	-5	-4	-3	0	0	0
Share of payment forfeited							
If withdraw before requesting the 2nd study		100%	100%	50%	100%	100%	50%
If withdraw later		100%	100%	100%	100%	100%	100%
If complete		100%	0%	0%	100%	0%	0%

Column (1) reports the as-is equilibrium in levels; columns (2)–(4) and (5)–(7) report changes relative to as-is under fees and deposits of $20,000 per MW of nameplate capacity (per-MW design) and $400,000 per project (flat design), respectively, at queue entry. The bottom rows report the share of that payment forfeited at each outcome. In column (2), the entry fee is not refundable. In column (3), the deposit is refundable when the project completes but forfeited if the project withdraws. In column (4), the deposit is 50% refundable if the project withdraws before requesting the second study; thereafter, the deposit is refunded if the project completes but forfeited if it withdraws. Columns (5)–(7) correspond to columns (2)–(4) under the flat design.

Figure F.3: Completed GW vs. Flat Fee Level



Each point is an equilibrium simulation. Three series: nonrefundable fee (solid), deposit (dashed), and staged deposit (dotted) where projects receive 50% of the deposit if they withdraw before requesting the second study, no refund if they withdraw after, and a full refund if they complete.